

Search and Negotiation with Biased Beliefs in Consumer Credit Markets*

Erik Berwart Sean Higgins Sheisha Kulkarni Santiago Truffa

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Abstract

How do biased beliefs about the interest rate distribution affect search, negotiation, and loan terms in consumer credit markets? In collaboration with Chile's financial regulator, we conducted a randomized controlled trial with 112,063 loan seekers. Randomly eliciting beliefs about interest rates led participants to search more and obtain lower rates. Most participants underestimated both interest rate levels and dispersion. Showing them a price comparison tool we built using administrative data on the universe of consumer loans caused them to update their beliefs, negotiate more, obtain lower rates without searching more, and be more likely to take out a loan.

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*Berwart: Comisión para el Mercado Financiero, eberwart@cmfchile.cl. Higgins: Department of Finance, Kellogg School of Management, Northwestern University, sean.higgins@kellogg.northwestern.edu. Kulkarni: Department of Finance, McIntire School of Commerce, University of Virginia, sk7nn@virginia.edu. Truffa: Facultad de Economía y Negocios, Universidad Adolfo Ibáñez, santiago.truffa@uai.cl. We gratefully acknowledge funding from the Think Forward Initiative, ING Bank; Financial Institutions and Markets Research Center, Avi Nash Fund, and Guthrie Center for Real Estate Research, Northwestern Kellogg; Household Finance Small Grant Program, Alfred P. Sloan Foundation and NBER; Sparkassenstiftung für Internationale Kooperation; University of Virginia; Fondecyt iniciación: 11200010; Digital Credit Observatory and Lab for Inclusive Fintech, UC Berkeley; and Consumer Protection Research Initiative, Innovations for Poverty Action. We are very grateful to our discussants Bronson Argyle, Deniz Aydın, Claire Célériér, Ben Charoenwong, Isis Durmeyer, John Grigsby, Chris Hansman, Clément Mazet-Sonilhac, Monica Paiella, Kelly Shue, Guosong Xu, and Constantine Yannelis. We thank seminar participants at AIEA, Bank of Canada, BI Norwegian, Central Bank of Chile, CFPB, Chicago Federal Reserve, CUHK, Emory, Erasmus, FCA, FDIC, HEC Montréal, Houston, ISB, MIT, Northwestern, NYU, Oxford, Peking, Philadelphia Federal Reserve, PUC Chile, UC Berkeley, Universidad de los Andes, Virginia, WashU, and Wayne State, as well as participants at numerous conferences including BEAM, CEPR Household Finance, NBER Summer Institute Household Finance, and Stanford SITE Financial Regulation for helpful comments. We thank Arturo Charleston, Vicente Corral, Andrés Cruz, Matías Fuentes, Jora Li, Nicolas Min, Erick Molina, Limin Peng, Carlos Restituyo, Agustín Valenzuela Pidal, Francisco Villarroel, and Honghao Wang for their excellent research assistance. This RCT was pre-registered (<https://www.socialscisceregistry.org/trials/8553>). IRB approvals: NBER #22_099, #23_154, #24_138, Northwestern University #STU00213001, and Universidad de los Andes #CEC201946.

Consumer credit markets feature large amounts of *within-consumer* price dispersion (Stango and Zinman, 2016; Ponce, Seira, and Zamarripa, 2017). Even though many consumers pay substantial costs by borrowing at higher rates than they could, this price dispersion can persist in equilibrium if consumers engage in limited search and negotiation (Stahl, 1989; Hortaçsu and Syverson, 2004; Allen, Clark, and Houde, 2014).¹ Why, then, do consumers not search or negotiate more? The existing literature has focused on *costs* that prevent search or negotiation, including time and travel costs, high rejection rates, cognitive effort to compare complex offers, and the costs of gathering additional quotes to use in a negotiation.² We focus instead on the expected *benefits*, and test whether biased beliefs about the interest rate distribution constrain search and negotiation.

Motivated by a model of sequential search and negotiation with potentially biased beliefs, we conducted a randomized controlled trial (RCT) in close collaboration with Chile’s financial regulator, the Comisión para el Mercado Financiero (CMF). The RCT, conducted with 112,063 Chileans searching for loans, tests how correcting biased beliefs about the distribution of interest rates affects search, negotiation, and loan terms in consumer credit markets. We randomize both whether we measure beliefs about the interest rate distribution and whether we show loan seekers a price comparison tool designed to correct biased beliefs. We built the price comparison tool using administrative data from CMF on the universe of consumer loans merged with borrower characteristics. The tool shows participants the conditional distribution of interest rates based on similar loans obtained recently by similar borrowers.

Biased beliefs can affect both search and negotiation in our model, as they affect the perceived benefits of search and thus the consumer’s reservation rate. In a market without negotiation, consumers who underestimate the first moment of the distribution of interest rates (as we find most consumers do) will search more than is optimal because they will overestimate the expected benefit from another draw from the distribution of rates, while consumers who underestimate the second moment of the distribution (as we also find that most consumers do) will search less than is optimal because they underestimate the expected benefit from another draw. In a market with negotiation, however, a consumer’s reservation rate—which is a function of their beliefs and their search cost—is no longer a sufficient statistic for search. Thus, correcting biased beliefs may lead consumers to obtain lower interest rates by *negotiating more successfully*, rather than by searching more.

¹Consumers incur a substantial cost to borrow at higher rates than they could obtain with more search or negotiation. In the US auto loan market, the average borrower pays \$488 more in present value for a \$17,000 car (Argyle, Nadauld, and Palmer, 2023). In mortgage markets, Woodward and Hall (2012) estimate that borrowers pay \$1,000 more for a \$100,000 mortgage, and Bhutta, Fuster, and Hizmo (2026) estimate that borrowers pay \$6,250 more for a median \$250,000 mortgage.

²On physical search costs across branches, see Allen, Clark, and Houde (2013) and Argyle, Nadauld, and Palmer (2023). On high rejection rates leading to higher per-offer search costs for less creditworthy borrowers, see Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao (2024). On the cognitive effort required to compare offers, see Galenianos and Gavazza (2022). On the costs of gathering quotes to negotiate in mortgage markets, see Allen and Li (2025).

Our model makes predictions about heterogeneous treatment effects of the price comparison tool on negotiation based on the direction and magnitude of consumers’ biases about the first and second moments of the interest rate distribution, which we test in our RCT. For consumers who overestimate or substantially underestimate the first moment of the distribution, our model predicts that the tool will not have a treatment effect on negotiation, while it will increase the probability of successfully negotiating for those who somewhat underestimate the first moment. For those who underestimate the second moment of the interest rate distribution, correcting these underestimates of the second moment should increase negotiation while not necessarily increasing search.

We test these predictions from the model in an RCT where we recruited participants through Google ads targeted to people searching for keywords related to consumer loans in Chile. After participants clicked on the Google ad and consented to participate in the study, we collected their contact information and national ID numbers, which we use to conduct follow-up surveys and track participants’ future loan outcomes in administrative data.³ We then had them fill out a baseline survey that randomized whether we ask them their beliefs about the distribution of interest rates (which we refer to as the “elicit beliefs” treatment). Merely eliciting participants’ beliefs about interest rates led them to search at 0.13 (3.9%) more institutions and to obtain lower interest rates—1.2% lower according to administrative data and 7.1% lower according to survey data—than participants who were not asked their beliefs.

After the baseline survey, we cross-randomized whether we show participants a price comparison tool, a simple tool showing our estimate of the cost savings from search in pesos, or a control video. We then elicit beliefs again from those assigned to the elicit beliefs treatment, to measure the extent to which participants update their beliefs after seeing the price comparison or simple tool. To measure real outcomes on search, negotiation, and loan terms, we use a combination of follow-up phone surveys and administrative data from CMF.

Participants had biased beliefs about both the first and second moments of the interest rate distribution. While there is significant heterogeneity in beliefs, the vast majority underestimated both the interest rate they would get on the loan they took out and the dispersion in rates. We measure whether participants underestimated the rate they would get by comparing actual interest rates obtained by participants after the study (according to administrative data) to their beliefs about the rate they would obtain. On the first moment, nearly three-quarters (73.2%) thought they would obtain an interest rate lower than what they actually obtained. On the second moment, 74.8% of participants underestimated dispersion in the interest rates banks could offer them compared to

³Chile’s national ID number, or *rol único tributario* (RUT), is commonly used in everyday life. For example, people are asked to give their national ID numbers when they check out at the grocery store. The data merge using national ID number was completed in accordance with the terms of the collaboration with CMF and with the study’s IRB protocols.

administrative data.⁴

We then test the effects of a price comparison tool designed to correct biased beliefs on (i) beliefs about interest rates, (ii) search behavior, (iii) negotiation, and (iv) loan terms. We built the price comparison tool using administrative data on loan and borrower characteristics for the universe of consumer loans in Chile, i.e., data from over 1.8 million loans to 1.2 million borrowers over two years. The tool shows treated participants the conditional distribution of interest rates that similar borrowers obtained for similar loans over the previous six months.

Immediately after seeing the price comparison tool, simple tool, or control video, participants assigned to the elicit beliefs treatment were again asked their beliefs about the distribution of interest rates. The price comparison tool caused participants to update and report expecting to receive a 16.2 pp *higher* interest rate on the loan they obtain, or a 55.4% increase compared to the control mean posterior belief of a 29.2% expected annual interest rate. The price comparison tool also led participants to increase their expectation of how much price dispersion they face in the market by 15.9 pp, or 68.7% relative to the control mean posterior of 23.2 pp dispersion in annual interest rates.⁵

To measure effects on search behavior, negotiation, and loan terms, we combine the administrative data on originated loans with rich data on participants' search histories that we collected in a follow-up phone survey conducted with a subset of 6,132 participants. The price comparison tool increased negotiation without increasing search. Specifically, the tool led consumers to be 3.7 pp (38.7%) more likely to negotiate, but had no effect on the number of institutions at which participants searched for information (which includes "soft search" such as visiting a bank website or branch to get a sense of the interest rate the bank would offer without formally applying), the number of institutions at which they applied for a loan, or the specific institutions at which they searched. While the lack of a treatment effect on search could be due to offsetting effects of updating inaccurate beliefs about the first and second moments of the distribution, even among subsamples that underestimated the second moment and did not underestimate the first moment we

⁴The questions we use to measure beliefs about dispersion follow those used to elicit uncertainty about future earnings (Guiso, Jappelli, and Pistaferri, 2002), returns to schooling (Attanasio and Kaufmann, 2014), and macroeconomic growth (Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024). We ask participants the lowest and highest rates a bank could offer them and the percent of offers with rates above the midpoint of the two to capture potential asymmetry in the distribution. Our preferred measure of dispersion—due to its simplicity—is the highest minus lowest rate a bank could offer them, which is proportional to the standard deviation assuming rates follow a symmetric triangular distribution (as in Attanasio and Kaufmann, 2014; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024; Bassi, Rasul, Veroux, and Vitali, 2026). We also show robustness to assuming a split triangular distribution, exploiting our asymmetry question (as in Guiso, Jappelli, and Pistaferri, 2002; Attanasio and Kaufmann, 2014; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024). Attempting to elicit a more detailed distribution led to higher dropout in our pilot, consistent with Weber, D'Acunto, Gorodnichenko, and Coibion (2022).

⁵Although we winsorize responses to these interest rate questions at the 95th percentile (as pre-registered), the distributions are nevertheless quite skewed and the medians are substantially lower than the means. The control median posterior of the annual rate people expect to get is 18%, while the control median posterior of dispersion is 10.7 pp.

do not find a statistically significant impact on search. Instead, as highlighted by our model, the lack of an effect on search is likely due to the possibility of negotiating, which breaks the sufficient statistic relationship between a consumer’s reservation rate and search.

By negotiating more successfully, treated participants obtained 13.1% more offers and 11.9% lower interest rates (measured in the follow-up survey). In a subsequent survey we conducted, we find that much negotiation occurs prior to receiving a formal offer, which explains why negotiating also affects the probability of receiving an offer: in many cases the lender first informally offers an interest rate, then only if terms are agreed on does the lender issue a formal offer. In administrative data, we observe that treated participants are 4.7% more likely to take out a loan, which causes a selection effect when we regress interest rates—which are only observed for those who take out a loan—on treatment. If the tool causes “always-takers” to receive lower interest rates and if “compliers” who only take out a loan if treated are more marginal borrowers who receive higher interest rates, the inclusion of these higher-rate compliers only in the treatment group will pull the true negative treatment effect toward zero. Indeed, we do not find a treatment effect on interest rates in administrative data; however, when we partition the sample into likely always-takers vs. compliers using a machine learning method based on Chernozhukov, Demirer, Duflo, and Fernández-Val (2025), we find that the tool causes a 3.5% reduction in interest rates for likely always-takers.

We next test two key predictions of our model. In the model, the lender does not observe the consumer’s beliefs when making its initial offer, but does receive a signal about the consumer’s beliefs during the negotiation. The first prediction is that the effect of the price comparison tool on negotiation should be non-monotonic in the bias about the first moment of the interest rate distribution. Intuitively, consumers who overestimated the first moment will not benefit from updating their beliefs closer to the truth: the lender will assume—based on the signal it receives about the consumer’s beliefs during the negotiation—that the consumer will likely accept the initial offer. Consumers who vastly underestimated the first moment also will not benefit: their partially updated beliefs will still be too far below the true distribution, and thus the lender will infer that it most likely cannot profitably lend to them, and will not incur the cost of negotiating. Meanwhile, consumers who somewhat underestimated the first moment will successfully negotiate more: their updated beliefs make it more likely that the lender can still profitably lend to them at a lower, negotiated rate, so the threat of the consumer walking away leads the lender to lower the rate. The second prediction is that the price comparison tool should cause consumers who underestimated dispersion to successfully negotiate more (but not necessarily search more).

Our results align with these two predictions. First, the treatment effect on negotiation is non-monotonic in bias about the first moment of the distribution: there is no treatment effect for those who overestimated the first moment, nor for those who vastly underestimated it, while the effect on negotiation is concentrated among those who somewhat underestimated the first moment. Second,

the effect of the price comparison tool on negotiation is concentrated among those who underestimated dispersion, increasing their probability of negotiating by 7.9 pp (77.7%). Meanwhile, the tool did not have an effect on negotiation for those who did not underestimate dispersion.

Our paper makes three main contributions. First, we examine the role of *biased beliefs* about the interest rate distribution as a constraint to search and negotiation in consumer credit markets, despite these markets featuring high within-borrower price dispersion, and thus high interest costs incurred by borrowers who do not search or negotiate much. High price dispersion faced by the same borrower (or, in some studies, observationally similar borrowers) is a hallmark of consumer credit markets around the world (Zinman, 2015), including markets for credit cards (Stango and Zinman, 2016; Ponce, Seira, and Zamarripa, 2017), consumer loans (Karlan and Zinman, 2019; Cuesta and Sepúlveda, 2021), auto loans (Argyle, Nadauld, and Palmer, 2023), and mortgages (Allen, Clark, and Houde, 2014; Gurun, Matvos, and Seru, 2016; Guiso, Pozzi, Tsoy, Gambacorta, and Mistrulli, 2022; Coen, Kashyap, and Rostom, 2024). Due to this price dispersion, not searching or negotiating much leads borrowers to pay substantially higher interest costs (Allen, Clark, and Houde, 2019; Argyle, Nadauld, and Palmer, 2023; Bhutta, Fuster, and Hizmo, 2026). Our treatments increase search or negotiation and thereby lower consumers' interest rates: eliciting interest rate beliefs increases search, while correcting biased beliefs increases negotiation.

We build on studies that have focused primarily on the role of *costs* that prevent search and negotiation, by relaxing the assumption—made due to data limitations—that consumers have correct beliefs about the distribution of prices from which they are drawing, and thus know the expected *benefits*. For search, these costs include physical search costs across branches (Allen, Clark, and Houde, 2013; Argyle, Nadauld, and Palmer, 2023), high rejection rates (Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao, 2024), and the cognitive effort of comparing offers (Galenianos and Gavazza, 2022)—especially since financial products are often complex (Célérier and Vallée, 2017; Kulkarni, Truffa, and Iberti, forthcoming) and can include shrouded costs (Campbell, Jackson, Madrian, and Tufano, 2011; Stango and Zinman, 2014; Ferman, 2016; Alan, Cemalcilar, Karlan, and Zinman, 2018). For negotiation, costs include the disutility of negotiating (Jindal and Newberry, 2018), the time and effort of repeated back-and-forth bargaining (Backus, Blake, Larsen, and Tadelis, 2020), and potentially the need to gather competing offers to credibly threaten to walk away (Allen, Clark, and Houde, 2019; Allen and Li, 2025). We show that individuals do *not* have correct beliefs about the interest rate distribution they face, and that a price comparison tool shifts their beliefs and leads them to obtain better interest rates by negotiating more.⁶

⁶In other contexts beyond consumer financial markets, various papers have studied how beliefs affect search or negotiation. In labor markets, Conlon, Pilossoph, Wiswall, and Zafar (2018) and Bandiera, Bassi, Burgess, Rasul, Sulaiman, and Vitali (2025) study how beliefs affect search, Jäger, Roth, Roussille, and Schoefer (2024) study how beliefs affect search and negotiation intentions, and Caldwell, Haegele, and Heining (2026) show that workers signal their beliefs to employers by providing their salary expectations before they receive a job offer. In education markets,

Second, we show how biased beliefs about the first and second moment of the distribution each affect negotiation in theory, then test these model predictions empirically. We build on models of consumer credit markets that feature sequential search (Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao, 2024; Argyle, Nadauld, and Palmer, 2023) and negotiation (Allen, Clark, and Houde, 2013, 2019; Allen and Li, 2025), but depart from prior models in two ways. First, we introduce biased beliefs and show theoretically how correcting biased beliefs about each moment affects negotiation. Second, prior models of negotiation in mortgage markets assume that negotiation requires searching for additional offers; however, in our detailed survey data on search histories, initial offers, and negotiated rates, we observe that many consumers in our setting successfully negotiate without an outside offer. This is consistent with the finding in Grennan and Swanson (2020) that negotiation may only require accurate information about the price distribution. Our model predicts that the effect of (partially) correcting biased beliefs by showing consumers the price comparison tool is non-monotonic in the consumer’s initial bias about the first moment of the distribution; meanwhile, for consumers who underestimate dispersion, the tool will increase negotiation without necessarily affecting search. We then take these model predictions to the data, exploiting the randomized assignment of participants to the price comparison tool, and find evidence consistent with both predictions.

Third, we contribute to the literature on the effects of beliefs on financial decision-making by experimentally correcting biased beliefs about the distribution of interest rates a consumer faces. Local house price movements shape expectations about future house price growth (Kuchler and Zafar, 2019), and increased expectations about house price growth cause increases in real estate investments (Armona, Fuster, and Zafar, 2019; Liu and Palmer, 2026). Experiencing low stock market returns makes people more pessimistic about future stock market returns and less likely to participate in the stock market (Malmendier and Nagel, 2011), while experiencing inflation makes people expect more inflation in the future and choose fixed-rate mortgages (Malmendier and Nagel, 2016). Correcting biased beliefs about the dollar costs of payday loans reduces loan demand (Bertrand and Morse, 2011), while more experienced payday borrowers have more accurate beliefs (Allcott, Kim, Taubinsky, and Zinman, 2022). We show that experimentally correcting biased beliefs about the interest rate distribution leads consumers to obtain lower interest rates by negotiating more, without needing to search more.

1 Model

We present a simple theoretical framework to illustrate how biased beliefs affect search and negotiation and to generate testable predictions that motivate our RCT design and empirical analysis.

Agte, Allende, Kapor, Neilson, and Ochoa (2024) and Arteaga, Kapor, Neilson, and Zimmerman (2022) study how correcting biased beliefs about school quality and admission probabilities affects search, and Bleemer and Zafar (2018) show that correcting biased beliefs about college returns increases intended college attendance.

We build on models of consumer credit markets that feature sequential search (Agarwal, Grigsby, Hortagsu, Matvos, Seru, and Yao, 2024; Argyle, Nadauld, and Palmer, 2023), where consumers search across lenders because they do not know which lenders will offer which rates, and negotiation (Allen, Clark, and Houde, 2013, 2019; Allen and Li, 2025). We depart from these models in two ways.

First, we allow for consumers to have biased beliefs about the distribution of interest rates they face—which we measure empirically and shift experimentally with a price comparison tool. In contrast, standard models in the literature assume consumers have accurate beliefs due to data limitations, namely not observing consumers’ beliefs. These models would predict that showing consumers the true distribution should have no effect, since they already know it. In contrast, we find that the price comparison tool leads people to obtain lower interest rates by negotiating more.

Second, we include negotiation in a sequential search model. We model negotiation slightly differently than prior papers on search and negotiation in consumer credit markets (Allen, Clark, and Houde, 2013, 2019; Allen and Li, 2025), however, as in these other models negotiation only occurs if the consumer pays a search cost to collect offers from other banks. These models would therefore predict that search and negotiation always occur together and move in the same direction, whereas we observe in our detailed survey data that negotiation frequently occurs after the consumer receives an offer but *without* the consumer obtaining an offer or quote from another bank. Furthermore, our price comparison tool leads consumers to negotiate more without searching more, which implies that search and negotiation do not always occur together in our context. Because our survey collected both interest rate offers and originated rates, as well as directly asked consumers whether they negotiated and whether the bank lowered the rate in response, we develop a model that captures these negotiation dynamics directly.

1.1 Model Setup

Search A consumer i searches for loans sequentially from a continuum of banks, paying a search cost s_i to search at each additional bank. When a consumer meets a bank j , the bank can provide a loan at cost c_{ij} if the consumer and bank agree to originate the loan. Let $H_i(\cdot)$ denote the distribution of interest rates that consumer i could obtain across banks, which is conditional on the consumer’s characteristics and the characteristics of the loan they are looking for. Let $\hat{H}_i(\cdot)$ denote the consumer’s *perceived* distribution of interest rates that banks would offer them, based on their potentially biased beliefs. The consumer’s type, denoted by θ_i , describes how much their perceived distribution of interest rates $\hat{H}_i(\cdot)$ differs from the actual distribution $H_i(\cdot)$. In our baseline model, we assume that consumers do not update their beliefs based on the offers they receive during search, but we show in an extension that—under additional assumptions about prior beliefs—our predictions still hold when consumers update as Bayesians.

Negotiation The interest rate is decided in a two-stage bargaining game. The bank begins by posting a loan rate r_{ij}^1 , which can be conditional on observable characteristics of the consumer (and the characteristics of the loan they are looking for), but not on the consumer type θ_i as this is not observed by the bank. If the consumer is unsatisfied with the first-round offer, they initiate a negotiation with the bank. During this negotiation, the bank receives a noisy signal ξ_{ij} of the consumer's beliefs about the interest rate distribution (i.e., the consumer's type θ_i), but the consumer's search cost s_i and the bank's marginal cost c_{ij} remain private information even during the negotiation.

After the consumer initiates the negotiation, the bank decides whether to lower the interest rate. If the bank proposes a second-round rate $r_{ij}^2 < r_{ij}^1$ to the consumer, the bank incurs a fixed cost d_j , e.g., for the additional paperwork. In contrast, we model negotiation as costless to consumers; otherwise, the bank would be able to infer something about the consumer's search cost based on whether the consumer negotiates—rather than the search cost remaining private information—which would likely make the model intractable. The consumer then decides whether to accept the second-round offer. If the consumer and bank agree on a rate, the bank provides the consumer with a loan at the first- or second-round rate r_{ij} , depending on at which round the consumer accepted the offer, at the predetermined cost c_{ij} . Alternatively, the consumer can reject the offer and decide whether to continue searching.

Treatment Our treatment shows consumers the true distribution of interest rates $H_i(r)$ that the borrower faces, conditional on consumer characteristics and the characteristics of their desired loan. We assume that treatment leads to a *partial* correction of biased beliefs. Specifically, after seeing the true distribution $H_i(r)$, the consumer's type changes from θ_i to $\lambda_i \theta_i$ for some $\lambda_i \in (0, 1)$.

1.2 Model Predictions

We characterize the equilibrium in Appendix A, and now turn to the predictions from the model that we will take to the data. Formal statements of the predictions, as well as technical assumptions and proofs, are also in Appendix A.

Proposition 1. *In a market where consumers have potentially biased beliefs and cannot negotiate, a consumer's equilibrium search result is completely characterized by the consumer's reservation rate. In contrast, in a market with biased beliefs **and negotiation**, a consumer's equilibrium search result is **not** completely characterized by the consumer's reservation rate.*

In other words, in a market without negotiation, the consumer's reservation rate—determined by their search cost and beliefs about the interest rate distribution they face—is a sufficient statistic for the number of institutions at which they search. Thus, without negotiation, there are clear predictions of the effect of correcting biased beliefs on search. If a consumer underestimated the

first moment of the distribution (as we find most consumers do), the price comparison tool will increase their reservation rate, thus decreasing the number of institutions searched and increasing the interest rate they obtain. If the consumer underestimated the second moment (as we also find most consumers do), the price comparison tool will reduce their reservation rate, thus increasing the number of institutions searched and lowering the interest rate they obtain.

In contrast, when negotiation is possible—as we document empirically in our survey—the consumer’s reservation rate is no longer a sufficient statistic for search. Thus, we should no longer necessarily expect an effect of correcting consumers’ beliefs on search. The intuition is that beliefs now also affect negotiation, given our assumption that the lender observes a noisy signal about the consumer’s beliefs during the negotiation process. Partially correcting beliefs thereby changes the threat that the consumer will walk away if they believe they can likely get a better rate elsewhere, which increases the probability that the bank will lower its second-round rate.

We next turn to the predicted treatment effects of our price comparison tool in a setting with negotiation, which depend on the consumer’s biased beliefs.

Proposition 2. *If a consumer is biased about the first moment of the interest rate distribution, then the treatment effect of the price comparison tool on negotiation depends on the direction and magnitude of the bias. In particular:*

- (a) *For consumers who sufficiently under- or overestimate the first moment of the distribution, there will be no treatment effect of the price comparison tool on negotiation.*
- (b) *For consumers who moderately underestimate the first moment, treatment increases the probability of a successful negotiation.*

The intuition behind this prediction is that consumers who vastly underestimate the first moment will still substantially underestimate it after their biased beliefs are partially corrected by the tool, given the assumption that they still place some weight on their prior belief. If they still underestimate it sufficiently, when the lender receives a noisy signal about the consumer’s belief during the negotiation, it will infer that the consumer will likely reject its lowest profitable offer (because the consumer will still think they can get a better rate elsewhere). Because the lender incurs a cost to respond with a lower second-round rate, it will not do so. Similarly, for consumers who sufficiently *overestimate* interest rates, when the lender receives a noisy signal about this belief during the negotiation, it will determine that it does not need to lower the rate because the consumer will likely accept the first-round offer given their beliefs.

In contrast, for consumers who somewhat underestimate the first moment, updating their beliefs will increase the probability that they are now in the region where the lender determines it can lend to them profitably with a lower second-round rate that the consumer will likely accept. For these consumers, updating their beliefs also shifts their reservation rate upwards, which is a channel that

works in the opposite direction in terms of the effect of the tool on negotiation, but our proofs show that the first channel always dominates.

Proposition 3. *If a consumer underestimates the second moment of the distribution, then the price comparison tool increases the consumer’s probability of a successful negotiation with the bank.*

The driving force behind this prediction is that when the consumer sees the tool, they update their beliefs, learning that there is more dispersion than they initially thought. Their reservation rate then falls, as there is more expected utility to be gained from getting another draw from the distribution (and the symmetric change in their beliefs about the higher-rate part of the distribution does not affect expected utility since the consumer would not accept a higher rate on their next draw). Because the lender observes a noisy signal about the consumer’s beliefs during the negotiation, the lender also expects a lower reservation rate for consumers who now expect more dispersion, making it more likely that the lender optimally responds with a lower rate, rather than sticking to its first-round rate.

Given our empirical evidence that beliefs are biased and that the price comparison tool shifts beliefs, the model predicts that the price comparison tool will (i) shift negotiation without necessarily shifting search, (ii) increase negotiation for those who somewhat underestimated interest rates prior to seeing the tool but not for those who substantially underestimated or overestimated rates, and (iii) increase negotiation for those who underestimated dispersion. We exploit our experiment and rich data on search, negotiation, and loan terms to test these predictions.

2 Institutional Context

2.1 Chilean Consumer Loans

Consumer loans are a popular credit product offered by banks in Chile: 43% of Chilean households have outstanding consumer credit, with consumer loans and credit cards being equally popular forms of obtaining credit from banks (Banco Central de Chile, 2021). Consumer loans are typically uncollateralized, have fixed interest rates, and are paid in equal monthly installments up until the loan matures.

According to administrative data from the CMF on the universe of consumer loans obtained between November 2021 and February 2024 ($N = 1,863,087$ consumer loans), the mean and median annual interest rates are 25.8% and 23.9%, respectively. The median loan amount is \$4,488 USD, and the median maturity is 3 years. Based on our survey data, consumer loans are most commonly used to pay down other higher-interest debt (23.7% of borrowers), purchase or repair a car (16.3%), invest in their business (10.7%), make home improvements (5.2%), and purchase consumer durables (4.1%).

Various features of the Chilean consumer credit market are regulated. In 2012, a new law mandated that formal loan offers must be made through a standardized disclosure sheet in which the *carga anual equivalente* (CAE), which functions as the Chilean equivalent of the annual percentage rate (APR), is prominently displayed in large bold numbers in the upper right-hand corner; additional fees included in the CAE must also be itemized in the disclosure sheet (Kulkarni, Truffa, and Iberti, forthcoming). The Chilean consumer loan market also features interest rate caps, defined as 1.5 times the “current interest rate,” where the current interest rate is calculated as a volume-weighted average of interest rates on originated consumer loans, conditional on loan characteristics (Cuesta and Sepúlveda, 2021). Unlike in the US and many other countries, Chilean credit bureaus do not report continuous credit scores; rather, they report binary flags of whether people have defaulted on prior loans.

2.2 Search for Consumer Loans

Chileans search for loans a number of ways: 92.3% of our follow-up survey respondents visited at least one bank website during their search, 37.3% used a mobile banking app, 33.9% visited a branch in person, 32.9% communicated with a bank by email, and 26.5% communicated with a bank by phone. Soft search (i.e., searching for information without formally applying for a loan) plays an important role: while control participants formally applied to 1.3 institutions on average, they searched across 3.4 institutions.

In our baseline and follow-up surveys, we asked participants what features of a loan were most important to them to better understand search behavior (Figure B.1). In our baseline survey (when participants were looking for a loan), the three most important features of the loan were all functions of the interest rate: the total loan cost was the most important feature for 25.1% of participants, the monthly payment for 21.7%, and the interest rate or APR for 19.9%. Similarly, in our follow-up survey, the most common reason for choosing a particular lender was a lower interest rate, with 44.4% of participants giving this answer.

Another potentially important feature is the probability of being approved for a loan, as consumer credit markets feature high rates of rejection and consumers need to “search for approval” (Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao, 2024). In our context, approval rates conditional on formally applying for a loan are 54.6%. Furthermore, 48.7% of survey respondents reported that the bank gave them some indication of whether their application would be approved before or without formally applying. In our baseline survey, 15% of participants named getting approved for the loan as the most important feature of the loan for which they were searching. In our follow-up survey, 28.2% of borrowers chose a particular lender because they were quickly approved by that institution, while 19.2% did so because that was the only offer they received.⁷

⁷Being quickly approved and only receiving one offer were the second- and third-most common reasons for

Less important features that participants reported in the baseline survey included whether the bank branch was nearby (the most important feature for 9.4% of participants) and whether it is a bank in which they already had an account (7.8%). In the follow-up survey, the less important features for choosing among offers included whether the loan payment could be automatically deducted from payroll (5.9%), whether they were a client of that bank (5.7%), trust in the institution (4.5%), and getting approved for a higher loan amount (3.4%).

We also asked participants what strategy they employed while searching for their loan to better understand the prevalence of sequential vs. simultaneous search (De Los Santos, Hortaçsu, and Wildenbeest, 2012). We find that while both sequential and simultaneous search are common, sequential search is more common (Figure B.2): 60.3% of participants reported having a target interest rate (consistent with sequential search), while 42% said they expected to search at a target number of banks or until receiving a target number of offers (consistent with simultaneous search).⁸

Online Search Tools There are two types of online tools available to people searching for consumer loans in Chile: (i) tools provided by particular banks on their websites and (ii) third-party comparison platforms. Nearly all (93.2%) of our participants reported in the follow-up survey using at least one bank website while looking for a loan, while only 12.5% of participants reported using a third-party comparison platform. We scraped data from as many of these sources as possible—conditional on the loan and borrower characteristics of our RCT participants—in order to quantify how accurate the information on these websites is compared to the loans participants actually received in the administrative data (see Appendix C for details). In short, neither bank websites nor third-party comparison websites provide accurate information (as we show in Section 5.2).

3 Experimental Design

3.1 Participant Recruitment

Figure 1 shows the design of the RCT and the funnel of participant recruitment. We recruited 112,063 participants to the RCT from November 2021 to June 2023 (see Appendix D for a detailed timeline of the experiment). We targeted Google ads from the CMF to people who searched for keywords related to consumer loans in Chile. Our Google ads campaign included 4,107,376 ads served from November 2021 to June 2023, and 18.5% of searches for keywords related to consumer loans in Chile were served our ad. Figure B.3 shows an example of one of the Google ads included in our campaign. The ads we served were clicked 612,945 times (14.9% of ads). Those who clicked on the ads were taken to a landing page with a description of our study and informed consent to participate. The following page asked for their national ID number—which is commonly given out

choosing a particular lender. Reasons were not mutually exclusive, as participants could name more than one.

⁸These survey questions were not mutually exclusive.

in Chile (e.g., for rewards programs at the grocery store)—and their contact information including email address and phone number.⁹ We then conducted a baseline survey, described in more detail in Section 4.2.

3.2 Elicit Beliefs Treatment

During the baseline survey, we randomly elicited beliefs about the interest rate distribution from 75% of participants, asking their expectations about (i) the rate they would get on the loan they ultimately took out, (ii) the lowest interest rate a bank could offer them, (iii) the highest interest rate a bank could offer them, and (iv) the fraction of offers that would have an interest rate above the midpoint between the lowest and highest rates. These questions follow the literature eliciting beliefs about future earnings (Guiso, Jappelli, and Pistaferri, 2002), returns to schooling (Attanasio and Kaufmann, 2014), and macroeconomic growth (Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024). In addition, we asked them at how many banks they expected to search, at which bank they would search first and second, whether they expected to be approved at these banks, and the rates they expected these two banks to offer them. We did not ask any of these expectations questions to a randomly selected 25% of the sample in order to test whether these survey questions have a treatment effect, motivated by evidence from Stango and Zinman (2014) and Zwane, Zinman, Van Dusen, Pariente, Null, Miguel, Kremer, Karlan, Hornbeck, Giné, Duflo, Devoto, Crepon, and Banerjee (2011). Indeed, we find that asking these questions led people to search more and obtain loans with lower interest rates.

Among those who clicked on the ad, 112,063 (18.3%) consented to participate and continued taking the baseline survey long enough to randomly be asked or not asked the questions on their expectations about interest rates and search; this is our sample for measuring the effects of eliciting beliefs. After viewing either the price comparison tool, simple tool, or a control video, we again asked the 75% of participants assigned to the elicit beliefs treatment the same interest rate expectation and search questions to test whether their expectations were affected by treatment.

3.3 Price Comparison Tool and Simple Tool Treatments

Price Comparison Tool Our price comparison tool (Figure 2a) showed participants a conditional distribution of interest rates that similar borrowers had received for similar loans over the past six months. We built the tool using administrative data on loan characteristics merged with borrower characteristics for the universe of consumer loans in Chile over two years (1.8 million loans to 1.2 million borrowers). Appendix E provides more details on how the histograms were created. We refreshed the data every month to show the previous six months of interest rate data, based on tests we conducted to determine the optimal time period of data to show (trading off showing accurate

⁹We use the national ID number not only to observe outcomes in administrative data but also to prevent the same individual from participating more than once and potentially being assigned to more than one treatment arm.

information and sufficient data points). Appendix E.1 provides more detail on this trade-off and describes the rationale behind why we showed participants data on loans from the prior six months.

Walking through each component of the price comparison tool shown in Figure 2a, the borrower and loan characteristics that the participant already answered in the baseline survey were loaded automatically in the top panel of the tool (“1. Verify that your data are correct”), but these values could be modified by the user. The second panel of the tool (“2. Look at the information”) showed the user the distribution of interest rates that similar borrowers had obtained for similar loans in the past six months. Based on the findings from focus groups we conducted on a prototype of the tool, we created a tool-tip that participants could see by hovering over the histogram bars, to help those that may not be familiar with interpreting data from graphs and histograms. Specifically, the tool-tip told the participant the number of loans that had that interest rate, gave a cumulative distribution function interpretation of the bar (what percent of loans had an interest rate at or below that rate), and converted the interest rate to a monthly and total loan cost in Chilean pesos based on the loan amount and maturity entered by the participant. In addition, we created a tutorial video that the user could watch (by clicking “watch tutorial”) to better understand how to use the tool.

In the third panel of the comparison tool (“3. Compare the impact of different interest rates for your wallet”), we compared two interest rates in the histogram to show the participant the implications of these different rates for their monthly and total loan costs (in Chilean pesos). The inclusion of this part of the price comparison tool was inspired by research that it is important to translate differences in APRs into dollar costs (Bertrand and Morse, 2011), that borrowers target monthly payments rather than interest rates (Argyle, Nadauld, and Palmer, 2020), and more broadly that market participants respond more to dollar amounts than percentages (Shue and Townsend, 2021). By default, the highest and lowest interest rates were compared, but the user could drag the two triangle markers on the x-axis of the histogram or manually enter interest rate values to compare the cost implications of different rates. Participants experimenting with this feature would see the potentially substantially different costs for their loan depending on the interest rate they obtain.

One concern is that the dispersion shown in the tool could reflect both within-borrower and across-borrower dispersion based on characteristics that are unobservable to us but observable to lenders. To test this, we first calculate the within-borrower standard deviation of interest rates for each participant receiving multiple offers in the survey. We then conduct 1,000 simulations where we randomly draw interest rates from the participant’s conditional distribution in the administrative data (i.e., the same distribution we show them in the price comparison tool if they are in that treatment arm). In each simulation, we draw an equal number of interest rates as the number of offers the participant reported receiving. We then calculate, for each simulation, the standard deviation in interest rate offers “obtained” by that borrower from their conditional distribution.

For each participant we then average their standard deviation in interest rate offers across their 1,000 simulations. If the conditional distribution in administrative data includes substantial across-borrower dispersion, the simulated standard deviations should be higher than the within-borrower standard deviations from the survey. In contrast, the standard deviations are almost identical: the average standard deviation in administrative data is 0.01 pp *lower* than the within-borrower standard deviation in survey data. We conclude that the dispersion shown in the price comparison tool is an accurate representation of within-borrower dispersion.

Simple Tool on Benefits of Search Participants in the simple tool treatment arm viewed a simpler tool that provided the user with just two numbers on the estimated benefits of search (Figure 2b), which we estimated by simulating random draws from their conditional distribution (see Appendix E.2). This treatment was designed to be simpler to interpret and to avoid information overload. The borrower and loan characteristics that the participant already provided in the baseline survey again loaded automatically in the top panel of the tool (“1. Verify that your data are correct”). The bottom panel (“2. Look at the information”) said: “Using real data from loans granted to people similar to you, we estimate that shopping at 1 additional bank would lower your monthly payment by \$X and the total cost of your loan by \$Y, on average.” The number of additional banks could be modified using a drop-down menu, which the participant could use to determine the expected benefits of searching at up to five additional banks.

Control Video The control video was a 1 minute and 35 second long animated video created by the CMF describing key credit terms. The video was designed to provide information related to loans that would *not* be useful for search or negotiation. The video defined what a lender, debtor, and loan contract are, as well as key loan terms like maturity and principal. Figure B.4 shows a screenshot from our control video.

Among participants who consented, 46,051 consumer loan seekers (41.1%) continued taking the baseline survey long enough to reach the module where we randomized whether they saw the price comparison tool, simple tool, or control video; this is our sample for measuring the impact of the price comparison and simple tools. In all treatment arms including the control video, participants were required to stay on the treatment module page for one minute prior to clicking “Next” to proceed to the following module.¹⁰

¹⁰We include all 46,051 participants who made it to the module in which we randomized a tool treatment or the control video in our analyses of the treatment effects of the tools, regardless of whether they continued to the following module or closed the browser at this stage. Despite the smaller sample size of 46,051 for measuring the effects of the price comparison tool and simple tool, compared to the sample size of 112,063 for measuring the effect of the elicit beliefs treatment, the research design is internally valid. We did not randomize participants into one of the price comparison tool, simple tool, or control arms until they reached that module of the online survey. As a result, we can simply remove those who did not make it to the tool treatment module from the sample for estimating the effects of the tools, and still have balance across these treatment arms (both in theory and, as we show, in practice).

4 Data

In this section, we describe our administrative and survey data sets. Appendix F provides more detail on data cleaning, links to the survey questionnaires, and a timeline of the surveys.

4.1 Administrative Loan Data

We use administrative data on the universe of consumer loans from 2015–2024 from the Chilean financial regulator, the CMF. We observe the following borrower characteristics that banks use to determine whether to offer loans: age, marital status, gender, income, and municipality of residence.¹¹ Importantly, credit bureaus in Chile do not report continuous credit scores, but rather report binary flags if the borrower has defaulted on prior loans; thus, the interest rates that banks offer are not conditional on continuous credit scores. As for loan characteristics, we see each loan’s amount, interest rate, and maturity, as well as an anonymized code for the lender. We use these data in our construction of the conditional distribution of interest rates for both the price comparison tool and the simple tool on the benefits of search, as well as to measure whether participants had loans prior to participating in the RCT and to measure treatment effects on loan outcomes.

By obtaining participants’ national ID numbers, we are able to merge their treatment status and survey responses with administrative data to measure treatment effects on the loans they obtained after participating. In total, 21,522 out of 112,063 participants from our RCT took out a consumer loan between the time they participated in our RCT and one year later. Of these, 8,988 participants among the sample of 46,051 participants for measuring the effect of the price comparison tool and simple tool took out a consumer loan within one year of participating (Figure B.5).

To assess external validity, we compare our RCT participants who took out consumer loans to the universe of Chilean consumer loan borrowers between the start of our RCT and one year after it ended. Figure B.6 shows that borrowers in our RCT are—unsurprisingly—not perfectly representative of the overall population of borrowers in Chile; nevertheless, there is a large amount of overlap in the distributions of borrower characteristics.¹² The two groups are relatively similar on gender (all borrowers: 38% women vs. RCT sample: 37.4%), the percentage who live in the capital Metropolitan Region (all borrowers: 51.4% vs. RCT sample: 50%). First-time borrowers (defined as those who did not have a previous consumer loan in the administrative data prior to the RCT) make up 37.5% of the RCT sample compared to 41% of the overall population of borrowers. Borrowers in our RCT are relatively better-off than the overall population of borrowers—as the distribution of annual income for RCT borrowers is shifted right of that of all borrowers—though there is extensive overlap in the support of the distributions. The variable with the starkest differ-

¹¹Note that if applicants already have other products at the bank where they are applying for a loan, the bank might also use that information in its lending decision, and we do not observe these bank-specific data.

¹²We exclude borrowers who participated in our RCT from the “all borrowers” group (i.e., overall population of borrowers in Chile) in order to compare two mutually exclusive groups.

ences between the RCT sample and the overall population of borrowers is age, where participants in our RCT are younger than the general population of borrowers (median age of all borrowers: 38 vs. RCT sample: 33), which is unsurprising given the online nature of our recruitment process.

The distributions of loan terms (interest rate, loan amount, and maturity) obtained by our RCT participants and all borrowers in Chile also exhibit differences but have a large degree of overlap (Figure B.7). In general, borrowers in our RCT obtain slightly larger, longer-maturity, lower-interest rate loans. For example, the average loan maturity in our sample is 37 months, compared to 34 months in the overall population.

4.2 Baseline Survey

The baseline survey was conducted online after participants who searched for keywords related to loans clicked on a Google ad from the financial regulator and consented to participate in the study. In addition to the questions about beliefs for those assigned to the elicit beliefs treatment (described in Section 3.2), we asked participants about their sociodemographic characteristics and detailed questions regarding their existing banking relationships and other financial products.

We also asked participants questions to understand how they form beliefs about interest rates. Specifically, we asked them if they had ever obtained a quote for a consumer loan from a bank website, if they had seen an ad for a consumer loan advertising an interest rate, or if someone they know had told them what interest rate they got for a consumer loan. If they answered yes to any of these questions, we asked them how long ago this was and what interest rates were given by the bank website, advertisement, or person they know. We then asked whether they had searched for a loan before, and if so how long ago it was, how many offers they had received, and the range of interest rates of those offers. Finally, we asked questions on financial literacy (Lusardi and Mitchell, 2011), behavioral biases (e.g., procrastination), and a set of simple questions used to measure cognitive ability (Frederick, 2005), which are all related to search and the formation of beliefs (D’Acunto, Hoang, Paloviita, and Weber, 2023).

Table 1 reports means for characteristics from the baseline survey, and tests for balance between the 75% of participants who were randomly assigned to the elicit beliefs treatment vs. the 25% of participants who were not asked these questions. Participants in our sample are roughly 36 years old on average with an average monthly income of 1,125,959 pesos (1,142 USD at market exchange rates). Participants have a wide range of education: 3.7% did not complete high school, 36% completed only high school, 21.3% completed a 2-year post-secondary program (equivalent to an associate’s degree), and 39% completed a 5-year degree program or higher (equivalent to a bachelor’s degree). As for financial experience, 67.8% of our participants had a bank account, and 70.2% had some type of outstanding loan (consumer loan, mortgage, auto loan, credit card, or cash advance).

As expected due to randomization, our sample of 112,063 participants who were randomized to either receive or not receive the elicit beliefs questions is balanced: the p -value of the omnibus F-test regressing the elicit beliefs dummy on all baseline survey characteristics is 0.463. Furthermore, only one variable—the probability of having a loan already at baseline—is not balanced, as could be expected by chance: those assigned to the elicit beliefs treatment are 0.6 pp less likely to have a prior loan (statistically significant at the 5% level).

Table 2 tests for balance across the price comparison tool, simple tool, and control arms.¹³ We again find that the sample is balanced across treatment arms, as expected due to randomization. The p -value for our omnibus F-test of whether characteristics jointly predict the price comparison tool treatment is 0.279, and that for the simple tool treatment is 0.207. The only variable that has a statistically significant difference between the treatment arms and the control arm is having a bank account: participants are 1.6 and 1.3 pp more likely to have a bank account in the price comparison tool and simple tool arms compared to the control group (statistically significant at the 5% level).

4.3 Endline Phone Survey

We surveyed participants via phone at least six months after they participated in the RCT. We attempted to contact 39,713 participants (35.4% of our 112,063 sample), and ultimately collected 6,132 completed surveys, for a 15.3% response rate. Table B.1 shows that response rates are balanced across treatment arms, while Tables B.2 and B.3 show that the survey subsamples are balanced on covariates across the elicit beliefs and tool treatment arms, respectively.

The primary objective of the endline phone survey was to collect rich data on participants' search histories and negotiation. Data on search are poorly captured in most administrative data sets: in many administrative data sets including the CMF data, only originated loans are recorded. Even if all applications were recorded, true search behavior also involves informal quote requests, or even inquiring about the probability of approval at a particular lender (Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao, 2024). Data on negotiation are also absent in nearly all administrative data. For each bank at which consumers searched for information, we ask detailed questions about both search and negotiation: how they searched (e.g., using the bank's website or mobile banking app, going to a branch in person, emailing, calling by phone), whether they informally received any information about their probability of acceptance or an estimate of the interest rate they would receive, whether they formally applied, whether they were accepted or rejected, what loan terms they were formally offered if accepted, whether they negotiated the offer, the loan terms they were offered after this negotiation, and whether they took out the loan.

¹³The loan characteristics variables included in Table 2 are not included in Table 1 because they are asked in the same module as the elicit beliefs treatment, and the elicit beliefs treatment caused some participants to stop participating in the survey. (Note that participants who abandoned the survey during the elicit beliefs module are still tracked in the administrative data and included in the sample to estimate the effect of eliciting beliefs.)

4.4 WhatsApp Survey

We collected WhatsApp surveys with more detailed questions about the timing of negotiation, based on our finding that the price comparison tool increased negotiation. We attempted to survey 15,510 participants and received 2,344 responses, for a response rate of 15.1%.

5 Results

5.1 Eliciting Beliefs Leads to More Search and Lower Rates

Randomizing whether we elicited priors about the interest rate distribution and the number of institutions at which participants expected to search—rather than asking these questions to all participants—was motivated by evidence from Stango and Zinman (2014) and Zwane, Zinman, Van Dusen, Pariente, Null, Miguel, Kremer, Karlan, Hornbeck, Giné, Duflo, Devoto, Crepon, and Banerjee (2011) that survey questions can have treatment effects on real-world behavior and outcomes. We run the following regression:

$$y_i = \alpha + \beta \mathbb{1}(\text{Elicit Beliefs})_i + \varepsilon_i, \quad (1)$$

where y_i are search, negotiation, and loan term outcomes for participant i . The treatment dummy $\mathbb{1}(\text{Elicit Beliefs})_i$ equals one if the participant was assigned to the elicit beliefs treatment and zero otherwise.

Table 3, panel A, shows effects on search, negotiation, and loan outcomes according to survey data, while Table 4, panel A, shows effects on loan outcomes according to administrative data. We find that merely asking these questions led consumers to search at 0.13 more institutions, or a 3.9% increase compared to the control mean of 3.4 institutions (statistically significant at the 1% level). This led borrowers to obtain lower interest rates according to both survey and administrative data. Specifically, eliciting beliefs caused participants to receive 7.1% lower interest rates according to survey data (significant at the 10% level) and 1.2% lower interest rates according to administrative data (significant at the 5% level). The effect of eliciting beliefs on interest rates is not accompanied by changes to other loan terms such as the loan amount or maturity (Table B.4). Eliciting beliefs had no effect on the number of formal applications, probability of negotiating, number of offers, or probability of taking out a loan.

Eliciting beliefs about the distribution of interest rates and about participants' expected search may have increased search and improved loan outcomes through various mechanisms. First, it may have led them to form beliefs when they previously had not (Andre, Chopra, Michels, and Wohlfart, 2026), or similarly to retrieve their beliefs when they otherwise would not have during their loan search due to retrieval failures (Augenblick, Jack, Kaur, Masiye, and Swanson, forthcoming; Breza and Kaur, forthcoming). Second, it could have increased search by helping consumers form

“implementation intentions” (Gollwitzer, 1999; Milkman, Beshears, Choi, Laibson, and Madrian, 2011; Rogers, Milkman, John, and Norton, 2015): specifically, by prompting them with several questions about search—e.g., at how many banks they expected to search, at which bank they expected to search first, and whether they thought the first bank where they expected to search would approve them—these participants may have developed a more concrete search strategy that they were then able to implement. Third, it could have made our tool treatments more salient: for example, people who first reported their beliefs and then were shown data about interest rates might have been more likely to find this information salient; however, we test and rule out this third mechanism, as there are no interaction effects between the elicit beliefs and price comparison tool treatments on search.¹⁴

5.2 Participants Underestimate Rate They Will Obtain and Dispersion

We find that prior to viewing the tool, most users thought interest rates were lower than they actually were, and also underestimated price dispersion. We compare the beliefs participants reported in our baseline survey to administrative data: for the rate they expected to obtain, we compare to the rate they subsequently obtained; for dispersion, we compare to the conditional distribution of rates that similar borrowers received recently for similar loans.

Figure 3 shows that participants have biased beliefs about the first moment of the distribution. It compares the interest rates that participants report expecting to receive on the loan they take out to the interest rates we observe in administrative data for the loan they actually obtained subsequently (restricting to those who did take out a loan after participating in the RCT). It shows that participants have inaccurate beliefs about the interest rate that they will ultimately receive on their loan: 73.2% of borrowers think they will receive an interest rate *lower* than the rate they later receive. Conditional on underestimating, these borrowers think they will get a loan that is 14.3 pp lower than the rate they ultimately receive; the average bias including those who are accurate or overestimate is −6.9 pp.

Figure 4 shows biased beliefs about the second moment: the majority of participants (74.8%) *underestimate* price dispersion, but there is also a long right tail of participants who substantially *overestimate* dispersion. The figure shows the distribution of the difference between a participant’s beliefs on dispersion—measured as the highest rate they think a bank could offer them minus the lowest rate they think a bank could offer them—and the corresponding measure in the administrative data (i.e., the highest minus lowest rates we observe in the conditional interest rate distribution, conditional on that participant’s characteristics and the characteristics of the loan they are looking

¹⁴Specifically, we regress search on dummies for both the eliciting beliefs and tool treatments and their interactions; we still find a positive and statistically significant effect of eliciting beliefs on search for those who were not treated with either the simple or price comparison tools, and we cannot reject that the treatment effect of eliciting beliefs on search was the same for the price comparison tool and control groups ($p = 0.742$).

for). We use administrative data for similar borrowers and loans over the past six months—i.e., the same data we would show the borrower if assigned to the price comparison tool arm. Unlike in the case of beliefs about the loan they will obtain, for dispersion we cannot compare to offers they actually received, as these would only be a subset of draws from the full distribution of interest rates if they were to search across all banks.

Robustness The finding that consumers tend to underestimate the first moment is robust to using a measure we observe for all participants rather than only those who took out a loan: Figure B.8 compares participants’ beliefs about the rate they would obtain to the median of the conditional distribution we observe in administrative data rather than the rate they obtained.

The finding that most participants underestimate dispersion is also robust to different functional form assumptions and potential skewness in the distribution (see Appendix G). In Figure B.9, we calculate the implied standard deviation of their beliefs about the distribution assuming either a symmetric or split triangular distribution, following Guiso, Jappelli, and Pistaferri (2002), Attanasio and Kaufmann (2014), and Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber (2024). The split triangular distribution uses the additional beliefs question on the fraction of offers that would be above the midpoint between the highest and lowest rates a bank could offer them. We then compare this implied standard deviation with the standard deviation of their corresponding interest rate distribution conditional on loan and borrower characteristics in administrative data, and find that 78.6% and 82.5% of participants underestimate the dispersion of interest rates under these two assumptions.

Why are beliefs biased? We first assess what sources of information people use to form their beliefs, as well as whether these sources provide accurate information. At the time we elicited their beliefs (i.e., in the baseline survey), 41.2% of participants report having seen advertisements by banks, 43.8% used bank websites that provide an estimated interest rate to the consumer, 12.5% used third-party comparison websites, and 22.8% asked friends and family about interest rates.¹⁵ To assess whether bank advertisements, bank websites, or third-party comparison websites might cause people to have biased beliefs, we compare rates our participants *would* have seen in each of these contexts with the rate they ultimately received in our administrative data (Figure 5).¹⁶

For advertisements, we cannot observe all bank advertisements but can observe those that banks serve on Google. We randomly sample combinations of search terms that led people to the Google ads for our experiment and municipalities of participants. We then conduct automated Google

¹⁵The percent who *eventually* used a bank website as part of their search, as reported in the follow-up survey, is much higher, at 93.2%.

¹⁶Our survey also asked what interest rates people’s family and friends told them they had received, but only 15 participants who obtained a loan after participating in the RCT reported the interest rates that family or friends told them they had received, leaving us with too small of a sample to compare these survey responses to the rates those participants obtained.

searches using that keyword and specifying the location from which the search originated, and scrape the resulting first page of Google results—including both Google ads and regular Google search results (see Appendix H). The difference between interest rates that are shown in Google ads or search results and the rates people actually obtain are heavily negatively biased: 73.1% of rates shown on Google were lower than what participants ultimately received from the same bank in our administrative data (Figure 5a).

For bank websites and third-party comparison websites, we use a script to input participant and loan characteristics into the website and scrape the resulting loan terms (see Appendix C). While there is substantial noise in the quotes from bank websites, they are not biased in one direction or the other: very close to half (49.4%) of participants would have been shown an interest rate that is lower than the rate they ultimately received (Figure 5b). As for the third-party comparison website, the difference between the rate shown and the rate the same consumer obtained from the same bank is negatively biased, with 74% of quotes being lower than the rate the borrower ultimately received (Figure 5c). The third-party comparison website results suggest that banks have an incentive to shade down the quotes they provide to an aggregator platform, as people searching on a platform are likely deciding which bank to apply to based on which bank claims to offer the lowest rate. Banks can subsequently bait-and-switch the customer and offer them a higher rate once they have gone to the bank and formally applied for a loan.

We conclude that bank advertisements and third-party comparison websites are likely partially responsible for loan seekers' downward-biased beliefs about the first moment of the interest rate distribution. Bank websites, on the other hand, are not biased on average, but provide very noisy quotes. Because obtaining accurate information about the rate a bank would offer requires actually applying for a loan at that bank, this is a context in which it is costly for borrowers to correct their biased beliefs.

What predicts biased beliefs? Finally, we assess which characteristics predict biased beliefs using machine learning (see Appendix I). Individuals with more biased beliefs tend to be younger, be looking for a smaller loan amount, and have lower incomes, less education, and less experience with financial products (i.e., bank accounts and loans).

Alternative explanations for biased beliefs We next assess alternative explanations for why beliefs about the interest rate distribution are biased. One possibility is that biased beliefs could be explained by changes in interest rates over time. On average, interest rates increased over the 19 months that we conducted the experiment, with the within-month median interest rate among the universe of loans in administrative data ranging from 19.6% to 26.7% over this time period (Figure B.10).

If interest rates increased shortly before a participant was surveyed and they are slow to update

their beliefs, this could explain why they underestimate the first moment. To test whether consumers are slow to update, we regress beliefs about the rate a consumer will obtain on the median interest rate during the month in which that consumer was surveyed. We find that when median consumer loan interest rates are 1 pp higher, beliefs about the rate people expect to obtain are 1.3 pp higher (Table B.5), indicating that people do update as interest rates change over time and that their underestimating the first moment is not caused by slow updating. Furthermore, Figure B.10 shows that while interest rates rise over our sample period, consumers' interest rate beliefs are persistently lower than the level of rates in the market but do reflect the upward trend in rates, visually confirming the regression results from Table B.5.

Even if participants had accurate beliefs about interest rates as of the date they were surveyed, if interest rates increased between the baseline survey and loan origination, this could generate observed bias in the first moment. We show, however, that even when we restrict the sample to the first quartile of the number of days between participating in the RCT and obtaining a loan—those who obtained a loan within 21 days of participating—the underestimation of the first moment looks very similar, with 74.9% underestimating rates (Figure B.11). We conclude that consumers' underestimates of the first moment cannot be explained by changes to interest rates over time.

Another possibility is that we are overestimating dispersion in the administrative data if the conditional distribution combines within-borrower dispersion and across-borrower dispersion based on characteristics unobservable to us but observable to banks. However, we already showed in the exercise described in Section 3.3 that the dispersion we measure in administrative data is an accurate representation of within-borrower dispersion: true within-borrower dispersion (across offers the same borrower reported in the survey) is very similar to the dispersion of simulated draws from their corresponding conditional distribution in administrative data. An additional concern is that measuring dispersion as the highest minus lowest rate could be particularly susceptible to overestimation if our administrative data distribution were capturing substantial across-borrower dispersion. However, in addition to the evidence above that the dispersion in our tool is an accurate representation of within-borrower dispersion, our finding that people underestimate dispersion is robust to comparing beliefs about dispersion to the standard deviation of the administrative data distribution, which is less sensitive to its endpoints (Appendix G).

5.3 Price Comparison Tool Leads to Large Updates in Beliefs

After seeing the price comparison tool, did participants update their beliefs about the distribution of interest rates? To test this, we estimate the following specification:

$$Posterior_i - Prior_i = \beta_1 \mathbb{I}(\text{Simple Tool})_i + \beta_2 \mathbb{I}(\text{Price Comparison Tool})_i + \lambda_{b(i)} + \epsilon_i, \quad (2)$$

where $Prior_i$ is the interest rate expectation participant i reported prior to seeing the tool or control video and $Posterior_i$ is the interest rate expectation they reported after seeing it. In our main specification, interest rates are annualized and measured in percentage points (e.g., an expected interest rate of 18% per year would be coded as 18). The treatment dummies $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ equal one if the participant was assigned to that treatment arm and zero otherwise, and $\lambda_{b(i)}$ are bin density fixed effects. The bin density fixed effects are deciles of the number of observations in the tool that were shown or would have been shown to the participant, to control for the fact that people in higher-population municipalities or with more borrowers with similar characteristics would see more observations in the price comparison tool and might infer that there is more dispersion than those seeing fewer observations.¹⁷

Table 5 shows the results on belief updating. On average, the price comparison tool caused participants to increase their beliefs about the rate they would obtain by 16.2 pp, or 55.4% relative to the control mean posterior of 29.2%.¹⁸ Treated participants' beliefs about dispersion also increase by 16 pp compared to a control mean posterior of 23.2 pp of dispersion, an increase of 68.7%. Tables B.6 and B.7 show that the same conclusions hold if we use levels of posteriors as the dependent variable (without subtracting priors), with or without controlling for priors on the right-hand side. Tables B.8 and B.9 show the same pattern when we log-transform beliefs about interest rates. Finally, Table B.10 shows that the effect of the tool on beliefs about dispersion are robust to using a scale-invariant measure of normalized dispersion and to using the Guiso, Jappelli, and Pistaferri (2002) and Attanasio and Kaufmann (2014) measures of implied beliefs about the standard deviation.

In contrast, the simple tool quantifying the benefits of search but providing no direct information on the distribution of interest rates hardly affected priors about interest rates: the point estimate of the simple tool's effect on the interest rate the participant would obtain is less than 1 pp, while the coefficient on dispersion is very close to zero (0.01 pp), and neither is statistically significant.

Heterogenous treatment effects on beliefs Appendix J tests for heterogeneous treatment effects of the price comparison tool on interest rate beliefs using the machine-learning methodology proposed by Chernozhukov, Demirer, Duflo, and Fernández-Val (2025). We find heterogeneity in the tool's treatment effects on the rate participants expect to obtain and the highest rate a bank could offer them, while we do not find heterogeneity in effects on the lowest rate a bank could offer them or dispersion (Table J.1). Despite this heterogeneity, beliefs about both moments shift upwards

¹⁷For those in the simple tool and control groups, the bin density fixed effect is based on how many observations are in the price comparison tool histogram that *would have been shown* to the participant had they been assigned to the price comparison tool arm.

¹⁸Although we winsorize responses to these interest rate questions at the 95th percentile, the distributions are nevertheless quite skewed and the medians are substantially lower than the means. The control median posterior of the annual rate people expect to get is 18%.

even for participants in the quintile with the lowest predicted treatment effects (Table J.2). We find that the same characteristics that predict more biased beliefs also predict a larger treatment effect of the tool for both belief outcomes that exhibit heterogeneous treatment effects. In particular, the treatment effect of the tool on belief updating is larger for participants who are younger, are looking for smaller and shorter-maturity loans, and have lower incomes, less education, and less experience with financial products (Table J.3).

5.4 Effects of Tool on Search, Negotiation, and Loan Terms

Based on the findings from the previous two subsections that consumers have biased beliefs and update their beliefs after seeing the tool, we revisit predicted treatment effects from our model. There are three key insights from the model about the effects of (partially) correcting biased beliefs on search and negotiation. First, negotiation breaks the sufficient statistic relationship between reservation rates and search, so we should not necessarily expect an effect of shifting beliefs on search (Proposition 1). Second, the effect of the tool on those with biased beliefs about the first moment is non-monotonic: it should have no effect on negotiation for those who sufficiently overestimated or vastly underestimated the first moment, but should increase negotiation for those who somewhat underestimated it (Proposition 2). Third, the tool should lead those who underestimated dispersion to negotiate more (Proposition 3).

We run the following regression:

$$y_i = \alpha + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \varepsilon_i, \quad (3)$$

where y_i are search, negotiation, and loan term outcomes for participant i . The treatment dummies $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ equal one if the participant was assigned to that treatment arm and zero otherwise.

Table 3, panel B, shows the effect of our treatments on search behavior, negotiation, and loan outcomes in survey data. The price comparison tool did not lead people to search at more institutions or to formally apply for loans at more institutions, but instead made them 3.7 pp (39%) more likely to negotiate. The tool also did not lead people to be more likely to search or apply at their home bank or at a bank where they had expected to search, nor did it affect the probability of searching or applying at any particular institution (see Appendix K).¹⁹ The null average treatment effect on search is not masking offsetting effects in different parts of the distribution: we cannot reject that the distributions of the number of institutions searched are the same across treatment arms (Figure B.12). Furthermore, it does not appear to be masking offsetting effects on search for

¹⁹We define their home bank as any bank at which the consumer already had a financial product, and a bank where they expected to search using the baseline survey questions about the first and second institutions at which the consumer expected to search.

those predicted to search less vs. more in a setting without negotiation, as we find null effects on search even for these two groups (Table B.11). This is consistent with beliefs not being a sufficient statistic for search in a setting with negotiation (Proposition 1).

Despite not applying at more institutions, those treated with the price comparison tool receive 0.069 (13.1%) more offers, and Figure B.13 shows that this effect on the number of offers reflects a treatment effect on both the extensive and intensive margins. In a subsequent WhatsApp survey we conducted to understand the timing of negotiation, we find that negotiation often happens prior to the bank issuing a formal offer, which explains the effect on the number of formal offers received. In other words, the lender will often suggest an interest rate without making a formal offer, and only if the negotiation is successful and the lender and consumer agree on an interest rate does the lender incur the cost of formally issuing a loan offer. As a result of negotiating more, participants receive 11.9% lower interest rates according to the survey data.

Table 4, panel B, shows treatment effects of the tools in administrative data. The tool makes borrowers 0.9 pp (4.7%) more likely to take out a loan.²⁰ The positive treatment effect on taking up a loan causes a selection effect when we regress interest rates on treatment: the price comparison tool causes some people who would not have otherwise obtained a loan in the absence of treatment to obtain one, and only consumers who obtain a loan can be included in the interest rate regression. In particular, if the “compliers” (those who only obtain a loan if they are treated) are more marginal borrowers who thus obtain higher interest rates than the “always-takers” (those who would take out a loan regardless of treatment status), this selection effect could offset a negative treatment effect of the tool on the always-takers. We find evidence that suggests that this indeed drives the lack of an average treatment effect on interest rates in the administrative data and the discrepancy compared to the effect of the tool on interest rates in survey data.

To evaluate this possibility, we would need to separate likely always-takers from likely compliers, and separately test whether the tool has a negative treatment effect on interest rates for likely always-takers and a positive (selection) effect on interest rates among likely compliers. Since always-taker and complier status cannot be directly observed, we use machine learning methods from Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) to predict it, then estimate treatment effects on interest rates for these two groups.²¹ The results are shown in Table 4, panel C, and details are included in Appendix L. The method effectively isolates likely always-takers and compliers: for likely always-takers we observe zero treatment effect of the tool on the probability

²⁰According to our survey data, 65% of borrowers who did not take out a loan did not make the purchase or investment for which they were seeking the loan.

²¹Medina and Pagel (2025) use a similar conceptual approach. In particular, they use machine learning to estimate predicted individual treatment effects of a nudge on saving, then isolate the quintile that is predicted to have the highest treatment effect. For this quintile, they then estimate treatment effects on a separate outcome—borrowing—to study whether experimentally-induced saving causes borrowing.

of taking out a loan, while for likely compliers the coefficient on treatment is 1.7 pp (statistically significant at the 1% level; for the difference between the two groups, $p = 0.008$).

We then estimate treatment effects on interest rates for these two groups, finding that the tool causes a 3.5% decrease in interest rates for likely always-takers (statistically significant at the 5% level), whereas there is a 2.2% *positive* coefficient on interest rates for likely compliers (statistically significant at the 10% level; for the difference between the two groups, $p = 0.008$). Taken together, these results show that the compliers pull the estimated average treatment effect of the tool towards zero, creating a selection effect since only participants who received a loan can be included in the interest rate regression.

The simple tool that showed only the estimated interest cost savings from search did not have effects on any search, negotiation, or loan outcomes (Table 3, panel B, and Table 4, panel B), which is not surprising given that the simple tool did not lead people to update their beliefs about the interest rate distribution.

Heterogeneous treatment effects by beliefs about each moment Next, we turn to two key predictions from the model about heterogeneous effects of the tool on negotiation, based on consumers' biased beliefs. The first prediction is that the tool's effect on negotiation is non-monotonic in the bias about the first moment. In particular, there should be no effect of the tool for those who (prior to seeing the tool) sufficiently overestimated the first moment of the distribution: they will still overestimate it after partially correcting, and the lender will observe a noisy signal of this in the negotiation and will realize that the consumer is likely to accept the initial offer even if the lender refuses to lower the rate. There should also be no effect for those who vastly underestimated the first moment: they will still sufficiently underestimate it after partially correcting, so the lender will determine that it cannot profitably lend to that consumer and will not incur the cost of lowering their initial offer. In contrast, for those who somewhat underestimated rates, the tool should make them more likely to negotiate successfully: after partially correcting, their beliefs are more likely to be in the range where the lender will determine that it can profitably lend, so it will lower the offered rate as the consumer may walk away otherwise.

This inverse-U shape in the treatment effect of the tool by bias about the first moment is exactly what we find: those who overestimated or substantially (by more than 30 pp) underestimated rates do not negotiate more in response to the tool, while those who somewhat (by less than 30 pp) underestimated rates are more likely to negotiate (Figure 6). To formally test whether this non-monotonic inverse-U shape is statistically significant, we implement the Wolak (1989) inverse-U test and find that it is significant at the 10% level (see Appendix M).

The second prediction is that the tool will increase negotiation for consumers who underestimated the second moment of the distribution. We separately estimate specification (3) for those

who underestimated the second moment and for all others in Table 6.²² For those who underestimated dispersion, the tool led to a 7.9 pp (77.7%) increase in negotiation. It did not, however, lead to an increase in search, consistent with the insight from our model that introducing negotiation breaks the sufficient statistic relationship between reservation rates and search. For those who did not underestimate dispersion, there is no effect on negotiation, with a small and not statistically significant point estimate.

Heterogeneous treatment effects by other characteristics Appendix J.2 tests for heterogeneous treatment effects of the price comparison tool on search, negotiation, and loan terms, by other baseline characteristics, again using machine learning (Chernozhukov, Demirer, Duflo, and Fernández-Val, 2025). Across all outcomes, we find no detectable heterogeneity in treatment effects (Table J.4). This suggests that the null effects on search are not masking positive effects on search for some individuals and negative effects for others. Furthermore, it suggests that treatment effects of the tool on negotiation, the probability of taking out a loan, and interest rates tend to be spread evenly across participants.

5.5 Mechanisms Behind Effect of Tool on Negotiation

Lastly, we test whether the tool’s treatment effect on negotiation is explained by the tool shifting beliefs. While we have shown that the tool shifted beliefs and also caused an increase in negotiation, this does not necessarily imply that shifting beliefs is the mechanism that led borrowers to negotiate more and obtain lower interest rates. Alternative mechanisms include salience, attention, or consumers directly showing the tool to lenders as credible information that they could get a better rate elsewhere.

To test whether shifting beliefs is indeed the mechanism that explains the treatment effect of the tool on negotiation, we use the Kwon and Roth (2026) mechanism test (see Appendix N). We find that we cannot reject the null hypothesis that the entire effect of the tool on negotiation can be explained by the tool’s effect on beliefs ($p = 1.000$). To ensure that this is not merely an underpowered test, we conduct a placebo using search as the mechanism, as some models posit that searching for outside offers is necessary to negotiate. In this placebo test, we reject the null hypothesis that the tool’s effect on negotiating can be fully explained by the tool’s effect on search at the 10% significance level ($p = 0.081$). We conclude that the effect of the price comparison tool on negotiation is largely explained by its effect on beliefs about the interest rate distribution. These tests also highlight the importance of modeling negotiation as a distinct process from search in some contexts, and of modeling and empirically measuring biased beliefs.

²²We define underestimating dispersion as beliefs about dispersion prior to treatment being at least 1 pp lower than the observed dispersion in the administrative data. The breakdown of our participants’ accuracy in beliefs about both rates and dispersion are presented in Figure B.14. The majority of participants from whom we elicited interest rate beliefs underestimate both the levels of rates and dispersion in rates.

6 Conclusion

We document that consumers have biased beliefs about both the first and second moment of the distribution of interest rates that lenders would offer them. Most participants thought they would obtain an interest rate lower than what they actually received, and most also underestimated the second moment of the distribution. The presence of biased beliefs suggests that consumers are likely to have different search and negotiation behavior than that predicted by models where agents are assumed to perfectly know the distribution of rates from which they are drawing.

We designed a price comparison tool to correct biased beliefs and test its effects on search, negotiation, and loan terms in an RCT. The tool showed participants a histogram of interest rates that borrowers with similar characteristics obtained on similar loans in the last six months. We built the tool in collaboration with Chile’s financial regulator using administrative data on the universe of consumer loans. We recruited participants online through Google ads and both surveyed and treated them online. We measure outcomes in administrative data and a follow-up phone survey we conducted to collect rich data on participants’ search and negotiation behavior.

The price comparison tool caused participants to update their beliefs about both moments of the interest rate distribution. In terms of real effects of these changes in beliefs, the tool led participants to receive more offers and lower interest rates by negotiating more. These improved loan terms were obtained *without* searching more. Thus, we find that negotiation is an important action in consumer credit markets, in addition to search (which has received substantially more attention in the literature). Nevertheless, one of our treatments—merely asking borrowers their beliefs about interest rates—did lead them to search more and obtain lower rates through search.

While correcting biased beliefs can help consumers negotiate successfully for lower interest rates, we find that it is extremely difficult for consumers to correct biased beliefs themselves. In particular, rates shown by Google, bank websites, and third-party comparison tools are all biased or very noisy. Getting accurate interest rate quotes requires actually applying at a lender, which involves substantial search costs. Thus—with the caveat that we leave estimates of the potential general equilibrium effects of price comparison tools to future work—there is a role for financial regulators to require that lenders report loan-level information that includes both loan terms and borrower characteristics to the regulator, as Chile’s financial regulator does, and to synthesize and provide this information to consumers, as we did in this experiment.

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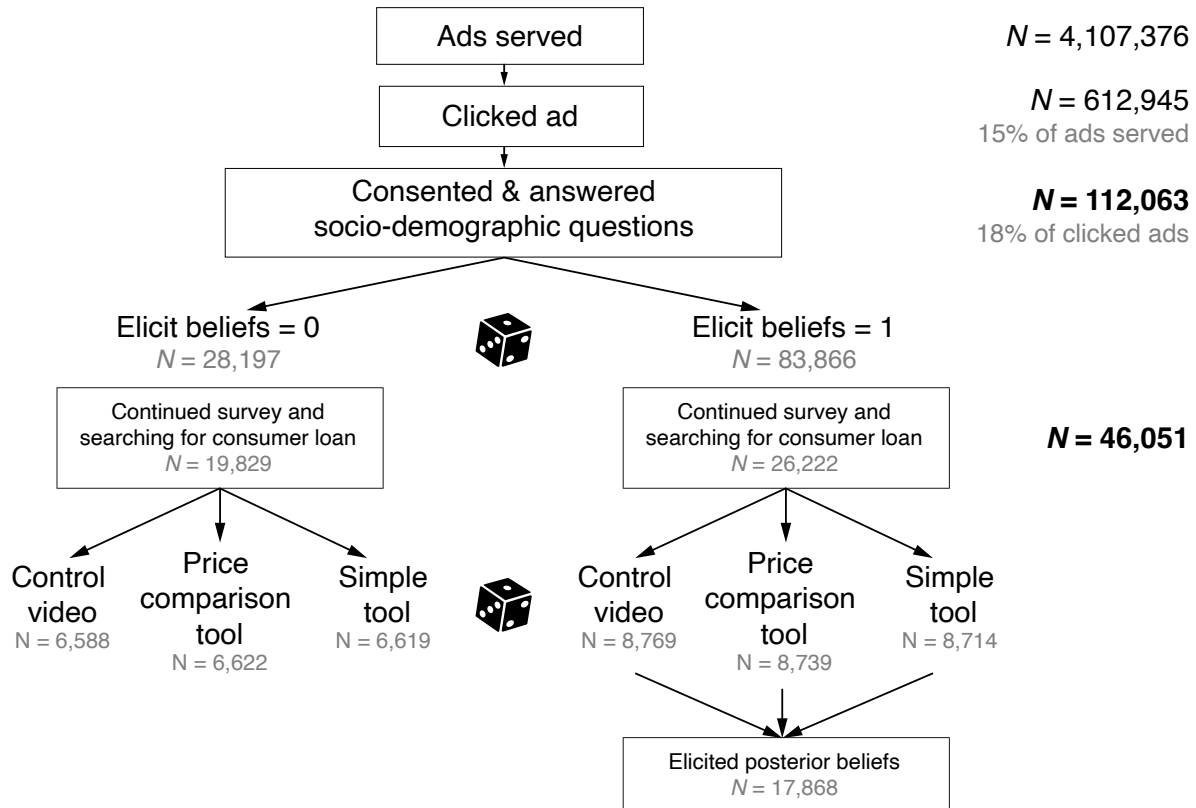
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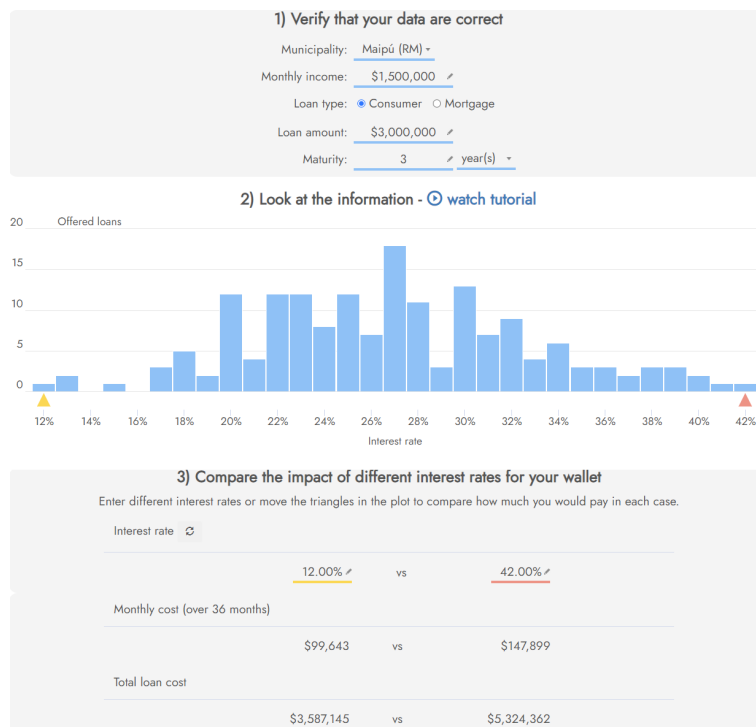
Figure 1: Experimental Design



This figure shows the progression of the participants through our study. They are randomized at two key points: when they are assigned either “Elicit beliefs = 0” or “Elicit beliefs = 1” and subsequently when they are cross-randomized to one of our three treatment arms: the control video, price comparison tool, or simple tool.

Figure 2: Price Comparison Tool and Simple Tool Treatments

(a) Interest Rate Price Comparison Tool



(b) Simple Tool

1) Verify your data are correct

Municipality: Maipú (RM) ▾

Monthly income: \$1,500,000 ✓

Loan type: ☒ Consumer ☐ Mortgage

Loan amount: \$3,000,000 ✓

Maturity: 3 ✓ year(s) ▾

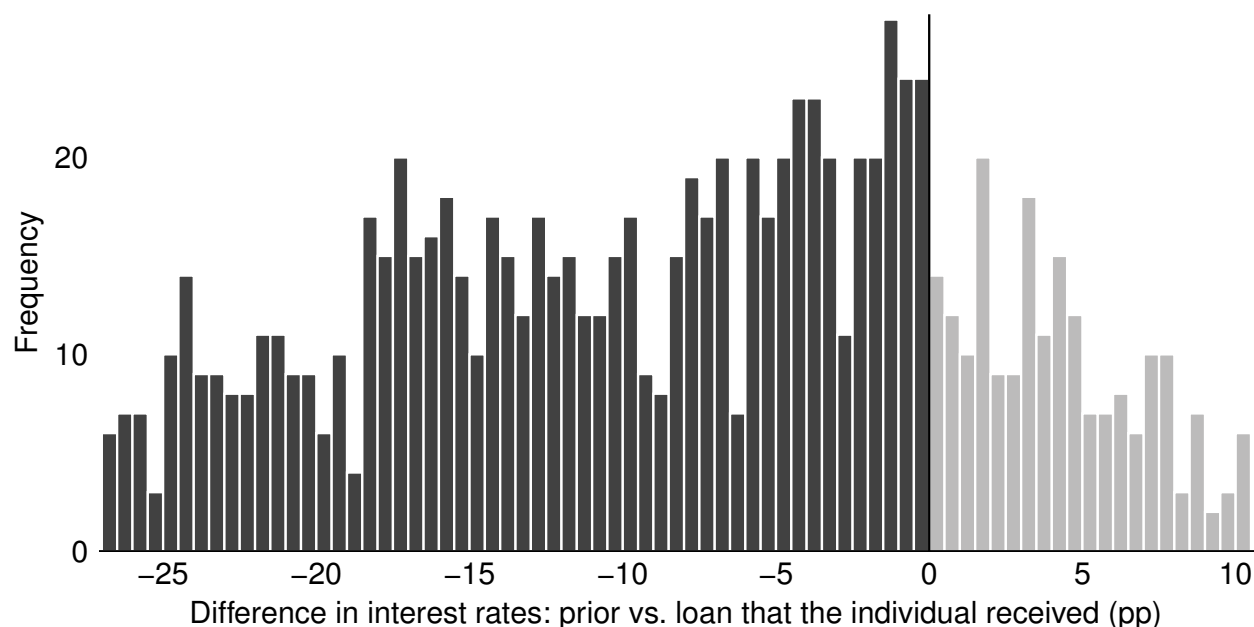
2) Look at the information

Using real data from loans granted to people similar to you, we estimate that shopping at additional bank would lower your monthly payment by \$5,954 and the total cost of your loan by \$214,343, on average.

[More details](#)

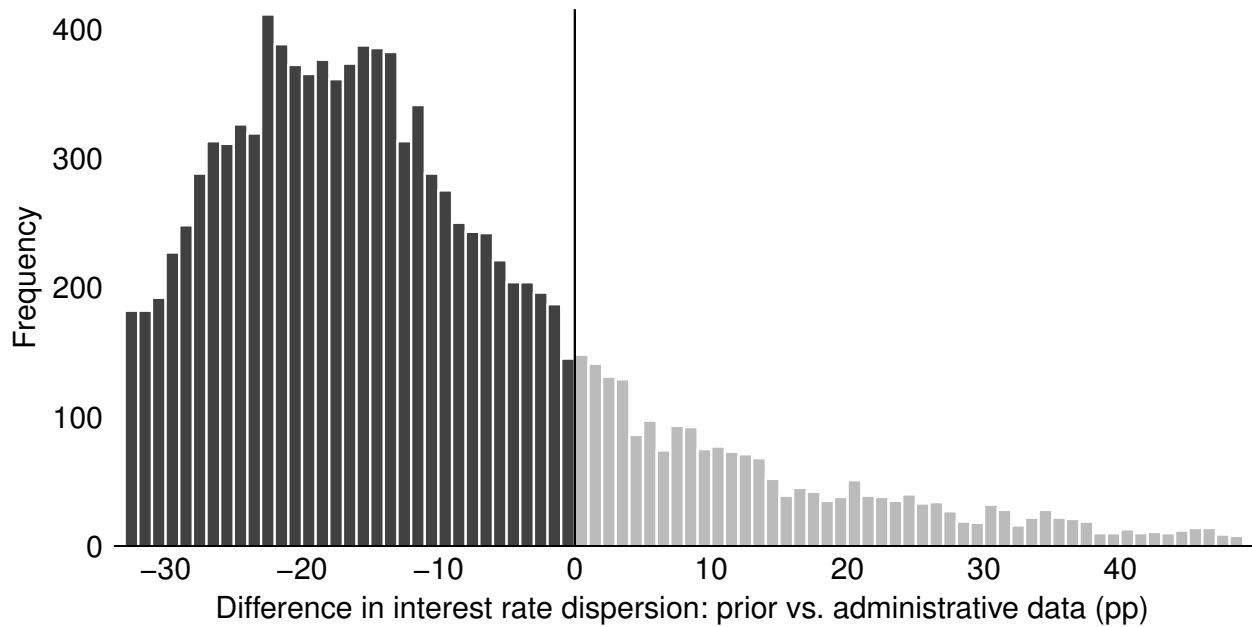
This figure shows a screenshot of English translations of our price comparison tool in panel (a) and simple tool highlighting the benefits of search in panel (b). For both tools, consumers had already entered the borrower and loan characteristics in the top panel of the tool in our baseline survey; this information is automatically populated for them. Participants can also change this information, in which case the tool is automatically refreshed to show the corresponding data. For the price comparison tool, participants can hover over the histogram bars for more information that helps them interpret and understand the information in the histogram. Participants can also move the triangles along the x-axis to see the implications on monthly and total loan costs. More details are in Appendix E.1. For the simple tool, participants can select from a drop-down menu the number of additional banks they plan to search (up to six banks). The simple tool then displays the amount of money they could save on the monthly and total cost by searching at that many additional banks. More details are in Appendix E.2.

Figure 3: Difference in Interest Rates Between Beliefs and Loan the Individual Received (pp)



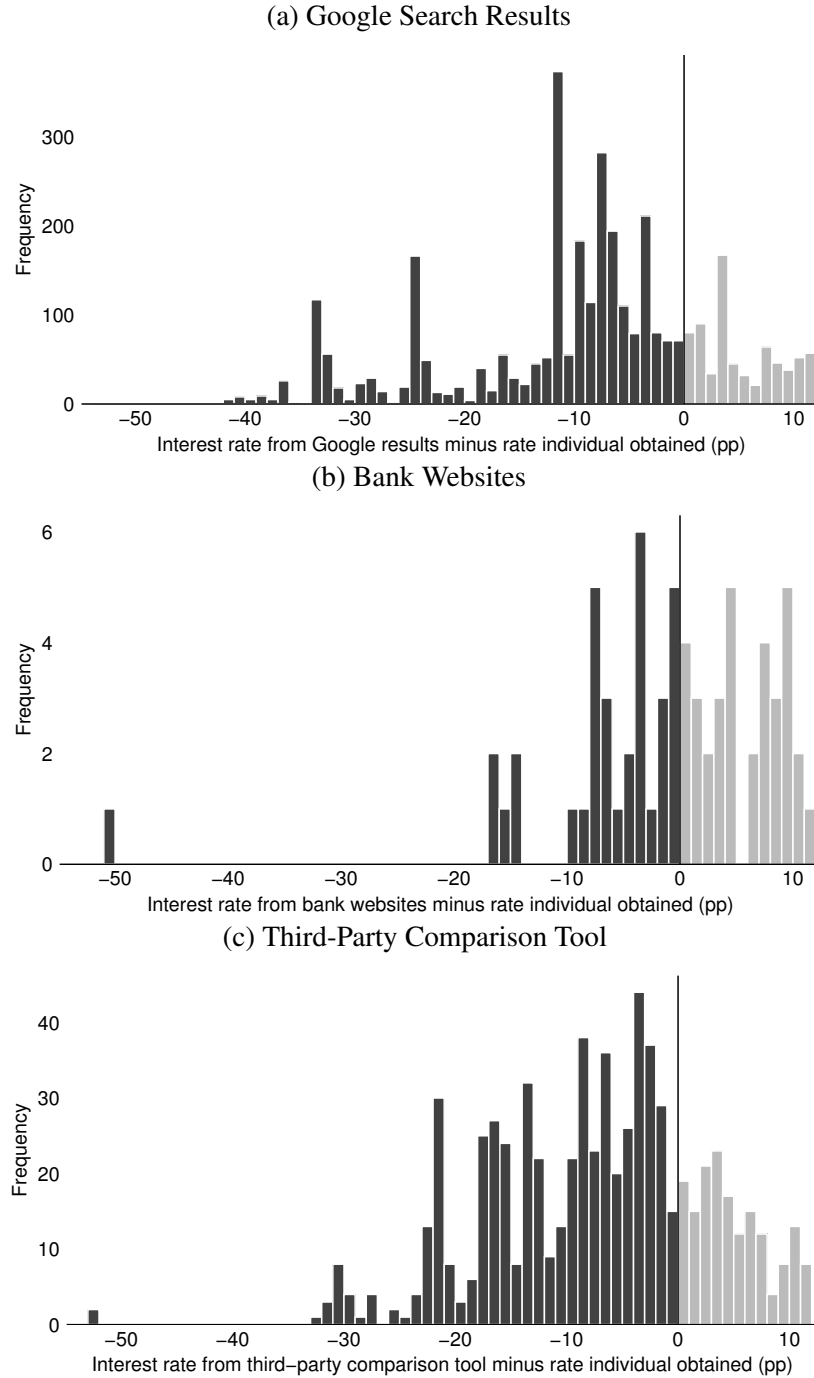
This figure shows that most participants underestimate the interest rate they ultimately obtain. The figure is a histogram of the difference between a participant's beliefs about the interest rate they will get on the loan they take out (as reported in our baseline survey) and the interest rate they ended up receiving on the loan they took out in our administrative data. We restrict to the subset of participants in the control group who took out a loan after participating and compare the belief they had reported in the baseline survey to the interest rate they obtained in the administrative data. For participants who obtained more than one loan after participating, we restrict to the first loan they obtained after participating. We remove observations beyond the 10th and 90th percentiles in the graph for legibility. The number of observations is 1,198. The percentage of people who underestimated the rate they would receive, i.e., the percentage of the sample in the negative portion of the histogram, is 73.2%. Both of these numbers include the observations not shown in the graph for legibility.

Figure 4: Difference in Interest Rate Dispersion Between Beliefs and Administrative Data (pp)



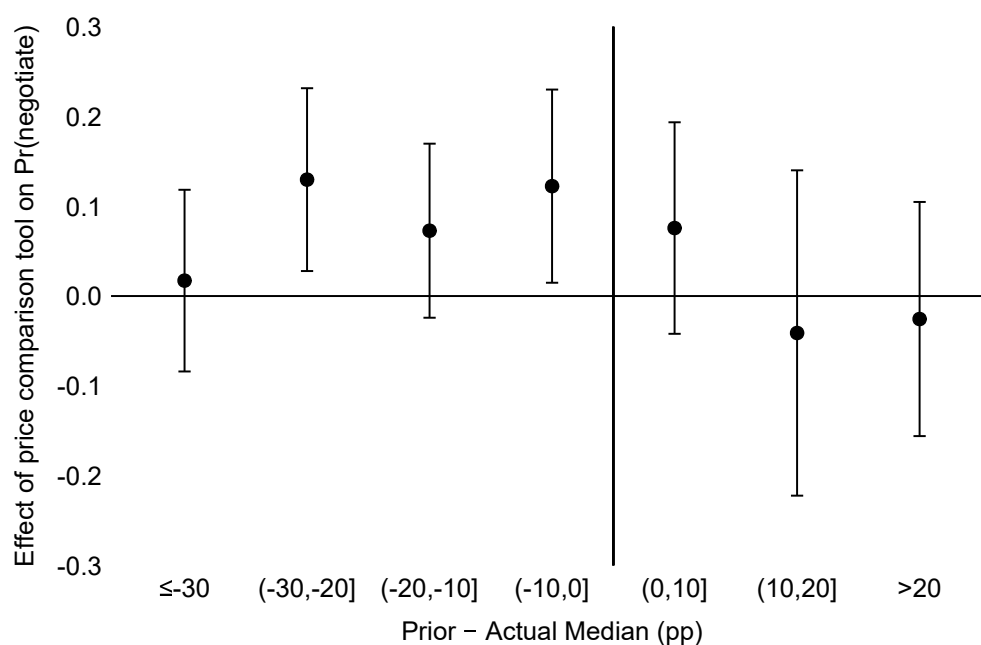
This figure shows that most participants underestimate dispersion. The figure is a histogram of the difference between a participant's prior expectations about the dispersion in interest rates that a bank could offer them, measured as the highest rate a bank could offer them minus the lowest rate a bank could offer them, compared to the dispersion we observe based on their characteristics in the administrative data (i.e., the dispersion they would have seen in the price comparison tool if assigned to that treatment arm). We remove observations beyond the 10th and 90th percentiles in the graph for legibility. The number of observations is 14,890. The percentage of people who underestimated dispersion, i.e., the percentage of the sample in the negative portion of the histogram, is 74.8%. Both of these numbers include the observations not shown in the graph for legibility.

Figure 5: Difference in Interest Rates Between Sources and Loan the Individual Obtained (pp)



This figure shows histograms of differences between interest rates shown by various sources and the actual interest rate received by participants in the administrative data. The various sources are Google search results in panel (a), bank websites in panel (b), and the most popular third-party comparison tool in panel (c); see Appendices C and H for more detail. In panel (a), there are 3,714 observations, of which 73.1% are negative. In panel (b), there are 87 observations, of which 49.4% are negative. In panel (c), there are 747 observations, of which 74% are negative. We trim the graphs at the 10th and 90th percentiles for legibility, using the lowest 10th percentile and the highest 90th percentile across the three graphs as common bounds. We restrict the analysis to loans obtained during months that overlap with the months in which we scraped data, to ensure that differences are not driven by changes in interest rates over time, which explains the small sample sizes in panels (a) and (b).

Figure 6: Heterogeneous Effects of Tool on Negotiation by Beliefs about First Moment



This figure shows heterogeneous treatment effects of the price comparison tool on the probability of negotiating, by how biased participants' beliefs about the first moment were. It shows results from specification (3) estimated separately by bins of the degree to which participants over- or underestimated the first moment of the interest rate distribution. The number of observations is 1,289. The x-axis measures the difference between the participant's belief about the interest rate they would obtain measured prior to being treated and the observed median of the interest rate distribution they face in administrative data, in percentage points. Bins are each 10 percentage points wide, except the two extreme bins.

Table 1: Balance of Pre-Treatment Characteristics by Elicit Beliefs Treatment

	Elicit Beliefs = 0 Mean (1)	Elicit Beliefs (2)	N (3)
<i>Personal characteristics</i>			
Age	35.939*** (0.059)	-0.106 (0.068)	112,063
log(Income)	13.625*** (0.007)	0.001 (0.008)	109,665
Incomplete high-school	0.037*** (0.001)	-0.001 (0.001)	108,809
Complete high-school	0.358*** (0.003)	0.003 (0.003)	108,809
Complete 2-year program	0.214*** (0.002)	-0.002 (0.003)	108,809
Complete 5-year program or higher	0.391*** (0.003)	0.000 (0.003)	108,809
<i>Financial products</i>			
Bank account	0.677*** (0.003)	0.002 (0.003)	106,220
Any loan	0.707*** (0.003)	-0.006** (0.003)	107,127
Omnibus F-statistic		0.979 [0.463]	112,063
Number of participants by arm	28,197	83,866	112,063

This table tests the balance of pre-treatment characteristics by elicit beliefs treatment. We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta \mathbb{1}(\text{Elicit Beliefs})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Elicit Beliefs})_i$ is a dummy indicating whether participant i was assigned to the elicit beliefs treatment. Column (1) shows $\hat{\alpha}$ which is the mean for the $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ group. Column (2) shows $\hat{\beta}$ which is the difference in means between the $\mathbb{1}(\text{Elicit Beliefs})_i = 1$ and $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ groups. Column (3) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression: $\mathbb{1}(\text{Elicit Beliefs})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -value of the omnibus F-statistic is included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Unlike in Table 2, we cannot include characteristics of the loan they are searching for in the balance tests since these questions were asked in the same module as the elicit beliefs treatment (rather than a prior module), and the elicit beliefs treatment affected whether participants continued in the survey. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Balance of Pre-Treatment Characteristics by Tool Treatment Arm

	Control Mean	Difference relative to control mean			N
		Price Comparison Tool	Simple Tool	Joint test F-stat	
	(1)	(2)	(3)	(4)	(5)
<i>Personal characteristics</i>					
Age	35.773*** (0.082)	-0.145 (0.116)	0.057 (0.116)	1.616 [0.199]	46,051
log(Income)	13.460*** (0.010)	0.000 (0.014)	0.004 (0.014)	0.06 [0.942]	44,978
Incomplete high-school	0.041*** (0.002)	0.001 (0.002)	0.002 (0.002)	0.426 [0.653]	44,615
Complete high-school	0.425*** (0.004)	-0.008 (0.006)	-0.007 (0.006)	1.068 [0.344]	44,615
Complete 2-year program	0.222*** (0.003)	0.006 (0.005)	0.005 (0.005)	0.865 [0.421]	44,615
Complete 5-year program or higher	0.312*** (0.004)	0.000 (0.005)	0.000 (0.005)	0.002 [0.998]	44,615
<i>Financial products</i>					
Bank account	0.618*** (0.004)	0.016*** (0.006)	0.013** (0.006)	4.566** [0.01]	43,272
Any loan	0.668*** (0.004)	0.002 (0.006)	0.006 (0.006)	0.526 [0.591]	43,675
<i>Desired loan characteristics</i>					
log(Loan Amount)	14.737*** (0.012)	0.020 (0.017)	0.017 (0.017)	0.883 [0.413]	43,775
log(Maturity (years))	1.320*** (0.005)	-0.003 (0.007)	0.009 (0.008)	1.334 [0.263]	40,920
<i>Omnibus F-statistic</i>					
Price Comparison Tool		1.179 [0.279]			30,718
Simple Tool			1.277 [0.207]		30,690
Number of participants by arm	15,357	15,361	15,333		46,051

This table tests balance of pre-treatment characteristics across treatment arms for those who reached the module in which they were assigned to one of the tool treatment arms or the control group. We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ are dummies indicating whether participant i was assigned to the simple tool or price comparison tool arms, respectively. Column (1) shows $\hat{\alpha}$ which is the mean for the control group. Columns (2) and (3) show $\hat{\beta}_2$ and $\hat{\beta}_1$ which are the differences in means between the price comparison tool treatment arm or simple tool treatment arm and the control group. Column (4) shows the F-statistic from a joint test of $\beta_1 = \beta_2 = 0$. Column (5) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression for the participants assigned to simple tool and control groups and for those assigned to the price comparison tool and control groups, separately: $\mathbb{1}(\text{Treatment})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$, where $\mathbb{1}(\text{Treatment})_i$ is either $\mathbb{1}(\text{Simple Tool})_i$ or $\mathbb{1}(\text{Price Comparison Tool})_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -values of the joint test and omnibus F-statistics are included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Desired loan characteristics” refer to characteristics of the loan the participant is searching for. “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Treatment Effects on Survey Data Outcomes

	N of inst. searched (1)	N of inst. applied (2)	Pr(negotiate) (3)	N of offers (4)	Pr(take loan) (5)	Log interest rate (6)
<i>Panel A: Effects of Eliciting Beliefs</i>						
Elicit Beliefs	0.130*** (0.048)	−0.031 (0.038)	0.011 (0.010)	−0.003 (0.024)	0.001 (0.015)	−0.073* (0.042)
Control Mean (in levels)	3.357	1.192	0.111	0.579	0.360	38.5%
Observations	5,774	5,565	5,465	5,525	5,516	1,241
<i>Panel B: Effects of Simple Tool and Price Comparison Tool</i>						
Simple Tool	0.053 (0.071)	0.018 (0.052)	0.013 (0.013)	0.019 (0.032)	0.013 (0.021)	0.000 (0.074)
Price Comparison Tool	0.017 (0.071)	0.025 (0.051)	0.037*** (0.014)	0.069** (0.033)	0.036* (0.021)	−0.127** (0.062)
Control Mean (in levels)	3.450	1.121	0.097	0.531	0.369	30.8%
Observations	3,283	3,167	3,114	3,147	3,143	555

This table shows the effects of the elicit beliefs, simple tool, and price comparison tool treatments on search, negotiation, and loan terms using survey data. It shows results from specification (1) in panel A and specification (3) in panel B. Column (1) is a count variable of the number of institutions at which participants report searching for a consumer loan; the sample excludes participants who did not know or refused to answer how many institutions they searched. Column (2) is a count variable of the number of institutions at which participants applied for a loan at any time after participating in the RCT; compared to column (1), this sample also excludes participants who did not know or refused to answer whether they applied for a loan at all of the institutions where they searched. Column (3) is a dummy variable equal to 1 if the participant tried to negotiate with at least one institution where they applied; compared to column (2), the sample also excludes participants who did not know or refused to answer whether the loan was approved for all their loan applications, and participants who did not know or refused to answer whether they negotiated for all their loan offers. Column (4) is a count variable of the number of approved loan applications at any time after participating in the RCT; compared to column (2), it excludes participants who did not know or refused to answer whether the loan was approved. Column (5) is a dummy variable equal to 1 if the participant reported taking out a loan at any time after participating in the RCT; compared to column (4), the sample also excludes participants who did not know or refused to answer whether they took out the loan for all loan offers they received. Column (6) is the natural logarithm of the interest rate offered post-negotiation; compared to column (4), the sample excludes searches that did not generate an offer and offers for which the participant did not report the offered interest rate. For column (6), each observation is a loan offer. We test for balance across treatment arms for the panel A survey subsample in Table B.2 and for the panel B survey subsample in Table B.3. Continuous variables are winsorized at the 95th percentile by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses for columns at the individual level, while standard errors clustered at the individual level are reported for columns at the loan level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Treatment Effects on Administrative Data Outcomes

	Pr(take loan) (1)	Log interest rate (2)
<i>Panel A: Effects of Eliciting Beliefs</i>		
Elicit Beliefs	-0.0042 (0.0027)	-0.0121** (0.0059)
Control Mean (in levels)	0.1952	25.6%
Observations	112,063	21,522
<i>Panel B: Effects of Simple Tool and Price Comparison Tool</i>		
Simple Tool	0.0058 (0.0045)	0.0048 (0.0099)
Price Comparison Tool	0.0089** (0.0045)	0.0039 (0.0099)
Control Mean (in levels)	0.1903	25.7%
Observations	46,051	8,988
<i>Panel C: Effects of Price Comparison Tool for Likely Always-Takers and Compliers</i>		
Price Comparison Tool Effect for Always-Takers	-0.0080 (0.0076)	-0.0352** (0.0179)
Price Comparison Tool Effect for Compliers	0.0171*** (0.0056)	0.0217* (0.0119)
p-value of Difference between Always-Takers and Compliers	[0.008]	[0.008]
Control Mean (in levels)	0.1874	25.6%
Observations	30,718	5,982

This table shows the effects of the elicit beliefs, simple tool, and price comparison tool treatments on the probability of taking a loan and log interest rates using administrative data. Panels A and B show results from specifications (1) and (3), respectively. Panel C uses the method in Appendix L to separate likely always-takers from likely compliers, and reports results from specification (15): specifically, it reports $\hat{\beta}_1$ in the row for always-takers, $\hat{\beta}_1 + \hat{\beta}_3$ in the row for compliers, and the p -value on the interaction term, i.e., on a test of $\beta_3 = 0$. Column (1) is a dummy variable equal to 1 if the participant obtained a consumer loan within 1 year after participating in the RCT according to administrative data from the CMF. Column (2) is the natural logarithm of the interest rate on the loan the participant took out according to administrative data from the CMF; compared to column (1), the column excludes those who did not take out a loan in administrative data within 1 year after participating. We test for balance across treatment arms for the panel A, column (1) full sample in Table 1, for the panel B, column (1) full sample in Table 2, and for the column (2) subsamples who obtained a loan within 1 year of participating in Tables B.12 and B.13. Continuous variables are winsorized at the 95th percentile by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effects of Price Comparison Tool and Simple Tool on Beliefs

	Expected rate (1)	Lowest rate (2)	Highest rate (3)	Dispersion (4)
Simple Tool	0.70 (0.43)	0.84** (0.35)	−0.19 (0.79)	0.01 (0.66)
Price Comparison Tool	16.18*** (1.18)	10.89*** (0.93)	30.35*** (2.24)	15.93*** (1.45)
Observations	6,817	6,760	6,661	6,272
Control Mean Posterior	29.22	22.65	47.45	23.18
Control Median Posterior	18	12	25	10.68
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs. It shows results from specification (2). Each column shows $\hat{\beta}_1$ and $\hat{\beta}_2$ for one of the following outcome variables: (1) the interest rate the participant expects to get on the loan they take out; (2) the lowest interest rate the participant expects a bank could offer them; (3) the highest interest rate the participant expects a bank could offer them; (4) the difference between the highest and the lowest interest rates the participant expects a bank could offer them. The sample sizes for each column differ based on the number of participants who responded to the corresponding questions. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Tables B.6-B.10 show alternative specifications including using the prior as a control for the posterior rather than subtracting the prior on the left-hand side, using the posterior on the left-hand side without controlling for the prior, taking the natural logarithm of the interest rate expectations, taking the natural logarithm of the interest rate expectations without controlling for priors, and using alternative measures of dispersion. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Heterogeneous Effects of Tools by Beliefs about Dispersion

	Underestimated dispersion				All others			
	N of inst. searched	N of inst. applied	Pr(negotiate)	N of offers	N of inst. searched	N of inst. applied	Pr(negotiate)	N of offers
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Simple Tool	0.015 (0.123)	0.013 (0.099)	0.031 (0.026)	−0.065 (0.061)	0.289** (0.134)	0.109 (0.088)	0.007 (0.021)	0.104** (0.053)
Price Comparison Tool	0.084 (0.140)	0.033 (0.096)	0.079*** (0.028)	0.050 (0.062)	0.003 (0.124)	0.035 (0.084)	0.028 (0.022)	0.068 (0.054)
Control Mean (in levels)	3.562	1.188	0.101	0.622	3.391	1.034	0.078	0.444
Observations	965	939	925	935	1,063	1,026	1,013	1,021

This table shows the heterogeneous effect of the price comparison tool on search and negotiation using follow-up survey data. It shows results from specification (3) estimated separately for those who underestimate dispersion in columns (1)–(4) and all others in columns (5)–(8). Underestimating dispersion is defined as beliefs about dispersion being at least 1 pp lower than the observed dispersion faced by that consumer in the administrative data. The outcomes and samples in each column are as follows, and the sample in all columns excludes survey respondents who did not reach the module in which the tool treatments were assigned. Column (1) is a count variable of the number of institutions at which participants report searching for a consumer loan; the sample excludes participants who did not know or refused to answer how many institutions they searched. Column (2) is a count variable of the number of institutions at which participants applied for a loan at any time after participating in the RCT; compared to column (1), this sample also excludes participants who did not know or refused to answer whether they applied for a loan at all of the institutions where they searched. Column (3) is a dummy variable equal to 1 if the participant tried to negotiate with at least one institution where they applied; compared to column (2), the sample also excludes participants who did not know or refused to answer whether the loan was approved for all their loan applications, and participants who did not know or refused to answer whether they negotiated for all their loan offers. Column (4) is a count variable of the number of approved loan applications at any time after participating in the RCT; compared to column (2), it excludes participants who did not know or refused to answer whether the loan was approved. Columns (5)–(8) repeat the outcomes from columns (1)–(4). Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Supplemental Appendix

Appendix A Model Details and Proofs

This appendix characterizes the equilibrium of the model described in Section 1 and includes formal statements of any technical assumptions, model predictions, and their proofs. We assume search costs, funding costs, and the lender's cost of negotiating are drawn from continuous distributions, with $s_i \sim F(\cdot)$, $c_{ij} \sim G(\cdot)$, and $d_j \sim K(\cdot)$. We drop the i and j subscripts for notational convenience.

A.1 Equilibrium Behavior

Consumer behavior In our baseline model, we assume consumers do not update their beliefs in response to the offers they receive; in Appendix A.4, we relax this assumption and show that—under additional assumptions about consumers' prior beliefs—our propositions still hold.

Since the negotiation process is costless to consumers and they expect a possibly lower rate after the negotiation, the consumer will always choose not to accept the first-round offer immediately and to instead attempt to negotiate. In the second round, the consumer then makes decisions as in a standard sequential search model, i.e., by comparing the benefit of accepting the offer and the cost of searching further. Thus, in the model, the consumer always negotiates, and our predictions can be thought of as predictions about whether the negotiation is successful, i.e., whether the bank offers a strictly lower second-round rate $r^2 < r^1$ and the consumer accepts this lower rate r^2 . An advantage of our rich survey data is that we observe r^1 , r^2 , and whether the consumer accepts the offer for each offer they received.

A consumer who has an offer in hand from bank j with interest rate r (after attempting to negotiate) will search at another bank if and only if their search cost is lower than the expected gain of getting an additional draw by searching at an additional institution, given their potentially biased beliefs:

$$s \leq \int_{-\infty}^r (r - \tilde{r}) d\hat{H}(\tilde{r}; \theta).$$

Then the consumer's optimal strategy can be described by a reservation rate r^* defined as the rate at which the consumer is indifferent between searching at another bank or accepting the loan offer in hand. Thus a consumer with search cost s and beliefs $\hat{H}$ has a reservation rate r^* that satisfies

$$\begin{aligned} s &= \int_{-\infty}^{r^*} (r^* - \tilde{r}) d\hat{H}(\tilde{r}; \theta) \\ &= \hat{H}(\tilde{r}; \theta)(r^* - \tilde{r}) \Big|_{-\infty}^{r^*} + \int_{-\infty}^{r^*} \hat{H}(\tilde{r}; \theta) d\tilde{r} \\ &= \int_{-\infty}^{r^*} \hat{H}(\tilde{r}; \theta) d\tilde{r}. \end{aligned}$$

This determines the reservation rate as a function of the consumer's search cost s and belief type θ , i.e., $r^*(s, \theta)$.

Since $F(\cdot)$ is the distribution of search costs, the distribution of reservation rates is²³

$$\Phi(r; \theta) = \mathbb{P}_s[r^*(s, \theta) \leq r] = \mathbb{P}\left[s \leq \int_{-\infty}^r \hat{H}(\tilde{r}; \theta) d\tilde{r}\right] = F\left(\int_{-\infty}^r \hat{H}(\tilde{r}; \theta) d\tilde{r}\right). \quad (4)$$

This describes the aggregate behavior of consumers with type θ .

Bank behavior The bank's behavior in the two-stage bargaining game is solved backwards.

After the consumer engages in negotiation, the bank receives a noisy signal ξ of the consumer's type θ . Intuitively, this can be thought of in a couple of ways. First, the initial offer may be made to the consumer before the consumer comes physically into a branch, but then the consumer goes to the branch to negotiate and something about how they look provides a noisy signal about their beliefs to the bank. Alternatively, what the consumer says in the negotiation could provide the noisy signal. The consumer could of course try to strategically obfuscate their beliefs in the negotiation—which we do not model directly—but we merely require there to be some truthful, noisy signal about their beliefs that the bank can nevertheless extract during the negotiation.

Conditional on the consumer's type θ_i , the realized signal ξ_{ij} (subscripts included here for clarity) is assumed to be independent of other characteristics of both the consumer and the bank, as well as the signal received by the other banks meeting the same consumer. Then, the bank forms a belief on the consumer's type θ , which is summarized by the posterior distribution $\hat{\Psi}(\theta|\xi)$. Since the reservation rate is a function of search costs *and* beliefs, and the uncertainty about the consumer's search cost is captured by the distribution $\Phi(r, \theta)$, the bank's posterior belief about the distribution of potential reservation rates of the consumer is

$$\hat{\Phi}(r, \xi) = \int \Phi(r, \theta) d\hat{\Psi}(\theta|\xi).$$

When deciding whether to lower the offer rate, the bank has to compare the cost d with the benefits of enlarging its market share (i.e., increasing the probability that the consumer accepts the offer) by offering a lower second-round rate.

Following the analysis of Agarwal, Grigsby, Hortaçsu, Matvos, Seru, and Yao (2024), who model sequential search with rejection but without biased beliefs or negotiation, the residual demand faced by the bank is

$$q(r, \xi) = \int_r^{+\infty} \frac{1}{H(\tilde{r}; \xi)} d\hat{\Phi}(\tilde{r}; \xi).$$

²³We use subscripts of $\mathbb{P}$ and $\mathbb{E}$ to indicate over which variable(s) these probabilities and expectations are calculated.

The bank then attempts to set the rate through the profit maximization problem

$$\max_{r \leq r^1} \pi(r; \xi, c) = q(r, \xi)(r - c),$$

where r^1 is the first-round offered rate. The bank lowers the rate if and only if

$$\max_{r \leq r^1} \pi(r; \xi, c) - \pi(r^1; \xi, c) > d.$$

We define the negotiation as successful if the bank lowers the rate and the consumer accepts the bank's offer.

Let the monopoly interest rate r^{2*} be the unconditionally optimal solution to the profit maximization problem

$$r^{2*}(\xi, c) = \arg \max_r \pi(r; \xi, c),$$

and let the optimal threshold $r^{2\circ}$ be the largest solution $r < r^1$ to

$$\pi(r; \xi, c) - d = \pi(r^1; \xi, c).$$

Then the optimal second-round rate is

$$r^2(\xi, c, d, r^1) = \begin{cases} r^{2*}(\xi, c), & r^{2*}(\xi, c) < r^{2\circ}(\xi, c, d, r^1), \\ r^1, & r^{2*}(\xi, c) \geq r^{2\circ}(\xi, c, d, r^1). \end{cases}$$

Later, we also write $r^2(\xi, c, d)$ as an abbreviation for $r^2(\xi, c, d, r^1(c, d))$. The first-round rate r^1 is then chosen as

$$r^1(c, d) = \arg \max_{r^1} \mathbb{E}_\xi \left[(\pi(r^{2*}; \xi, c) - d) \mathbb{1}[r^{2*} < r^{2\circ}] + \pi(r^1; \xi, c) \mathbb{1}[r^{2*} \geq r^{2\circ}] \right].$$

Thus, the optimal first-round rate is determined by the (marginal) distribution of the signal of the consumer's type, the costs of financing this consumer c , and the cost of lowering the interest rate in a negotiation d .

Equilibrium search and interest rates obtained The consumer accepts the bank's offer if and only if

$$r^2(\xi, c, d) \leq r^*(s, \theta). \tag{5}$$

Moreover, the consumer successfully negotiated if and only if the consumer not only accepts the offer but also obtains an offer lower than the initial rate, i.e.,

$$r^2(\xi, c, d) < r^1(c, d) \text{ and } r^2(\xi, c, d) \leq r^*(s, \theta).$$

Therefore, the probability of the consumer accepting a bank's offer is

$$\mathbb{P}_{\xi, c, d}[r^2(\xi, c, d) \leq r^*(s, \theta)],$$

and the probability of a successful negotiation is

$$\mathbb{P}_{\xi, c, d}[r^2(\xi, c, d) < r^1(c, d) \text{ and } r^2(\xi, c, d) \leq r^*(s, \theta)].$$

The expected number of banks at which the consumer will search is given by

$$\frac{1}{\mathbb{P}_{\xi, c, d}[r^2(\xi, c, d) \leq r^*(s, \theta)]}. \quad (6)$$

The expected interest rate they will ultimately obtain is given by

$$\mathbb{E}_{\xi, c, d}[r^2(\xi, c, d) | r^2(\xi, c, d) \leq r^*(s, \theta)]. \quad (7)$$

Both of these equilibrium outcomes are determined not only by the consumer's reservation rate $r^*(s, \theta)$ but also by the consumer's type θ .

A.2 Sufficient Statistic Predictions

In this section, we include a proof for Proposition 1 from Section 1.

Proposition 1. *In a market where consumers have potentially biased beliefs and cannot negotiate, a consumer's equilibrium search result is completely characterized by the consumer's reservation rate. In contrast, in a market with biased beliefs **and negotiation**, a consumer's equilibrium search result is **not** completely characterized by the consumer's reservation rate.*

Proof. We first prove the case without negotiation. When there is no negotiation, the bank does not observe a signal ξ about the consumer's belief type θ , as this is only revealed during the negotiation. Thus, the bank believes that the consumer has a distribution of reservation rates

$$\bar{\Phi}(\tilde{r}) = \mathbb{E}_{\theta}[\Phi(\tilde{r}; \theta)].$$

The residual demand faced by the bank is then given by

$$\bar{q}(r) = \int_r^{+\infty} \frac{1}{\bar{H}(\tilde{r})} d\bar{\Phi}(\tilde{r}).$$

Therefore, the bank will offer the interest rate to maximize its profit:

$$r^*(c) = \arg \max_r \hat{\pi}(r; c) = \arg \max_r \bar{q}(r)(r - c).$$

This indicates that the bank's offered rate is solely determined by their cost of providing a loan, c , and the aggregate distribution of reservation rates. As a result, a consumer accepts the current offer if and only if

$$r^*(c) \leq r^*(s, \theta).$$

The expected number of institutions at which the consumer will search is given by

$$\frac{1}{\mathbb{P}_c[r^*(c) \leq r^*(s, \theta)]}$$

and the expected interest rate they will ultimately obtain by

$$\mathbb{E}_c[r^*(c) | r^*(c) \leq r^*(s, \theta)].$$

The consumer's search cost s and type θ only enter these equilibrium quantities through their effect on r^* . In other words, for a given reservation rate $r^*(s, \theta)$, these quantities are uniquely determined and independent of the search cost s or the consumer's belief type θ .

We now turn to the case with negotiation. In this case, the bank observes a noisy signal ξ about the consumer's belief type θ during the negotiation, and the equilibrium for this case was derived in Section A.1. A consumer with reservation rate r^* accepts the current offer by the bank if and only if (5) holds. The consumer's type θ enters into this expression not only through its effect on the reservation rate $r^*(s, \theta)$ but also through the second-round rate $r^2(\xi, c, d)$ since ξ is assumed to be a noisy signal that increases in θ . This can also be seen by noting that the equilibrium expected number of banks at which the consumer will search in (6) and the expected interest rate they will ultimately obtain in (7) are functions of $r^2(\xi, c, d)$. Intuitively, since the bank can learn the consumer's type during the negotiation, the bank's profit function in the second round $\pi(r; \xi, c)$ is computed based on the perceived residual demand faced by the bank, $q(r, \xi)$, which has incorporated the bank's knowledge about the consumer's potentially biased beliefs.

□

A.3 Treatment Effect Predictions

We next build up the lemmas required for Propositions 2 and 3, and state technical assumptions needed for the lemmas to hold. We then formally state and prove the propositions, which follow directly from the lemmas. For expositional convenience and to take our predictions to the data, we characterize the consumer's type θ as their bias about the first and second moment of the interest rate distribution, $\theta = (\Delta\mu, \Delta\sigma^2)$.

Biased beliefs about the first moment We begin with analyzing the effect of biases about the first moment of the interest rate distribution. Throughout this section, we fix $\Delta\sigma^2 = 0$ and take $\theta = (\Delta\mu, 0)$. Hence, the parameter θ only captures first-order biases in the consumer's belief.

The main results depend on the following distributional assumptions.

Assumption 1. The distribution of search costs $F(\cdot)$ has a continuous, log-concave, and decreasing density function.

Examples of distributions satisfying Assumption 1 include exponential distributions and half-normal distributions.

Assumption 2. The cumulative distribution function of the consumer's belief $\hat{H}(r; \theta)$ is log-supermodular in (r, θ) .

Assumption 2 indicates that when θ increases, $\hat{H}(r; \theta)$ decreases more slowly (in the sense of first-order stochastic dominance, FOSD) for larger r .

Finally, let $\Psi(\xi|\theta)$ denote the conditional distribution function of the signal.

Assumption 3. The conditional distribution of signals $\Psi(\xi|\theta)$ satisfies the monotone-likelihood-ratio (MLR) property, i.e., ξ increases with θ in the MLR sense.

After observing the signal ξ , the bank infers the consumer's type using Bayes' rule:

$$\hat{\psi}(\theta|\xi) = \frac{\psi(\xi|\theta)v(\theta)}{\int \psi(\xi|\theta)v(\theta)d\theta}.$$

Here, lowercase letters denote density functions of the corresponding uppercase-letter distribution functions, and $v(\theta)$ denotes the density function of the bank's prior belief on the consumer's type θ . Under Assumption 3, the posterior distribution of the consumer's type θ on observing a signal ξ , $\hat{\Psi}(\theta|\xi)$, also satisfies the MLR property. This is a standard assumption in the signaling literature.

Lemma 1. *Under Assumptions 1 and 2, if $\hat{H}(r; \theta)$ increases in θ in the sense of FOSD, then the density function $\phi(r; \theta)$ is also log-supermodular in (r, θ) , i.e., the distribution of reservation rates $\Phi(r; \theta)$ increases in θ in the sense of a monotone-likelihood-ratio (MLR) shift.*

Proof. Note that

$$\log \phi(r; \theta) = \log f \left(\int_{-\infty}^r \hat{H}(\tilde{r}; \theta) d\tilde{r} \right) + \log \hat{H}(r; \theta).$$

It holds that

$$\frac{\partial^2 \log \phi(r; \theta)}{\partial r \partial \theta} = (\log f)'' \cdot \hat{H}(r; \theta) \int_{-\infty}^r \hat{H}(\tilde{r}; \theta) d\tilde{r} + (\log f)' \cdot \hat{H}_\theta(r; \theta) + \frac{\partial^2 \log \hat{H}(r; \theta)}{\partial r \partial \theta}. \quad (8)$$

Since $F(\cdot)$ has a log-concave and decreasing density, $(\log f)'' < 0$ and $(\log f)' < 0$. The FOSD increase of $\hat{H}$ in θ implies $\hat{H}_\theta(x; \theta) \leq 0$ for any x , as well as $\int_{-\infty}^r \hat{H}_\theta(\tilde{r}; \theta) d\tilde{r} \leq 0$. Also, the log-supermodularity of $\hat{H}(r; \theta)$ indicates that the last term is positive. Therefore, each term on the right-hand side of equation (8) is positive, so $\partial^2 \log \phi(r; \theta) / \partial r \partial \theta$ is also positive. This indicates that $\phi(r; \theta)$ is also log-supermodular in (r, θ) .

Meanwhile, when $\hat{H}(r; \theta)$ increases in θ in the sense of FOSD, $\int_{-\infty}^r \hat{H}(\tilde{r}; \theta) d\tilde{r}$ also decreases in θ for all r . Thus, $\Phi(r; \theta)$ decreases in θ for all r . The log-supermodularity of $\phi(r; \theta)$ is thus equivalent to an MLR increase of $\Phi(r; \theta)$ in θ . $\square$

Lemma 2. *Under Assumption 3, if the distribution of reservation rates $\Phi(r; \theta)$ increases in θ in the MLR sense, then the monopoly interest rate r^{2*} also increases in ξ .*

Proof. By Assumption 3 and the assumption of this lemma, both $\hat{\Psi}(\theta|\xi)$ and $\Phi(r; \theta)$ satisfy the MLR property. Therefore, by Karlin's (1968) composition theorem, the posterior distribution of the reservation rate $\hat{\Phi}(r; \xi)$ also satisfies the MLR property.

Then, it is true that the residual demand function $q(r, \xi)$ is also log-supermodular in (r, ξ) . Note that

$$\begin{aligned} \frac{\partial^2}{\partial r \partial \xi} \log q(r; \xi) &= \frac{\partial}{\partial \xi} \frac{q_r(r; \xi)}{q(r; \xi)} \\ &= \frac{q_{r\xi}(r; \xi)q(r; \xi) - q_r(r; \xi)q_\xi(r; \xi)}{(q(r; \xi))^2} \\ &= \frac{1}{(q(r; \xi))^2} \left(-\frac{\hat{\phi}_\xi(r; \xi)}{H(r)} \int_r^{+\infty} \frac{\hat{\phi}(\tilde{r}; \xi)}{H(\tilde{r})} d\tilde{r} + \frac{\hat{\phi}(r; \xi)}{H(r)} \int_r^{+\infty} \frac{\hat{\phi}_\xi(\tilde{r}; \xi)}{H(\tilde{r})} d\tilde{r} \right) \\ &= \frac{1}{(q(r; \xi))^2} \int_r^{+\infty} \frac{\hat{\phi}(r; \xi)\hat{\phi}_\xi(\tilde{r}; \xi) - \hat{\phi}(\tilde{r}; \xi)\hat{\phi}_\xi(r; \xi)}{H(r)H(\tilde{r})} d\tilde{r} \\ &= \frac{1}{(q(r; \xi))^2} \int_r^{+\infty} \frac{\hat{\phi}(r; \xi)\hat{\phi}(\tilde{r}; \xi)}{H(r)H(\tilde{r})} \left(\frac{\hat{\phi}_\xi(\tilde{r}; \xi)}{\hat{\phi}(\tilde{r}; \xi)} - \frac{\hat{\phi}_\xi(r; \xi)}{\hat{\phi}(r; \xi)} \right) d\tilde{r} \\ &= \frac{1}{(q(r; \xi))^2} \int_r^{+\infty} \frac{\hat{\phi}(r; \xi)\hat{\phi}(\tilde{r}; \xi)}{H(r)H(\tilde{r})} \left(\int_r^{\tilde{r}} \frac{\partial^2}{\partial \xi \partial x} \log \hat{\phi}(x; \xi) dx \right) d\tilde{r} > 0. \end{aligned}$$

Finally, the log-supermodularity of $q(r; \xi)$ implies a higher monopoly rate r^{2*} . Note that the monopoly rate is

$$\begin{aligned} r^{2*} &= \arg \max_r q(r; \xi)(r - c) \\ &= \arg \max_r \log q(r; \xi) + \log(r - c). \end{aligned}$$

Since we have established the log-supermodularity of $q(r; \xi)$, the theorem from Topkis (1998) applies. Thus, r^{2*} is monotonically increasing in ξ (i.e., an FOSD increase in Φ). $\square$

Next, we note that an extremely deviated prior (i.e., a large bias about the first moment of the interest rate distribution) leads to negotiation failure.

To establish this, we have to further assume that the signal is informative on the extreme cases so that the bank can recognize those consumers with highly deviated priors.

Assumption 4. For every $\theta_1 < \theta_2$, $\inf_{\xi} \frac{\psi(\xi|\theta_2)}{\psi(\xi|\theta_1)} = \inf_{\xi} \frac{\psi(\xi|\theta_1)}{\psi(\xi|\theta_2)} = 0$.

This is true for a signal structure with additive Gaussian noise. This unbounded belief assumption is common in the information economics literature (e.g., Smith and Sørensen, 2000; Acemoglu, Dahleh, Lobel, and Ozdaglar, 2011).

Lemma 3. Under Assumptions 3 and 4, when the mean of the reservation rate distribution goes too low or too high, the probability of the bank's lowering its rate after negotiation, i.e., $r^2(\xi, c, d) < r^1(c, d)$, also approaches zero.

Proof. Let $\mu_{\Phi}(\theta) = \int r d\Phi(r; \theta)$ be the mean of the reservation rate distribution and $\mu_{\hat{\Phi}}(\xi) = \int r d\hat{\Phi}(r; \xi)$ be the posterior mean perceived by a bank receiving a signal of ξ . Assumption 3 guarantees that ξ varies monotonically with θ and $\mu_{\Phi}(\theta)$ in the MLR sense and $\mu_{\hat{\Phi}}(\xi)$ varies monotonically with ξ . Assumption 4 guarantees that for sufficiently small and large ξ , the posterior mean $\mu_{\hat{\Phi}}(\xi)$ also approaches the lower and upper bounds of $\mu_{\Phi}(\theta)$.

To see the latter point, consider the behavior of the posterior mean

$$\mu_{\hat{\Phi}}(\xi) = \int \mu_{\Phi}(\theta) \hat{\psi}(\theta|\xi) d\theta$$

as ξ decreases. Let $[\underline{\theta}, \bar{\theta}]$ be the convex hull of the support of the distribution of θ . Fix some $\varepsilon > 0$. Define

$$\rho_{\underline{\theta}+\varepsilon}(\theta|\xi) = \frac{\hat{\psi}(\theta|\xi)}{\hat{\psi}(\underline{\theta}+\varepsilon|\xi)}.$$

Then, by the MLR property in Assumption 3, as ξ decreases, $\rho_{\underline{\theta}+\varepsilon}(\theta|\xi)$ increases for $\theta \in (\underline{\theta}, \underline{\theta} + \varepsilon)$ and decreases for $\theta \in (\underline{\theta} + \varepsilon, \bar{\theta})$. Moreover, by Assumption 4, $\rho_{\underline{\theta}+\varepsilon}(\theta|\xi) \rightarrow 0$. Therefore,

$$\int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \hat{\psi}(\theta|\xi) d\theta = \frac{\int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \rho_{\underline{\theta}+\varepsilon}(\theta|\xi) d\theta}{\int_{\underline{\theta}}^{\bar{\theta}} \rho_{\underline{\theta}+\varepsilon}(\theta|\xi) d\theta} \leq \frac{\int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \rho_{\underline{\theta}+\varepsilon}(\theta|\xi) d\theta}{\int_{\underline{\theta}}^{\bar{\theta}+\varepsilon} \rho_{\underline{\theta}+\varepsilon}(\theta|\xi) d\theta} \rightarrow 0,$$

since the denominator increases and the numerator converges to 0 by the monotone convergence theorem. Since

$$\mu_{\hat{\Phi}}(\xi) = \int_{\underline{\theta}}^{\underline{\theta}+\varepsilon} \mu_{\Phi}(\theta) \hat{\psi}(\theta|\xi) d\theta + \int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \mu_{\Phi}(\theta) \hat{\psi}(\theta|\xi) d\theta,$$

it holds that

$$\mu_{\Phi}(\underline{\theta}) \leq \mu_{\hat{\Phi}}(\xi) \leq \mu_{\Phi}(\underline{\theta} + \varepsilon) \left(1 - \int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \hat{\psi}(\theta|\xi) d\theta \right) + \mu_{\Phi}(\bar{\theta}) \int_{\underline{\theta}+\varepsilon}^{\bar{\theta}} \hat{\psi}(\theta|\xi) d\theta.$$

As proven before, as ξ decreases, the upper bound converges to $\mu_{\Phi}(\underline{\theta} + \varepsilon)$, which can be made arbitrarily close to $\mu_{\Phi}(\underline{\theta})$ by taking a small enough $\varepsilon > 0$. Therefore, $\mu_{\hat{\Phi}}(\xi)$ approaches $\mu_{\Phi}(\underline{\theta})$. Similarly, when ξ increases, it also approaches $\mu_{\Phi}(\bar{\theta})$.

With this established, we now examine the bank's behavior, which can be divided into two cases. First, consider the case where $\mu_{\Phi}(\theta)$ is too high. Since the monopoly interest rate $r^{2*}(\xi, c)$ increases with the mean of the reservation rate distribution, it will eventually be larger than $r^1(c, d)$ for sufficiently high $\mu_{\Phi}(\theta)$. The bank is only allowed to lower the interest rate, so the bank's optimal strategy is to keep the rate unchanged.

Next, consider the case where $\mu_{\Phi}(\theta)$ is too low. The bank finds it profitable to lower the offer rate if and only if there exists $r \in (c, r^1)$ such that

$$\pi(r; \xi, c) - \pi(r^1; \xi, c) > d.$$

However, for $r > c$, the demand function

$$\begin{aligned} q(r; \xi) &= \int_r^{+\infty} \frac{d\hat{\Phi}(\tilde{r}; \xi)}{H(\tilde{r}; \xi)} \leq \frac{1}{H(c; \xi)} \int_r^{+\infty} d\hat{\Phi}(\tilde{r}; \xi) \\ &= \frac{1}{H(c; \xi)} (1 - \hat{\Phi}(r; \xi)) \leq \frac{1}{H(c; \xi)} (1 - \hat{\Phi}(c; \xi)). \end{aligned}$$

Next, by Markov's inequality,

$$1 - \hat{\Phi}(c; \xi) \leq \frac{1}{c} \mu_{\hat{\Phi}}(\xi).$$

Therefore, for $r \in (c, r^1)$,

$$q(r; \xi) \leq \frac{1}{cH(c; \xi)} \mu_{\hat{\Phi}}(\xi),$$

so

$$\pi(r; \xi, c) \leq \frac{r^1 - c}{cH(c; \xi)} \mu_{\hat{\Phi}}(\xi).$$

For sufficiently low $\mu_{\hat{\Phi}}(\xi)$, the profit $\pi(r; \xi, c) \leq d$. In conclusion, we have proven that for $\mu_{\hat{\Phi}}(\xi)$ too low, the profit never exceeds d . Therefore, the bank will never find it profitable to lower the interest rate. $\square$

Combining these lemmas, we conclude that the optimal interest rate provided by the bank is weakly increasing with the mean of the reservation rate distribution except for extremely low means.

We next consider the effects of biased beliefs on the aggregate behaviors of a large number of bank-consumer pairs with belief $\hat{H}$. When faced with a consumer of belief $\hat{H}$, the probability of a successful negotiation is $\mathbb{P}[r^{2*} < r^{2\circ} \text{ and } r^{2*} \leq r^*]$. This requires us to take expectations of each individual-bank pair's quantities over all distributions F , G , and K .

Lemma 4. *Fix the variance of $\hat{H}$. When the bias in the first moment $\Delta\mu$ is too large in either direction, the probability of a successful negotiation approaches zero from above.*

Proof. The probability of a successful negotiation is given by

$$\mathbb{P}_{\xi, c, d}[r^2(\xi, c, d) < r^1(c, d) \text{ and } r^2(\xi, c, d) \leq r^*(s, \theta)].$$

Lemma 3 has established that when the bias in the first moment $\Delta\mu$ is too large, i.e., the mean of reservation rate distribution is either too low or too high, the conditional probability

$$\mathbb{P}_{\xi}[r^2(\xi, c, d) < r^1(c, d)]$$

approaches zero for any fixed c, d . Note that

$$\mathbb{P}_{\xi}[r^2(\xi, c, d) < r^1(c, d) \text{ and } r^2(\xi, c, d) \leq r^*(s, \theta)] \leq \mathbb{P}_{\xi}[r^2(\xi, c, d) < r^1(c, d)].$$

We conclude that the left-hand-side probability approaches zero as well for any c, d . Since the probability is bounded above by one, the unconditional probability

$$\mathbb{P}_{\xi, c, d}[r^2(\xi, c, d) < r^1(c, d) \text{ and } r^2(\xi, c, d) \leq r^*(s, \theta)]$$

also approaches zero by the dominated convergence theorem. $\square$

Biased beliefs about the second moment We next analyze the effect of consumer biases about the second moment of the interest rate distribution. Throughout this section, we fix $\Delta\mu = 0$ and take $\theta = (0, \Delta\sigma^2)$. Hence, the parameter θ only captures second-order biases in the consumer's beliefs.

We first show that expecting a lower variance leads to a higher reservation rate.

Lemma 5. *If $\hat{H}(\cdot)$ increases in the sense of second-order stochastic dominance (SOSD), then $\Phi(\cdot)$ increases in the sense of FOSD.*

Proof. In this case, the quantity

$$\int_{-\infty}^{\tilde{r}} \hat{H}(r) dr$$

decreases for each $\tilde{r} \in \mathbb{R}$. Also, the distribution $F(\cdot)$ is increasing, so $\Phi(\tilde{r})$ also decreases for each $\tilde{r} \in \mathbb{R}$. This implies that $\Phi(\cdot)$ increases in the sense of FOSD. $\square$

To proceed in the second-order case, we have to strengthen the conclusions of Lemma 5 by imposing an assumption on the distribution of reservation rates.

Assumption 5. The distribution of reservation rates $\Phi(r; \theta)$ belongs to a location family of distributions $F(x; \theta) = F_0(x - \mu(\theta))$, where $\mu(\cdot)$ is strictly increasing and F_0 is log-concave.

This assumption is satisfied, for example, when the distribution of reservation rates is a normal distribution with a fixed variance. Under this assumption, we are able to establish the following conclusion.

Lemma 6. *Under Assumption 5, if $\Phi(\cdot)$ increases in the sense of FOSD, the monopoly interest rate r^{2*} increases.*

Proof. When Assumption 5 holds, an FOSD increase in $\Phi(\cdot)$ is equivalent to an MLR shift in $\Phi(\cdot)$. To see this, note that

$$\frac{\partial^2 \log \phi(r; \theta)}{\partial r \partial \theta} = \frac{\partial^2 \log f_0(r - \mu(\theta))}{\partial r \partial \theta} = - \frac{\partial^2 \log f_0}{\partial x^2} \bigg|_{r - \mu(\theta)} \mu'(\theta).$$

Therefore, for log-concave distribution F_0 and strictly increasing μ , this is strictly positive, and thus $\Phi(r; \theta)$ experiences an MLR shift.

With this result, we then apply Lemma 2 to conclude the proof. $\square$

We next discuss the impact of a second-order deviation from the accurate prior.

Lemma 7. *Fix the mean of $\hat{H}$. As the variance of $\hat{H}$ decreases from the accurate prior, i.e., the consumer underestimates the second moment by more, the probability of a successful negotiation decreases.*

Proof. As the variance of $\hat{H}$ decreases (i.e., the consumer underestimates the second moment by more), $\hat{H}$ increases in the sense of SOSD, so by Lemma 5 the distribution of reservation rates shifts to the right. Since the variance of $\hat{H}$ is smaller than that of the accurate prior and the mean is correct, there is no chance for this scenario to fall into extreme cases. According to the prior lemmas, the probability of a successful negotiation decreases. $\square$

We now formally state the predictions around heterogeneous treatment effects of the price comparison tool by the consumer's bias about the first and second moments of the distribution, which follow directly from the prior lemmas.

Proposition 2. *If a consumer is biased about the first moment of the interest rate distribution, i.e., $\theta = (\Delta\mu, 0)$ for some $\Delta\mu \neq 0$, then the treatment effect of the price comparison tool on negotiation depends on the direction and magnitude of the bias. In particular:*

- (a) *For any $\varepsilon > 0$, there exist $L_1, L_2 > 0$ such that when $\Delta\mu < -L_1$ or $\Delta\mu > L_2$, the treatment effect on the probability of a successful negotiation is less than ε .*
- (b) *For $\Delta\mu$ in some intermediate range, treatment increases the probability of a successful negotiation.*

Proof. This is a direct result of Lemma 4. $\square$

Proposition 3. *If a consumer underestimates the second moment of the distribution, i.e., $\theta = (0, \Delta\sigma^2)$ for some $\Delta\sigma^2 < 0$, then the price comparison tool increases the consumer's probability of a successful negotiation with the bank.*

Proof. This is a direct result of Lemma 7. $\square$

A.4 Learning

Finally, we extend our baseline model in which we assumed consumers do *not* update their beliefs when they receive a draw from the distribution; we now relax that assumption and allow consumers to update from the offers they receive as Bayesians. We show that under additional assumptions

about the consumer's beliefs—necessary to make progress in a model with learning—our treatment effect predictions still hold for a small finite number of searches (reflecting our empirical context, where the median participant searches for information from three institutions).

Since the proofs of Propositions 2 and 3 compare the probability of successful negotiation for consumers with different prior beliefs, it suffices to understand the impact of learning on the probability of a successful negotiation. In particular, we examine the analog of Lemmas 4 and 7 with learning. We now denote the consumer's type after n draws from the distribution as θ_n , and their bias about the first and second moments as $\Delta\mu_n$ and $\Delta\sigma_n^2$, respectively.

Assumption 6. The consumer perceives the interest rates they observe as independent and identically distributed (i.i.d.) draws from a $\text{Normal}(\mu, \sigma^2)$ distribution with unknown mean μ and variance σ^2 .

Assumption 7. The consumer's prior belief about (μ, σ^2) follows a normal-inverse-gamma distribution $\text{NIG}(\mu_0, \kappa_0, \alpha_0, \beta_0)$, i.e.,

$$\mu \mid \sigma^2 \sim \text{Normal}(\mu_0, \sigma^2/\kappa_0), \quad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0),$$

and the consumer updates this belief using Bayes' rule after each observed rate.

Under Assumptions 6 and 7, the prior is conjugate, so the consumer's belief after observing n rates remains in the normal-inverse-gamma family, $\text{NIG}(\mu_n, \kappa_n, \alpha_n, \beta_n)$. After observing one additional rate r_n from the bank, the hyperparameters update as follows (Bernardo and Smith, 1994; Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin, 2014):

$$\begin{aligned} \kappa_n &= \kappa_{n-1} + 1, \\ \mu_n &= \mu_{n-1} + \frac{r_n - \mu_{n-1}}{\kappa_{n-1} + 1}, \\ \alpha_n &= \alpha_{n-1} + \frac{1}{2}, \\ \beta_n &= \beta_{n-1} + \frac{\kappa_{n-1}}{\kappa_{n-1} + 1} \frac{(r_n - \mu_{n-1})^2}{2}. \end{aligned}$$

Note that the consumer updates according to their *perceived* model in Assumption 6, even though the rates they observe are in equilibrium generated by banks' strategies. Under this specification, the consumer's perceived mean of the rate distribution is μ_n , so the first-moment bias $\Delta\mu_n$ corresponds to μ_n deviating from the true mean; likewise, the perceived variance is $(\kappa_n + 1)/\kappa_n \cdot \beta_n/(\alpha_n - 1)$, so the second-moment bias $\Delta\sigma_n^2$ corresponds to the deviation of this perceived variance from the true variance.

To show that Lemma 4 and thus Proposition 2 hold with learning over a small finite number of draws from the distribution, we consider the two extreme forms of bias considered by Lemma 4: a large positive and large negative bias in the first moment $\Delta\mu$. It is worth noting that after a sufficiently large number of searches, a consumer's belief will eventually converge to the true distribution. Therefore, the phenomenon described in the following lemmas vanishes as the number of searches becomes large. However, in the real world, consumers typically obtain a small and limited number of offers, so the short-term effect of learning that we study here is expected to be more relevant.

Lemma 8. *For a consumer with an extremely positive bias in the first moment, the probability of a successful negotiation still approaches zero as the bias grows, even after learning in the first search.*

Proof. After learning from a draw from the distribution, consumers with an extremely positive bias in the first moment $\Delta\mu$ only partially correct their bias according to the observed loan rate r_n , and r_n does not depend on the bias size itself. Therefore, as long as the bias is large enough, the consumer still possesses a sufficiently large bias after learning such that the consumer will tend to accept the first loan rate offered by the bank. When the bank receives a noisy signal about the consumer's posterior belief that still features a large positive bias about the first moment, the bank will conclude that the consumer will accept the first-round rate with high probability. Thus, the probability of a successful negotiation still approaches zero as the positive bias in the first moment grows, even with learning, and Lemma 4 still holds with learning for consumers with a large positive bias. $\square$

As for consumers with an extremely negative bias in the first moment, the search usually continues for a few rounds and the effect of learning is no longer trivial.

Lemma 9. *For a consumer with an extremely negative bias in the first moment, learning in the first search further decreases the reservation rate and thus lowers the probability of a successful negotiation.*

Proof. For a consumer with an extremely negative bias in the first moment, the bank's first-round rate is not conditioned on the consumer's beliefs and thus is much higher than the consumer's own belief, i.e., $r_n - \mu_{n-1}$ is a large positive number. Therefore, after observing r_n and updating, the consumer holds a belief where

$$\mathbb{E}_n[r] = \mu_n \in (\mu_{n-1}, r_n),$$

$$\text{Var}_n[r] = \frac{\kappa_n + 1}{\kappa_n} \frac{\beta_n}{\alpha_n - 1} = \frac{\kappa_n + 1}{\kappa_n} \left(\frac{\beta_{n-1}}{\alpha_n - 1} + \frac{1}{\alpha_n - 1} \frac{\kappa_{n-1}}{\kappa_{n-1} + 1} \frac{(r_n - \mu_{n-1})^2}{2} \right).$$

That is, the consumer's first-moment bias gets partially corrected after the first observation, while their second-moment belief becomes quadratically large in their initial bias. We now show that the second effect dominates in its scale, so the reservation rate of the consumer decreases after updating.

Specifically, under the posterior belief of the consumer, the distribution of r is a nonstandardized t -distribution, i.e., the quantity

$$\sqrt{\frac{\alpha_n \kappa_n}{\beta_n(\kappa_n + 1)}}(r - \mu_n) \sim t_{2\alpha_n}.$$

Then, according to (4), the distribution of reservation rates for consumers of type θ_n is

$$\begin{aligned}\Phi(r; \theta_n) &= F\left(\int_{-\infty}^r \hat{H}(\tilde{r}; \theta_n) d\tilde{r}\right) \\ &= F\left((\tilde{r} - \mu_n) \hat{H}(\tilde{r}; \theta_n) \Big|_{-\infty}^r - \int_{-\infty}^r (\tilde{r} - \mu_n) \hat{h}(\tilde{r}; \theta_n) d\tilde{r}\right) \\ &= F\left(\sqrt{\frac{\beta_n(\kappa_n + 1)}{\alpha_n \kappa_n}} \left(z T_{2\alpha_n}(z) - \int_{-\infty}^z \tilde{z} t_{2\alpha_n}(\tilde{z}) d\tilde{z}\right)\right) \\ &= F\left(\sqrt{\frac{\beta_n(\kappa_n + 1)}{\alpha_n \kappa_n}} \left(z T_{2\alpha_n}(z) + \frac{2\alpha_n + z^2}{2\alpha_n - 1} t_{2\alpha_n}(z)\right)\right) \\ &= F(s_n \Lambda_n(z)),\end{aligned}$$

where

$$s_n = \sqrt{\frac{\beta_n(\kappa_n + 1)}{\alpha_n \kappa_n}}, \quad \Lambda_n(z) = z T_{2\alpha_n}(z) + \frac{2\alpha_n + z^2}{2\alpha_n - 1} t_{2\alpha_n}(z),$$

and the standardized observation

$$\begin{aligned}z &= \sqrt{\frac{\alpha_n \kappa_n}{\beta_n(\kappa_n + 1)}}(r - \mu_n) \\ &= \sqrt{\frac{\alpha_n \kappa_n}{\kappa_n + 1}} \left(\beta_{n-1} + \frac{\kappa_{n-1}}{\kappa_{n-1} + 1} \frac{(r_n - \mu_{n-1})^2}{2} \right)^{-1/2} \left(r - \mu_{n-1} - \frac{r_n - \mu_{n-1}}{\kappa_{n-1} + 1} \right) \\ &\rightarrow z_n^\circ = -\sqrt{\frac{2\alpha_n}{\kappa_{n-1}(\kappa_n + 1)}}\end{aligned}$$

as $r_n - \mu_{n-1} \rightarrow +\infty$. Let $\Delta_n = r_n - \mu_{n-1}$. Note that

$$\begin{aligned}
t'_{2\alpha_n}(z) &= -\frac{2\alpha_n+1}{2}t_{2\alpha_n}(z) \left(1 + \frac{z^2}{2\alpha_n}\right)^{-1} \frac{2z}{2\alpha_n} = -\frac{(2\alpha_n+1)z}{2\alpha_n+z^2}t_{2\alpha_n}(z), \\
\Lambda'_n(z) &= T_{2\alpha_n}(z) + z t_{2\alpha_n}(z) + \frac{2z}{2\alpha_n-1}t_{2\alpha_n}(z) + \frac{2\alpha_n+z^2}{2\alpha_n-1}t'_{2\alpha_n}(z) = T_{2\alpha_n}(z), \\
\frac{\partial z}{\partial \Delta_n} &= -\frac{1}{s_n} \frac{\partial \mu_n}{\partial \Delta_n} - \frac{r-\mu_n}{s_n^2} \frac{\partial s_n}{\partial \Delta_n} = -\frac{1}{s_n} \left(\frac{\partial \mu_n}{\partial \Delta_n} + z \frac{\partial s_n}{\partial \Delta_n} \right), \\
\frac{\partial \mu_n}{\partial \Delta_n} &= \frac{1}{\kappa_{n-1}+1} = \frac{1}{\kappa_n}, \\
\frac{\partial s_n}{\partial \Delta_n} &= \frac{1}{2} \sqrt{\frac{\kappa_n+1}{\beta_n \alpha_n \kappa_n}} \frac{\partial \beta_n}{\partial \Delta_n} = \sqrt{\frac{\kappa_n+1}{\alpha_n \kappa_n}} \frac{\kappa_{n-1}}{2(\kappa_{n-1}+1)} \frac{\Delta_n}{\sqrt{\beta_n}} \\
&\rightarrow \sqrt{\frac{\kappa_n+1}{\alpha_n \kappa_n}} \frac{\kappa_{n-1}}{2(\kappa_{n-1}+1)} \sqrt{\frac{2(\kappa_{n-1}+1)}{\kappa_{n-1}}} = \frac{1}{\kappa_n} \sqrt{\frac{\kappa_{n-1}(\kappa_n+1)}{2\alpha_n}} = -\frac{1}{\kappa_n z_n^\circ} \quad (\Delta_n \rightarrow +\infty).
\end{aligned}$$

Then, taking derivatives of the distribution of reservation rates $\Phi(r; \theta_n)$ with respect to Δ_n leads to

$$\begin{aligned}
\frac{\partial \Phi(r; \theta_n)}{\partial \Delta_n} &= f(s_n \Lambda_n(z)) \left(\frac{\partial s_n}{\partial \Delta_n} \Lambda_n(z) + s_n \Lambda'_n(z) \frac{\partial z}{\partial \Delta_n} \right) \\
&= f(s_n \Lambda_n(z)) \left(\frac{\partial s_n}{\partial \Delta_n} \Lambda_n(z) - T_{2\alpha_n}(z) \left(\frac{\partial \mu_n}{\partial \Delta_n} + z \frac{\partial s_n}{\partial \Delta_n} \right) \right) \\
&= f(s_n \Lambda_n(z)) \left(\frac{\partial s_n}{\partial \Delta_n} \frac{2\alpha_n+z^2}{2\alpha_n-1} t_{2\alpha_n}(z) - T_{2\alpha_n}(z) \frac{1}{\kappa_n} \right),
\end{aligned}$$

which is eventually positive for a sufficiently large bias size $\Delta_n = r_n - \mu_{n-1}$, because the first factor is always positive and the factor in the parentheses

$$\frac{\partial s_n}{\partial \Delta_n} \frac{2\alpha_n+z^2}{2\alpha_n-1} t_{2\alpha_n}(z) - T_{2\alpha_n}(z) \frac{1}{\kappa_n} \rightarrow -\frac{1}{\kappa_n z_n^\circ} \frac{2\alpha_n+(z_n^\circ)^2}{2\alpha_n-1} t_{2\alpha_n}(z_n^\circ) - T_{2\alpha_n}(z_n^\circ) \frac{1}{\kappa_n} = -\frac{1}{\kappa_n z_n^\circ} \Lambda_n(z_n^\circ) > 0$$

is eventually positive as $\Delta_n \rightarrow +\infty$. Therefore, the distribution function of the reservation rate $\Phi(r; \theta_n)$ increases pointwise in the bias size $r_n - \mu_{n-1}$ for a sufficiently large bias. This means that the distribution of reservation rates decreases in the sense of FOSD.

The decrease in the consumer's reservation rate after updating further reduces the incentive of the bank to lower the loan rate in its second round when it observes a noisy signal about the consumer's belief, which further decreases the probability of a successful negotiation. For those with a sufficiently extreme negative bias about the first moment, this probability already approaches zero without learning (Lemma 4), so it continues to approach zero after decreasing further, and Lemma 4 still holds with learning for consumers with a large negative bias. $\square$

Lemmas 8 and 9 establish that even after a few rounds of learning, for any $\varepsilon > 0$, the probability of the consumer experiencing a successful negotiation still remains below ε when the bias in the first moment is large enough. This is exactly the analog of Lemma 4 in the learning setting. Furthermore, consumers who somewhat underestimate the first moment might benefit from the treatment as it will make their bias less extreme, and thus the probability of a successful negotiation rises from zero. Therefore, Proposition 2 still holds under the same conditions.

Lastly, we examine whether Proposition 3 holds with learning by assessing Lemma 7. To do so, we must make an additional assumption about how consumers who underestimate the second moment update their prior beliefs in response to treatment under the assumptions about priors in Assumption 7. Since the proposition considers consumers who are unbiased about the first moment but underestimate the second moment, we assume that seeing the true distribution in the price comparison tool leads these consumers to increase their belief about how much dispersion there is by increasing β_n , since $\sigma^2 \sim \text{Inv-Gamma}(\alpha_n, \beta_n)$:

Assumption 8. Consumers who underestimate the second moment, i.e., $\theta_n = (0, \Delta\sigma_n^2)$ for some $\Delta\sigma_n^2 < 0$, partially correct their beliefs by increasing β_n in response to seeing the price comparison tool, without changing their beliefs about the other hyperparameters.

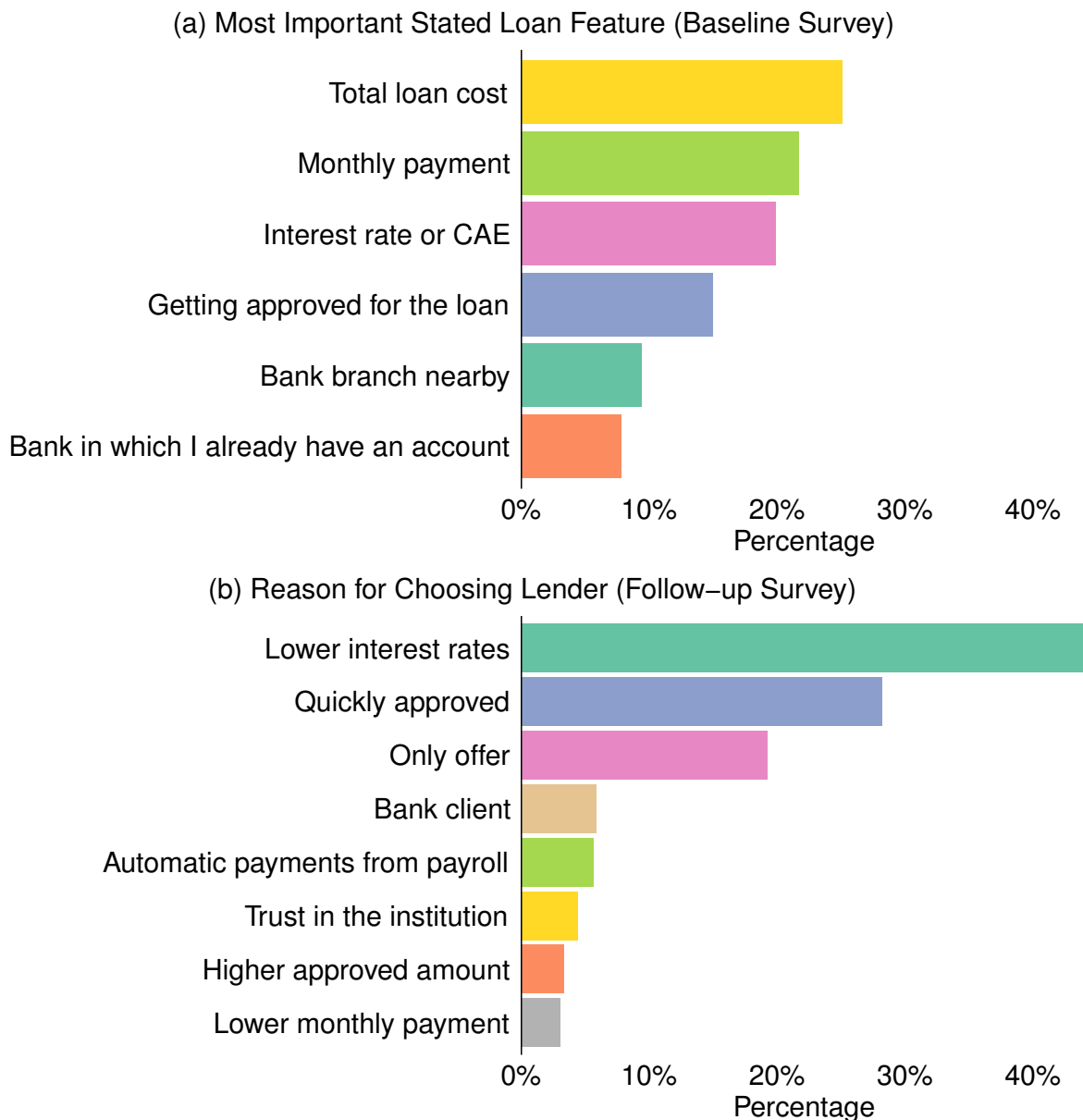
Lemma 10. *With learning, if a consumer underestimates the second moment of the distribution, i.e., $\theta_n = (0, \Delta\sigma_n^2)$ for some $\Delta\sigma_n^2 < 0$, then the price comparison tool still increases the consumer's probability of a successful negotiation with the bank.*

Proof. It suffices to compare two consumers of the same initial belief meeting the same bank, one of whom receives the treatment prior to meeting the bank and the other of whom does not. By Assumption 8 and the laws of motion for belief updating implied by Assumptions 6 and 7, the only difference in their beliefs lies in the β_n sequence, and the consumer receiving the treatment always has a higher β_n than the other. Therefore, the belief of the consumer receiving the treatment always has a higher variance, thus leading to a lower reservation rate. According to the proof of Lemma 7, this implies a higher probability of a successful negotiation. $\square$

Thus, Proposition 3 still holds with learning.

Appendix B Additional Figures and Tables

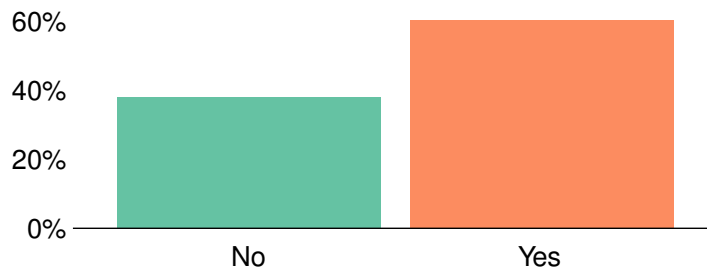
Figure B.1: Stated Importance of Loan Features



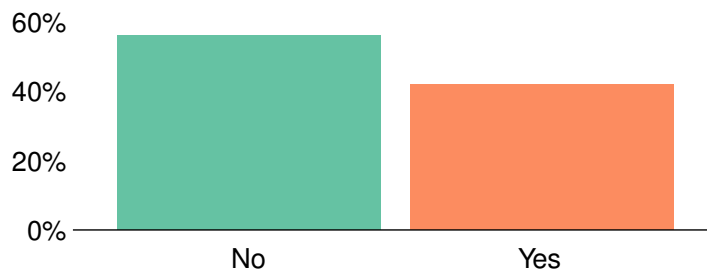
This figure shows the most important features of a loan that participants reported in the baseline survey and the reason participants chose to accept a particular loan offer in the follow-up survey. Panel (a) shows results from the baseline survey, conducted when participants were searching. It shows the reasons that participants ranked as most important in response to the question “What are the most important features of the loan you are looking for?” Panel (b) shows responses in our follow-up survey for the subset of participants who took out a loan. It shows responses to the question “Why did you take the loan from {Bank X} compared to offers you saw or received from other banks?” CAE refers to the *carga anual equivalente* which is analogous to an annualized percentage rate (APR).

Figure B.2: Sequential vs. Simultaneous Search

(a) Sequential Search: Target Interest Rate

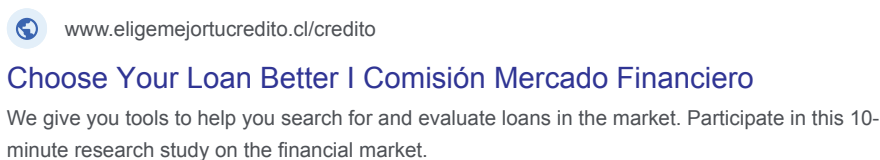


(b) Simultaneous Search: Target Number of Offers or Banks



This figure shows the results of asking participants in our follow-up phone survey questions about their search strategy. We asked three yes/no questions that underlie the two panels: (a) “Did you plan to search until you reached a target interest rate and then stop searching?”; (b) “Did you have a target number of offers you would like to receive from financial institutions to stop searching?” and “Did you have a target number of financial institutions from which you wanted to obtain information about loans?” For panel (a), we report the proportion that respond yes or no to the question, and for panel (b), we report the number of participants who answered “yes” to at least one of the two questions.

Figure B.3: Sample Google Advertisement for Participant Recruitment



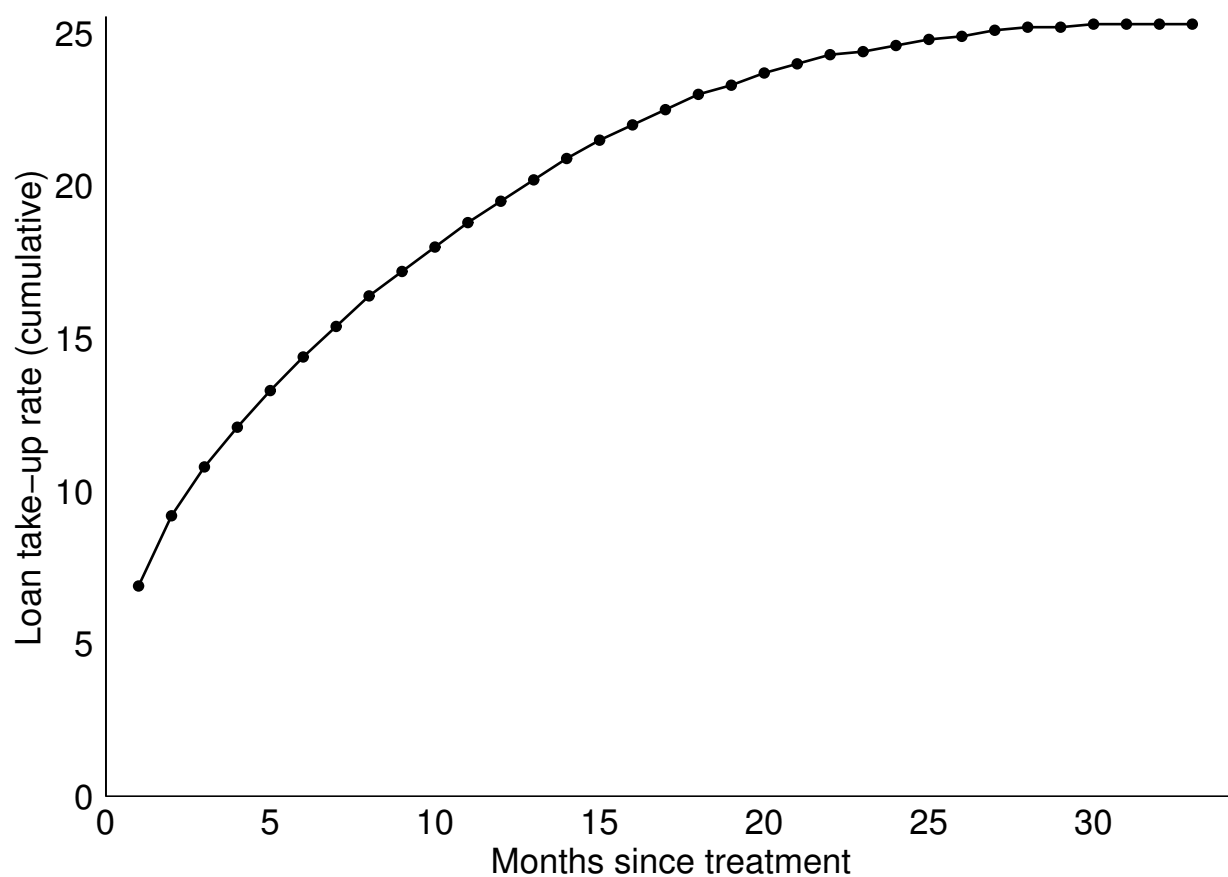
This figure shows an English translation of one of our Google advertisements that we targeted to people searching for keywords related to consumer loans in Chile to recruit them as participants in the RCT.

Figure B.4: Screenshot of Control Video



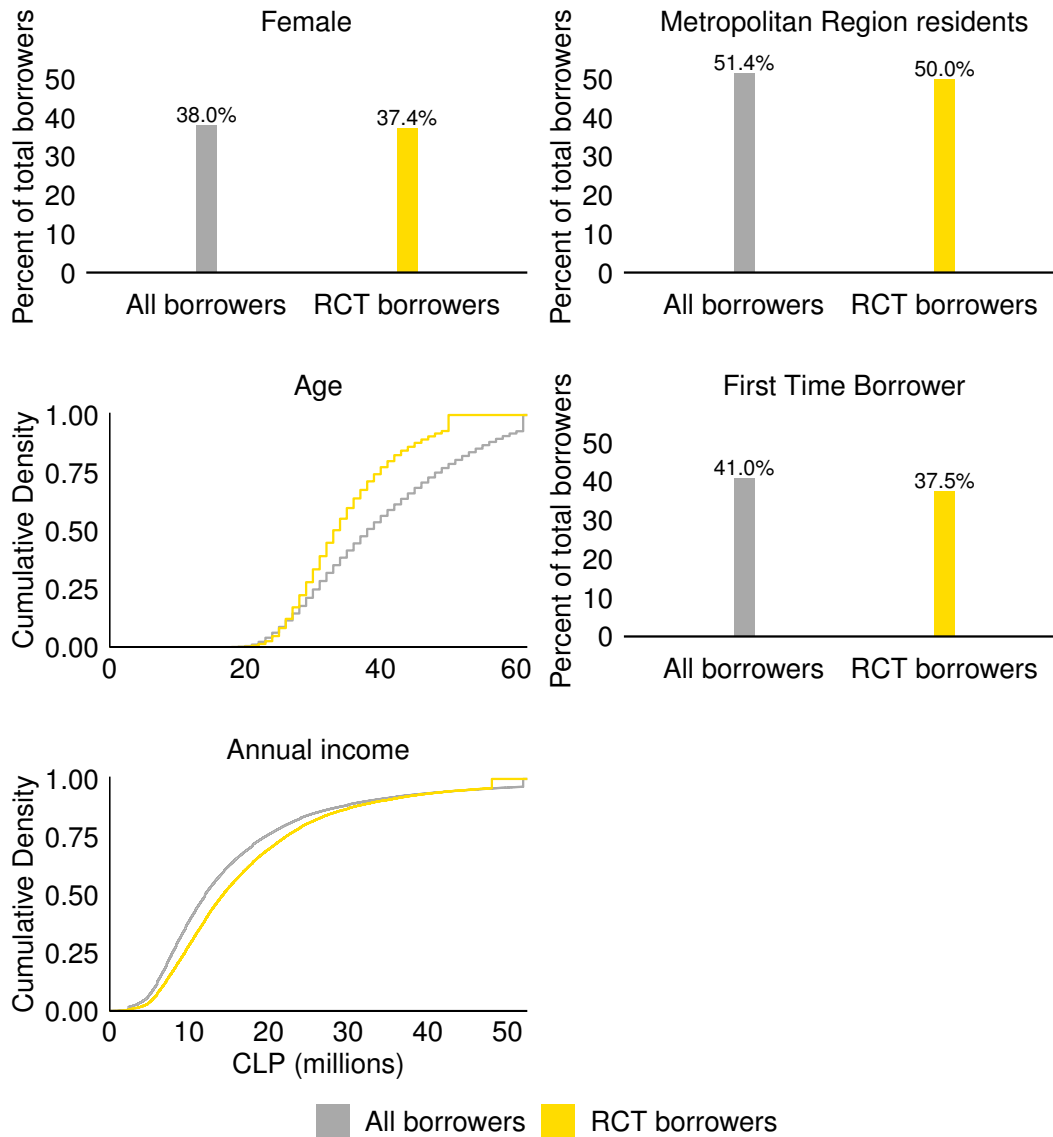
This figure shows a screenshot of the animated video shown to the control group. The video lasts 1 minute and 35 seconds and was developed by the Comisión Mercado Financiero (CMF) to provide basic loan terminology, but not provide information that would affect search or negotiation.

Figure B.5: Participant Loan Take-Up Rate Since Treatment



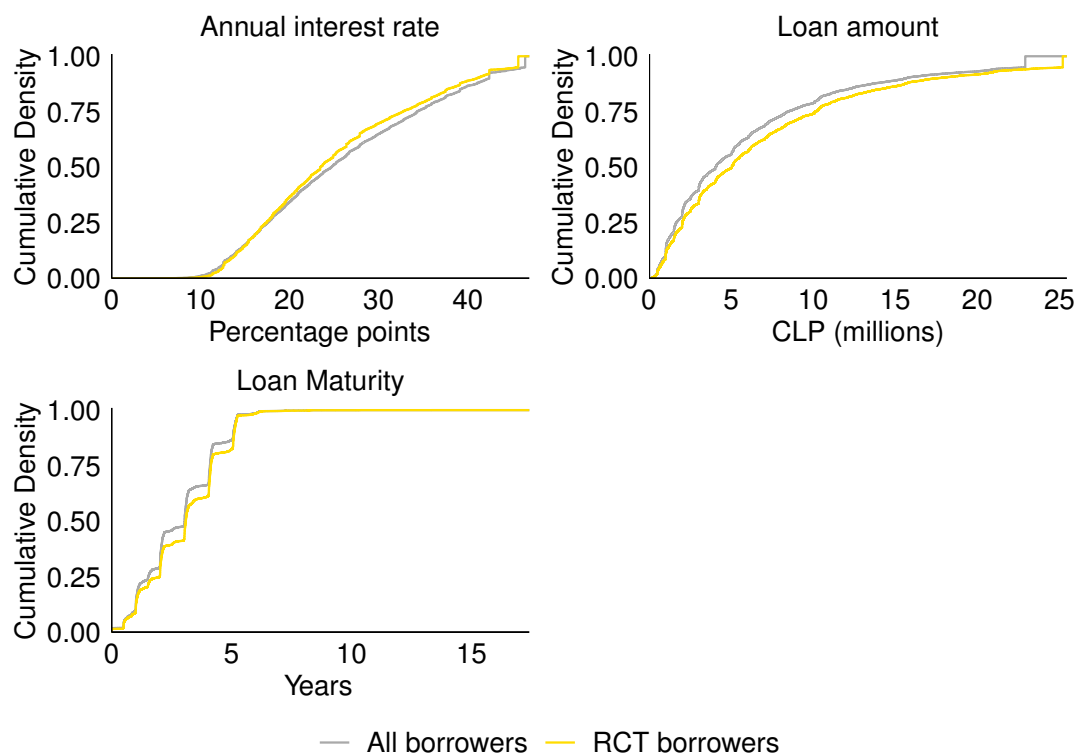
The figure shows the cumulative loan take-up rates of the 46,051 participants who were assigned to either our control video, price comparison tool, or simple tool. We define loan take up as the participants having a loan in our administrative data on bank consumer loans that was originated after the date they participated in our RCT.

Figure B.6: External Validity: Borrower Characteristics



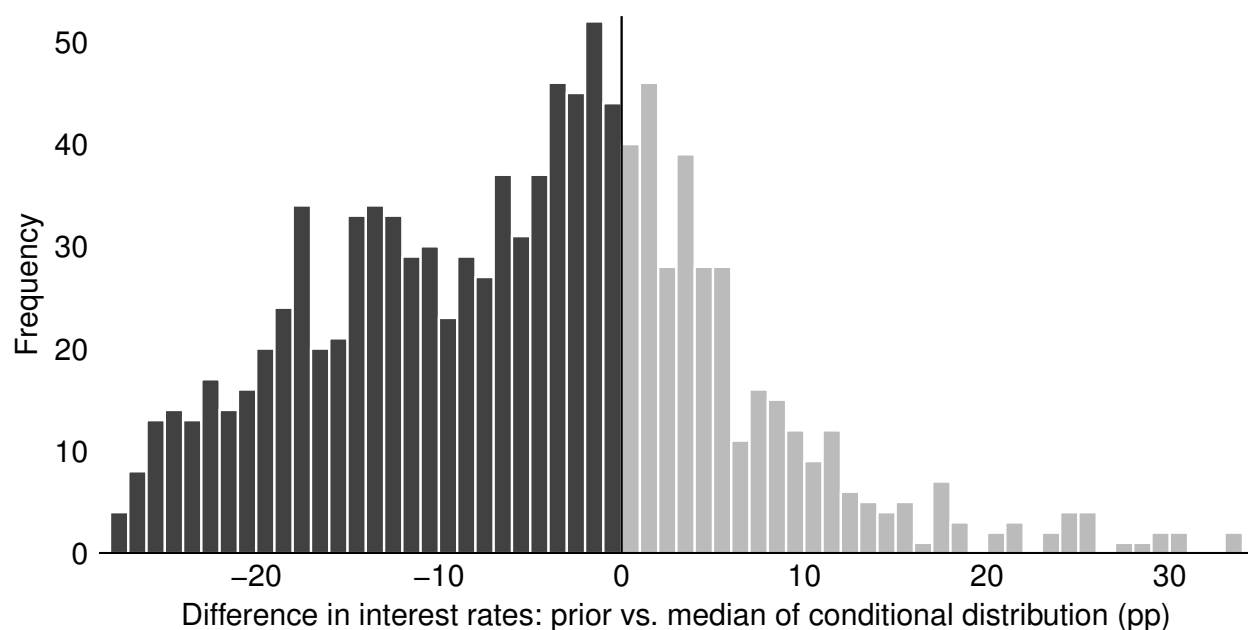
This figure compares borrower characteristics for participants in our RCT who subsequently obtained a loan in the administrative data (yellow bars), to those of all borrowers in the administrative data excluding RCT participants (gray bars). The sample is constructed from 31,148 loans taken out by RCT borrowers and 1,451,746 loans taken out by other borrowers from November 2021 to June 2024.

Figure B.7: External Validity: Loan Terms



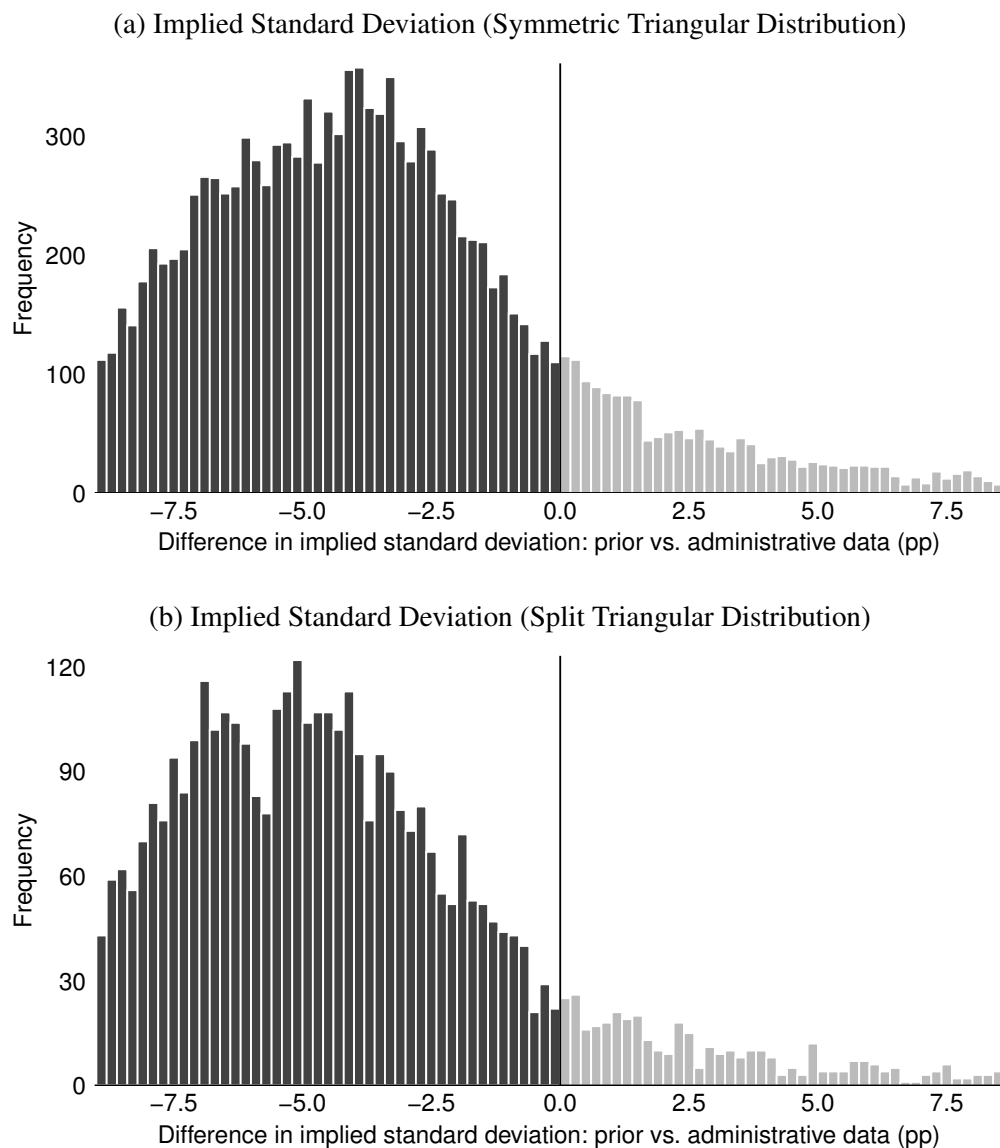
This figure compares loan terms for loans obtained by participants in our RCT who subsequently obtained a loan in the administrative data (yellow bars), to those of loans obtained by all borrowers in the administrative data excluding RCT participants (gray bars). The sample is constructed from 31,148 loans taken out by RCT borrowers and 1,451,746 loans taken out by other borrowers from November 2021 to June 2024.

Figure B.8: Difference in Interest Rate Beliefs and Median of Conditional Distribution (pp)



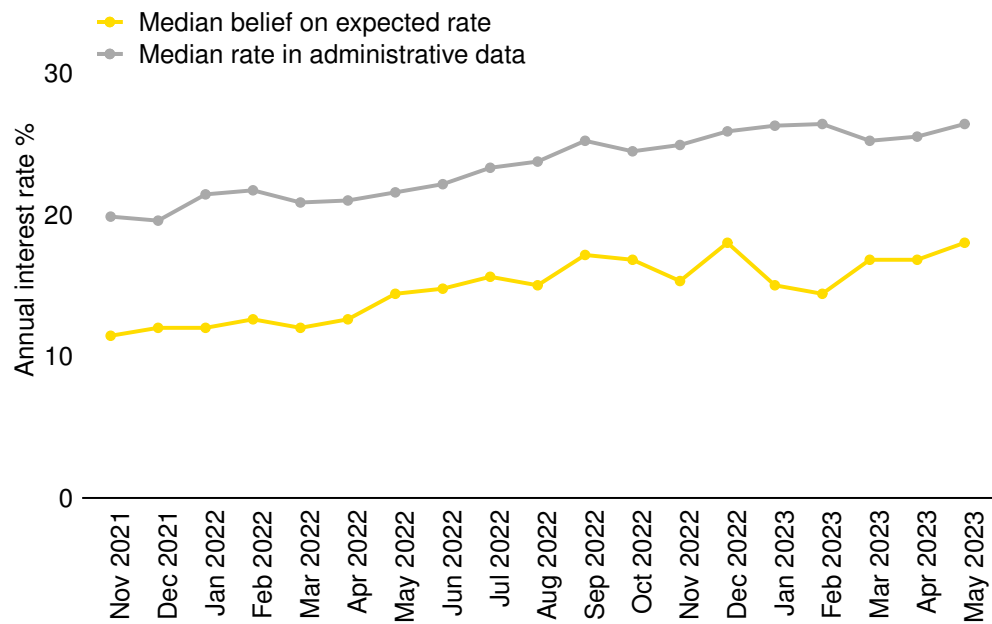
The figure is a histogram of the difference between a participant's beliefs about the interest rate they will get on the loan they take out (as reported in our baseline survey) and the median interest rate of the conditional distribution they would have seen in the tool. We restrict to the subset of participants in the control group who took out a loan after participating and compare the belief they had reported in the baseline survey to the median interest rate of the conditional distribution they would have seen in the tool. We remove observations beyond the 10th and 90th percentiles in the graph for legibility. The number of observations is 1,186. The percentage of the sample in the negative portion of the histogram is 66.4%. Both of these numbers include the observations not shown in the graph for legibility.

Figure B.9: Difference in Interest Rate Dispersion Between Beliefs and Administrative Data (pp)



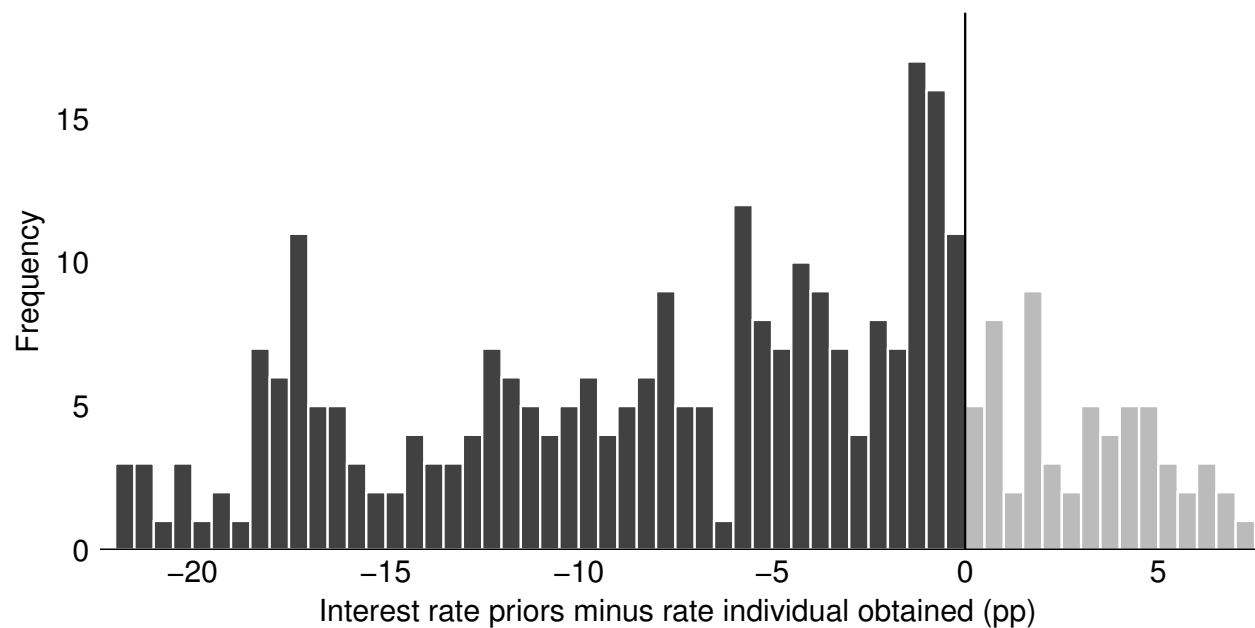
This figure shows the difference in interest rate dispersion between beliefs and administrative data using our survey questions to estimate the implied standard deviation of the consumer's beliefs about the interest rate distribution. In panel (a), we assume a symmetric triangular distribution, while in panel (b) we assume a split triangular distribution. We then subtract the standard deviation of the consumer's conditional interest rate distribution in administrative data from the implied standard deviation of their distribution of beliefs (see Appendix G for details). We trim the graphs at the 10th and 90th percentiles for legibility, using the lowest 10th percentile and the highest 90th percentile across the two graphs as common bounds. The number of observations in panel (a) is 14,890 and the percentage of people who underestimated the dispersion, i.e., the percentage of the sample in the negative portion of the histogram, is 78.6%. The number of observations in panel (b) is 4,797—lower than that of panel (a) since we added the fraction-over-midpoint question to the baseline survey partway through the experiment on October 24, 2022, and some participants after that date did not respond to the additional question—and the percentage of people who underestimated the dispersion is 82.5%. All of these numbers include the observations not shown in the graph for legibility.

Figure B.10: Interest Rates Over Time



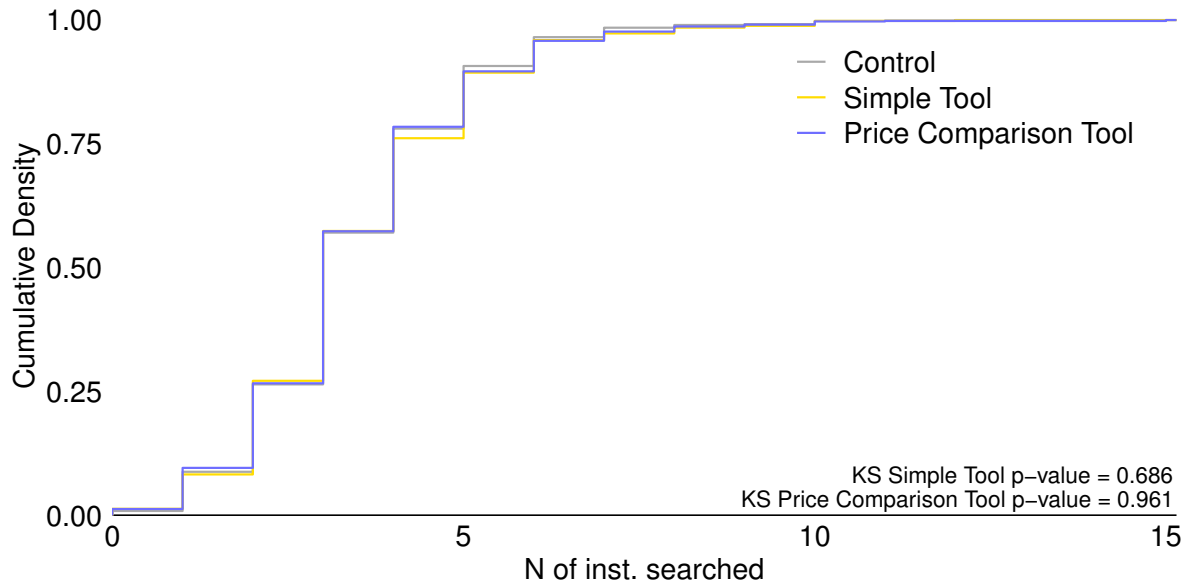
This figure shows the median annual interest rate from administrative data and the median belief about the expected annual interest rate reported by respondents in the baseline survey. For each month, we compute the median across all observed consumer loans (in administrative data) and across all respondents' beliefs (in survey data). Months are based on the loan origination date for administrative data and the response date for survey data.

Figure B.11: Difference in Interest Rates Between Beliefs and Rate the Individual Received (pp), Restricted to First Quartile of Time Between Participation and Obtaining Loan



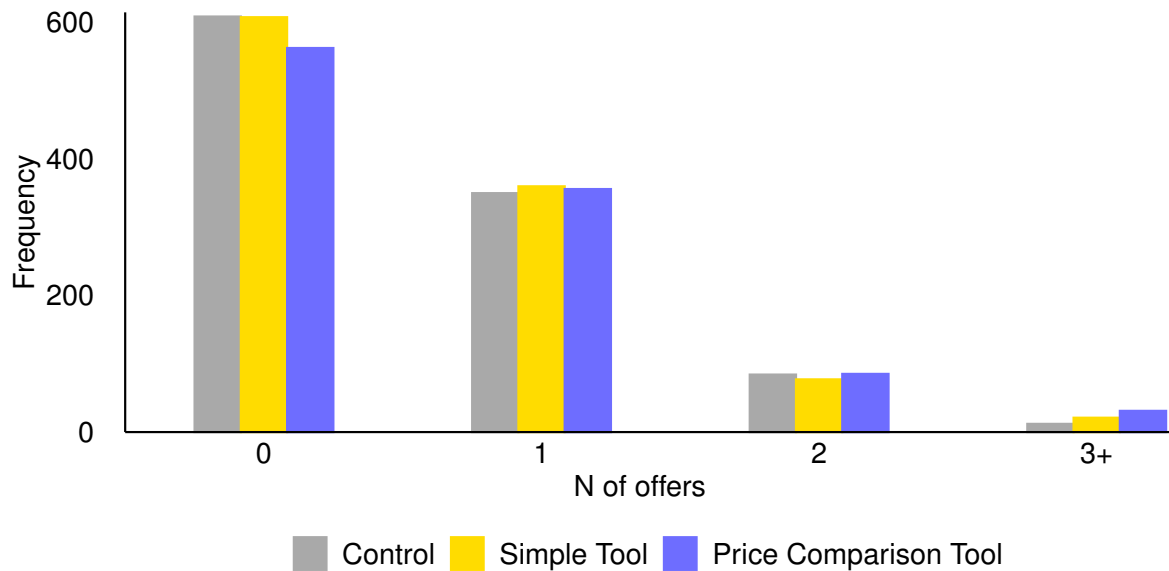
This figure is equivalent to Figure 3 but restricted to the first quartile of the number of days between participating in the RCT and obtaining a loan. It shows that even among participants who obtain a loan shortly after participating (within 22 days, which corresponds to the 25th percentile), most underestimate the interest rate they ultimately obtain. The figure is a histogram of the difference between a participant's beliefs about the interest rate they will get on the loan they take out (as reported in our baseline survey) and the interest rate they ended up receiving on the loan they took out in our administrative data. We restrict to the subset of participants in the control group who took out a loan within 22 days after participating and compare the belief they had reported in the baseline survey to the interest rate they obtained in the administrative data. For participants who obtained more than one loan after participating, we restrict to the first loan they obtained after participating. We remove observations beyond the 10th and 90th percentiles in the graph for legibility. The number of observations is 394. The percentage of people who underestimated the rate they would receive, i.e., the percentage of the sample in the negative portion of the histogram, is 74.9%. Both of these numbers include the observations not shown in the graph for legibility.

Figure B.12: Cumulative Distribution Function of Number of Institutions Searched



This figure shows the cumulative distribution function (CDF) of the number of institutions searched by treatment arm. $N = 3,283$. The Kolmogórov-Smirnov (KS) test is estimated comparing each tool treatment to the control group and is calculated by Monte Carlo simulations with 10,000 replications.

Figure B.13: Distribution of Number of Loan Offers Received by Treatment Arm



This figure shows a histogram of the number of loan offers received for consumer loans by treatment arm ($N = 3,147$). Bars are shown side-by-side for each category. Individuals that received 3 or more offers are grouped in the last bin. Results are based on values reported in the follow-up survey.

Figure B.14: Belief Heterogeneity

Prior – Actual Dispersion	Underestimate	N = 255	N = 1,923	N = 514	N = 8,137
	Equal	N = 5	N = 173	N = 31	N = 196
	Overestimate	N = 80	N = 2,152	N = 147	N = 1,125
	Don't know	N = 10,200	N = 362	N = 32	N = 865
		Don't know	Overestimate	Equal	Underestimate
		Prior – Actual Median			

This figure shows participants' beliefs about interest rate dispersion and the interest rate they expected to receive, compared to actual values observed in administrative data, conditional on each participant's characteristics ($N = 26,197$). "Prior – actual median" is the difference between a participant's beliefs about the interest rate they will get on the loan they take out (as reported in our baseline survey) and the median rate we observe based on borrower and loan characteristics in the administrative data (i.e., the median they would have seen in the price comparison tool if assigned to that treatment arm). "Prior – actual dispersion" is the difference between a participant's beliefs about the dispersion in interest rates that a bank could offer them—measured as the highest rate minus the lowest rate they believed they could be offered—and the actual dispersion we observe based on their characteristics in the administrative data. We allow for a tolerance of ± 1 percentage point when defining equality, as it is unlikely participants would report exact matches with the administrative data. "Don't know" includes those who responded "don't know," "don't want to answer," or skipped the question. Each cell reports the number of participants in each category. Colors reflect cell frequencies, with red indicating the cell with the highest numbers of observations. The sample includes participants who made it far enough in the baseline survey to be assigned to a tool treatment arm or to the control group.

Table B.1: Follow-Up Survey Response

	Pr(answer the survey)	
	(1)	(2)
(Intercept)	0.150*** (0.003)	0.154*** (0.004)
Elicit Beliefs	0.003 (0.004)	
Simple Tool		−0.004 (0.006)
Price Comparison Tool		−0.006 (0.006)
Observations	37,664	21,007

This table tests for differential response rates to the follow-up survey by elicit beliefs treatment status and by tool treatment status. It uses specifications (1) and (3), where y_i is a dummy variable equal to 1 if participant i responded to the follow-up survey. The sample in column (1) is restricted to participants whom we attempted to contact in the follow-up survey, and column (2) is further restricted to participants who made it far enough in the baseline survey to be assigned to a tool treatment arm or the control group. We define answering the survey as successfully completing (i.e., reaching the end of) the phone survey. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Balance by Elicit Beliefs Treatment for Survey Subsample in Table 3

	Elicit Beliefs = 0 Mean (1)	Elicit Beliefs (2)	N (3)
<i>Personal characteristics</i>			
Age	36.882*** (0.250)	-0.364 (0.293)	5,774
log(Income)	13.585*** (0.032)	0.039 (0.037)	5,667
Incomplete high-school	0.028*** (0.004)	0.000 (0.005)	5,634
Complete high-school	0.347*** (0.012)	-0.014 (0.014)	5,634
Complete 2-year program	0.211*** (0.010)	-0.002 (0.012)	5,634
Complete 5-year program or higher	0.413*** (0.013)	0.015 (0.015)	5,634
<i>Financial products</i>			
Bank account	0.681*** (0.012)	0.011 (0.014)	5,535
Any loan	0.734*** (0.011)	-0.013 (0.013)	5,582
Omnibus F-statistic		0.923 [0.517]	5,774
Number of participants by arm	1,577	4,197	5,774

This table tests the balance of pre-treatment characteristics by elicit beliefs treatment for the survey subsample in Table 3, panel A. We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta \mathbb{1}(\text{Elicit Beliefs})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Elicit Beliefs})_i$ is a dummy indicating whether participant i was assigned to the elicit beliefs treatment. Column (1) shows $\hat{\alpha}$ which is the mean for the $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ group. Column (2) shows $\hat{\beta}$ which is the difference in means between the $\mathbb{1}(\text{Elicit Beliefs})_i = 1$ and $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ groups. Column (3) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression: $\mathbb{1}(\text{Elicit Beliefs})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -value of the omnibus F-statistic is included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Unlike in Table 2, we cannot include characteristics of the loan they are searching for in the balance tests since these questions were asked in the same module as the elicit beliefs treatment (rather than a prior module), and the elicit beliefs treatment affected whether participants continued in the survey. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Balance by Tool Treatment Arm for Survey Subsample in Table 3

	Control Mean	Difference relative to control mean			N
		Price Comparison Tool	Simple Tool	Joint test F-stat	
	(1)	(2)	(3)	(4)	(5)
<i>Personal characteristics</i>					
Age	36.555*** (0.302)	-0.347 (0.424)	0.093 (0.432)	0.593 [0.553]	3,283
log(Income)	13.548*** (0.033)	-0.003 (0.051)	0.000 (0.049)	0.002 [0.998]	3,229
Incomplete high-school	0.027*** (0.005)	-0.003 (0.007)	-0.002 (0.007)	0.114 [0.893]	3,204
Complete high-school	0.380*** (0.015)	-0.027 (0.021)	-0.019 (0.021)	0.870 [0.419]	3,204
Complete 2-year program	0.206*** (0.012)	0.037** (0.018)	0.009 (0.018)	2.218 [0.109]	3,204
Complete 5-year program or higher	0.387*** (0.015)	-0.007 (0.021)	0.012 (0.021)	0.394 [0.674]	3,204
<i>Financial products</i>					
Bank account	0.647*** (0.015)	0.019 (0.021)	0.027 (0.021)	0.896 [0.408]	3,149
Any loan	0.697*** (0.014)	0.030 (0.020)	-0.004 (0.020)	1.795 [0.166]	3,176
<i>Desired loan characteristics</i>					
log(Loan Amount)	14.986*** (0.040)	0.022 (0.059)	0.017 (0.058)	0.080 [0.923]	3,110
log(Maturity (years))	1.362*** (0.019)	0.008 (0.027)	0.001 (0.026)	0.054 [0.948]	2,970
<i>Omnibus F-statistic</i>					
Price Comparison Tool		1.049 [0.400]			2,171
Simple Tool			1.361 [0.158]		2,212
Number of participants by arm	1,100	1,071	1,112		3,283

This table tests the balance of pre-treatment characteristics across treatment arms for the survey subsample in Table 3, panel B. We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ are dummies indicating whether participant i was assigned to the simple tool or price comparison tool arms, respectively. Column (1) shows $\hat{\alpha}$ which is the mean for the control group. Columns (2) and (3) show $\hat{\beta}_2$ and $\hat{\beta}_1$ which are the differences in means between the price comparison tool treatment arm or simple tool treatment arm and the control group. Column (4) shows the F-statistic from a joint test of $\beta_1 = \beta_2 = 0$. Column (5) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression for the participants assigned to simple tool and control groups and for those assigned to the price comparison tool and control groups, separately: $\mathbb{1}(\text{Treatment})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$, where $\mathbb{1}(\text{Treatment})_i$ is either $\mathbb{1}(\text{Simple Tool})_i$ or $\mathbb{1}(\text{Price Comparison Tool})_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -values of the joint test and omnibus F-statistics are included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Desired loan characteristics” refer to characteristics of the loan the participant is searching for. “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Effects of Price Comparison Tool and Simple Tool on Other Loan Terms

	Survey Data		Administrative Data		
	Log loan amount (1)	Maturity (years) (2)	Log loan amount (3)	Maturity (years) (4)	Days to take-up (5)
<i>Panel A: Effects of Eliciting Beliefs</i>					
Elicit Beliefs	0.098 (0.068)	0.175 (0.199)	0.015 (0.016)	-0.012 (0.024)	-1.616 (1.710)
Control Mean (in levels)	6.7M CLP	3.3	7.2M CLP	3.1	132
Observations	1,384	1,432	21,522	21,522	21,522
<i>Panel B: Effects of Simple Tool and Price Comparison Tool</i>					
Simple Tool	0.090 (0.086)	-0.194 (0.119)	0.016 (0.026)	-0.029 (0.038)	-0.192 (2.811)
Price Comparison Tool	-0.019 (0.089)	-0.015 (0.123)	0.013 (0.026)	-0.034 (0.038)	-2.796 (2.770)
Control Mean (in levels)	6.5M CLP	3.2	6.7M CLP	3.2	112
Observations	998	1,027	8,988	8,988	8,988

This table shows the effects of eliciting beliefs (panel A), the simple tool, and the price comparison tool (panel B) on loan terms other than the interest rate, using follow-up survey and administrative data. Column (1) is the log of the reported loan amount and column (2) is the reported maturity in years as reported in the follow-up survey. Both outcomes are at the loan level. Columns (3) through (5) use administrative data restricted to consumer loans taken within 1 year of participating in the RCT: column (3) is the log of the loan amount, column (4) is the loan maturity in years, and column (5) is the number of days between participation and loan take-up according to administrative data from the CMF. Continuous variables are winsorized at the 95th percentile by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses for columns at the individual level, while standard errors clustered at the individual level are reported for columns at the loan level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Correlation between Beliefs and Monthly Median Rate in Administrative Data

	Expected rate (1)	Lowest rate (2)	Highest rate (3)	Dispersion (4)
Median Rate _{<i>t</i>}	1.33*** (0.15)	1.29*** (0.12)	3.25*** (0.31)	1.67*** (0.19)
Observations	16,015	15,875	15,618	15,045

This table shows the correlation between beliefs and the monthly median rate during the month in which a consumer participated. Coefficients are from a regression of the participant i 's belief against the median rate during the month t in which participant i participated in the RCT (both measured in percentage points). The monthly median rate is calculated using the universe of consumer loans in administrative data for each month. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Effects of Tools on Beliefs, Controlling for Priors

	Expected rate (1)	Lowest rate (2)	Highest rate (3)	Dispersion (4)
Prior	0.80*** (0.02)	0.80*** (0.02)	0.68*** (0.02)	0.52*** (0.02)
Simple Tool	-0.26 (0.86)	0.44 (0.70)	1.07 (1.46)	-1.96** (0.80)
Price Comparison Tool	18.94*** (1.55)	14.36*** (1.23)	39.16*** (2.93)	20.61*** (1.78)
Observations	6,817	6,760	6,661	6,272
Control Mean Posterior	29.22	22.65	47.45	23.18
Control Median Posterior	18	12	25	10.68
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs. We estimate the following specification: $Posterior_i = \theta Prior_i + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \lambda_{b(i)} + \varepsilon_i$, where $Prior_i$ is the interest rate expectation participant i reported prior to seeing one of the tools or the control video, $Posterior_i$ is the interest rate expectation they reported after seeing it, and $\lambda_{b(i)}$ are bin density fixed effects. Each column shows $\hat{\theta}$, $\hat{\beta}_1$, and $\hat{\beta}_2$ for one of the following outcome variables: (1) the interest rate the participant expects to get on the loan they take out; (2) the lowest interest rate the participant expects a bank could offer them; (3) the highest interest rate the participant expects a bank could offer them; (4) the difference between the highest and the lowest interest rates. The sample sizes for each column differ based on the number of participants who responded to the corresponding questions. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.7: Effects of Tools on Beliefs, without Subtracting or Controlling for Priors

	Expected rate (1)	Lowest rate (2)	Highest rate (3)	Dispersion (4)
Simple Tool	−1.01 (1.19)	−0.02 (0.95)	−0.88 (2.00)	−2.95*** (0.98)
Price Comparison Tool	22.13*** (1.83)	17.22*** (1.48)	43.87*** (3.38)	23.38*** (1.93)
Observations	7,792	7,640	7,533	7,321
Control Mean Posterior	29.91	22.94	48.12	23.72
Control Median Posterior	17.88	12	25	10
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs. We estimate the following specification: $Posterior_i = \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \lambda_{b(i)} + \varepsilon_i$, where $Posterior_i$ is the interest rate expectation participant i reported prior to seeing one of the tools or the control video, and $\lambda_{b(i)}$ are bin density fixed effects. Each column shows $\hat{\beta}_1$ and $\hat{\beta}_2$ for one of the following outcome variables: (1) the interest rate the participant expects to get on the loan they take out; (2) the lowest interest rate the participant expects a bank could offer them; (3) the highest interest rate the participant expects a bank could offer them; (4) the difference between the highest and the lowest interest rates the participant expects a bank could offer them. The sample sizes for each column differ based on the number of participants who responded to the corresponding questions. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8: Effects of Tools on Beliefs in Logs, Controlling for Priors

	log(Expected rate) (1)	log(Lowest rate) (2)	log(Highest rate) (3)	log(Dispersion) (4)
log(Prior)	0.695*** (0.010)	0.701*** (0.010)	0.684*** (0.010)	0.578*** (0.013)
Simple Tool	-0.038* (0.023)	-0.008 (0.023)	-0.041* (0.024)	-0.091*** (0.032)
Price Comparison Tool	0.315*** (0.028)	0.273*** (0.027)	0.367*** (0.029)	0.335*** (0.038)
Observations	6,817	6,760	6,661	6,272
Control Mean Posterior (Levels)	29.22	22.65	47.45	23.18
Control Median Posterior (Levels)	18	12	25	10.68
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs. We estimate the following specification: $\log(\text{Posterior}_i + 1) = \theta \log(\text{Prior}_i + 1) + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \lambda_{b(i)} + \varepsilon_i$, where Prior_i is the interest rate expectation participant i reported prior to seeing one of the tools or the control video, Posterior_i is the interest rate expectation they reported after seeing it, and $\lambda_{b(i)}$ are bin density fixed effects. To account for any rate expectations of 0 (which do occur in the data), the transformation $\log(y_i + 1)$ was applied to the annualized interest rates in levels (where, for example, an 18% expected annual interest rate would be coded as 18). Each column shows $\hat{\theta}$, $\hat{\beta}_1$, and $\hat{\beta}_2$ for one of the following outcome variables: (1) the interest rate the participant expects to get on the loan they take out; (2) the lowest interest rate the participant expects a bank could offer them; (3) the highest interest rate the participant expects a bank could offer them; (4) the difference between the highest and the lowest interest rates. The sample sizes for each column differ based on the number of participants who responded to the corresponding questions. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.9: Effects of Tools on Beliefs in Logs, without Subtracting or Controlling for Priors

	log(Expected rate) (1)	log(Lowest rate) (2)	log(Highest rate) (3)	log(Dispersion) (4)
Simple Tool	−0.057* (0.033)	−0.031 (0.032)	−0.066* (0.035)	−0.128*** (0.039)
Price Comparison Tool	0.407*** (0.034)	0.376*** (0.034)	0.459*** (0.036)	0.398*** (0.043)
Observations	7,792	7,640	7,533	7,321
Control Mean Posterior (Levels)	29.91	22.94	48.12	23.72
Control Median Posterior (Levels)	17.88	12	25	10
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs. We estimate the following specification: $\log(\text{Posterior}_i + 1) = \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \lambda_{b(i)} + \varepsilon_i$, where Posterior_i is the interest rate expectation participant i reported prior to seeing one of the tools or the control video, and $\lambda_{b(i)}$ are bin density fixed effects. To account for any rate expectations of 0 (which do occur in the data), the transformation $\log(y_i + 1)$ was applied to the annualized interest rates in levels (where, for example, an 18% expected annual interest rate would be coded as 18). Each column shows $\hat{\beta}_1$ and $\hat{\beta}_2$ for one of the following outcome variables: (1) the interest rate the participant expects to get on the loan they take out; (2) the lowest interest rate the participant expects a bank could offer them; (3) the highest interest rate the participant expects a bank could offer them; (4) the difference between the highest and the lowest interest rates the participant expects a bank could offer them. The sample sizes for each column differ based on the number of participants who responded to the corresponding questions. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.10: Robustness of Effects of Tools on Beliefs about Second Moment

	Normalized Dispersion (1)	Implied SD (Symmetric) (2)	Implied SD (Split) (3)
Simple Tool	−0.02 (0.01)	0.00 (0.13)	−0.74*** (0.25)
Price Comparison Tool	0.03*** (0.01)	3.25*** (0.30)	2.73*** (0.45)
Observations	6,272	6,272	2,033
Control Mean Posterior	0.67	4.95	5.30
Control Median Posterior	0.67	2.18	1.98
Bin Density FEs	Yes	Yes	Yes

This table shows the effect of the simple tool and price comparison tool on beliefs about dispersion, using alternative measures of dispersion. It shows results from specification (2). Normalized dispersion is measured as the highest rate minus the lowest rate divided by the midpoint of the highest rate and lowest rate. The implied standard deviation (SD) is defined by assuming the belief distribution follows either a symmetric or split triangular distribution (see Appendix G). The number of observations in column (3) is lower than that of columns (1) and (2) since we added the fraction-over-midpoint question to the baseline survey partway through the experiment on October 24, 2022, and some participants after that date did not respond to the additional question. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values) by treatment arm. Heteroskedasticity-robust standard errors are reported in parentheses. SD = standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.11: Effects on Search by Predictions from a Model without Negotiation

	Predicted to search more		Predicted to search less		Ambiguous prediction	
	N of inst. searched	N of inst. applied	N of inst. searched	N of inst. applied	N of inst. searched	N of inst. applied
	(1)	(2)	(3)	(4)	(5)	(6)
Simple Tool	−0.182 (0.224)	−0.116 (0.174)	0.544 (0.344)	0.220 (0.232)	0.171* (0.103)	0.077 (0.075)
Price Comparison Tool	0.079 (0.299)	0.145 (0.187)	0.261 (0.288)	0.052 (0.215)	0.003 (0.102)	0.009 (0.071)
Control Mean	3.719	1.191	3.170	0.980	3.465	1.109
Observations	285	279	188	183	1,553	1,501

This table shows the effects of the simple tool and price comparison tool on the number of institutions searched and the number of institutions applied separately for those predicted by a model *without* negotiation to search more or less, or for whom the prediction is ambiguous, based on their priors about the first and second moment of the interest rate distribution. Respondents are grouped into three main categories according to their biased beliefs about the first and second moments. The “predicted to search more” group includes those who underestimated the second moment and either overestimated, were accurate, or did not answer about the first moment, together with those who overestimated the first moment and either had accurate beliefs or did not answer about the second moment. The “predicted to search less” group includes those who underestimated the first moment and either overestimated, had accurate beliefs, or did not answer about the second moment, together with those who overestimated the second moment and either had accurate beliefs or did not answer about the first moment. The “ambiguous prediction” group includes those who underestimated both the first and second moment, overestimated both the first and second moment, had accurate beliefs about both moments, had accurate beliefs about one moment and did not answer about the other moment, or did not answer about either moment. “Did not answer” in this table note refers to those who responded “don’t know” or “don’t want to answer” or skipped the question. We allow for a tolerance of ± 1 percentage point when defining accurate beliefs. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.12: Balance by Elicit Beliefs Treatment for Subsample Obtaining Loans in Table 4

	Elicit Beliefs = 0 Mean (1)	Elicit Beliefs (2)	N (3)
<i>Personal characteristics</i>			
Age	35.179*** (0.110)	0.012 (0.127)	21,522
log(Income)	14.040*** (0.009)	0.004 (0.010)	21,268
Incomplete high-school	0.007*** (0.001)	0.000 (0.001)	21,213
Complete high-school	0.207*** (0.006)	0.002 (0.006)	21,213
Complete 2-year program	0.200*** (0.005)	-0.002 (0.006)	21,213
Complete 5-year program or higher	0.585*** (0.007)	-0.001 (0.008)	21,213
<i>Financial products</i>			
Bank account	0.888*** (0.004)	0.007 (0.005)	21,238
Any loan	0.888*** (0.004)	-0.002 (0.005)	21,303
Omnibus F-statistic		0.401 [0.956]	21,522
Number of participants by arm	5,503	16,019	21,522

This table tests the balance of pre-treatment characteristics across treatment arms for the subsample obtaining loans in the administrative data in Table 4, panel A, column (2). We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta \mathbb{1}(\text{Elicit Beliefs})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Elicit Beliefs})_i$ is a dummy indicating whether participant i was assigned to the elicit beliefs treatment. Column (1) shows $\hat{\alpha}$ which is the mean for the $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ group. Column (2) shows $\hat{\beta}$ which is the difference in means between the $\mathbb{1}(\text{Elicit Beliefs})_i = 1$ and $\mathbb{1}(\text{Elicit Beliefs})_i = 0$ groups. Column (3) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression: $\mathbb{1}(\text{Elicit Beliefs})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -value of the omnibus F-statistic is included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Unlike in Table 2, we cannot include characteristics of the loan they are searching for in the balance tests since these questions were asked in the same module as the elicit beliefs treatment (rather than a prior module), and the elicit beliefs treatment affected whether participants continued in the survey. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.13: Balance by Tool Treatment Arm for Subsample Obtaining Loans in Table 4

		Difference relative to control mean			
	Control Mean	Price Comparison Tool	Simple Tool	Joint test F-stat	N
	(1)	(2)	(3)	(4)	(5)
<i>Personal characteristics</i>					
Age	35.082*** (0.156)	0.042 (0.219)	0.245 (0.220)	0.711 [0.491]	8,988
log(Income)	13.905*** (0.014)	0.019 (0.019)	0.022 (0.019)	0.777 [0.460]	8,865
Incomplete high-school	0.007*** (0.002)	0.002 (0.002)	0.001 (0.002)	0.260 [0.771]	8,832
Complete high-school	0.243*** (0.008)	-0.009 (0.011)	-0.015 (0.011)	0.978 [0.376]	8,832
Complete 2-year program	0.205*** (0.008)	0.011 (0.011)	0.015 (0.011)	1.023 [0.359]	8,832
Complete 5-year program or higher	0.544*** (0.009)	-0.003 (0.013)	0.000 (0.013)	0.039 [0.962]	8,832
<i>Financial products</i>					
Bank account	0.863*** (0.006)	0.016* (0.009)	0.004 (0.009)	1.775 [0.170]	8,844
Any loan	0.882*** (0.006)	0.000 (0.008)	0.003 (0.008)	0.069 [0.934]	8,875
<i>Desired loan characteristics</i>					
log(Loan Amount)	15.428*** (0.021)	0.054* (0.030)	0.041 (0.030)	1.779 [0.169]	8,604
log(Maturity (years))	1.425*** (0.011)	0.039** (0.015)	0.021 (0.015)	3.268** [0.038]	8,373
<i>Omnibus F-statistic</i>					
Price Comparison Tool		1.149 [0.305]			5,982
Simple Tool			0.661 [0.825]		5,928
Number of participants by arm	2,922	3,060	3,006		8,988

This table tests the balance of pre-treatment characteristics across treatment arms for the subsample obtaining loans in the administrative data in Table 4, panel B, column (2). We run the following regression separately for each baseline covariate k : $X_i^k = \alpha + \beta_1 \mathbb{1}(\text{Simple Tool})_i + \beta_2 \mathbb{1}(\text{Price Comparison Tool})_i + \varepsilon_i$, where X_i^k is a baseline covariate for participant i and $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ are dummies indicating whether participant i was assigned to the simple tool or price comparison tool arms, respectively. Column (1) shows $\hat{\alpha}$ which is the mean for the control group. Columns (2) and (3) show $\hat{\beta}_2$ and $\hat{\beta}_1$ which are the differences in means between the price comparison tool treatment arm or simple tool treatment arm and the control group. Column (4) shows the F-statistic from a joint test of $\beta_1 = \beta_2 = 0$. Column (5) shows the number of observations in each regression, which can change across covariates due to missing values when the respondent did not answer that question in the baseline survey. For the omnibus F-test to test whether the covariates jointly predict treatment, we run the following regression for the participants assigned to simple tool and control groups and for those assigned to the price comparison tool and control groups, separately: $\mathbb{1}(\text{Treatment})_i = \delta + \sum_{k=1}^K \gamma_k X_i^k + \varepsilon_i$, where $\mathbb{1}(\text{Treatment})_i$ is either $\mathbb{1}(\text{Simple Tool})_i$ or $\mathbb{1}(\text{Price Comparison Tool})_i$. The omnibus F-statistic is a test of $\gamma_1 = \dots = \gamma_K = 0$. To retain the full sample in the omnibus F-test, if a participant did not answer a particular question, we create a dummy variable indicating whether the variable was missing, replace the missing value with zero, and include the missing value dummy as an additional X^k covariate in the regression. The p -values of the joint test and omnibus F-statistics are included in square brackets. Continuous variables are winsorized at the 95th percentile (and 5th percentile if they take negative values). “Desired loan characteristics” refer to characteristics of the loan the participant is searching for. “Any loan” is a dummy variable equal to 1 if the participant had any of the following types of loan: consumer loan, mortgage, auto loan, credit card, or cash advance. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C Bank Websites and Comparison Websites

In our follow-up survey, 93.2% of participants reported using tools on bank websites known as “simulators” that provide an interest rate estimate and 12.5% reported using third-party comparison websites (aggregators) during their search. Thus, these channels are likely a way that some consumers form their beliefs about consumer loan interest rates. We investigate whether the simulator tools on banks’ own websites and the main third-party comparison tool provide accurate information. If not, these sources potentially contribute to participants’ biased beliefs. We scraped data from seven banks’ consumer loan simulators and the main third-party comparison website. For each participant in our RCT, we obtained data on individual and desired loan characteristics from their baseline survey. We then ran a script that fed these inputs into each website (including bank simulators and the third-party comparison website) and scraped the output. Next, for the same bank from which a participant received a loan in our administrative data, we compare the rate the participant actually received in the administrative data to the rate they would have seen on that bank’s website or for that bank in the third-party comparison website.

C.1 Description of Bank Websites

Many Chilean banks provide a “simulator” on their websites, which allows visitors to see what interest rate they could expect to receive on a loan. Consumers input their personal information along with desired loan amount, maturity, and other details. The simulator then shows the consumer the interest rate they can expect to obtain for this loan, as well as the total loan cost and monthly cost. The input variables required by each simulator are shown in Table C.1, panel A. All bank websites require information on loan amount and maturity. All but one bank also require the consumer’s RUT (national ID number), but in tests that we show below, we find that the interest rates that banks show in these simulators typically do not vary based on the RUT that is entered. Five out of twelve bank websites require the consumers’ income as an input. On the other hand, none of them require the users to enter their municipality of residence (*comuna*), despite this being an important variable that banks use to price loans.

We were able to scrape data from seven of the twelve bank simulators. We were unable to scrape the other five bank simulators due to firewalls, errors returned by the website when attempting to obtain a quote, and requirements that the user already have a bank account with that bank or pay for the quote.

C.2 Description of Comparison Websites

There are two main third-party comparison websites providing estimated loan terms from multiple banks, also known as aggregators. One is provided by a private company, using information reported to this company by banks, which are asked to report a schedule of rates they would offer

for different loans. Banks may have an incentive, however, to report downward-biased quotes to third-party comparison websites as a bait-and-switch technique, as shading down the rates they quote on comparison websites could direct traffic to them over other banks. The other third-party comparison website is run by a government consumer protection agency that is separate from the CMF. Table C.1, panel B, describes the inputs required by both third-party comparison websites. In both of these tools, consumers input their desired loan size and maturity and receive quotes for loans from different institutions. However, neither asks for any borrower characteristics, and thus the interest rate quotes they provide are not conditional on borrower characteristics. We scraped the private third-party comparison website but not the other government comparison website as the latter was down for several months when we were running our scrapers.

C.3 Obtaining Data from Bank and Comparison Websites

Our data collection period spanned from September 28, 2023 to October 9, 2023. We use the loan and consumer characteristics of each consumer-loan seeker in the baseline survey as inputs to the simulators, thereby replicating what our RCT participants would see should they use these tools. For identification-related inputs, such as RUT (national ID number) and contact information, we use random fake RUT numbers generated by adapting the code at <https://codepen.io/alisteroz/pen/KEoqgQ> for Python. To test whether the outputs shown by the bank websites depend on the RUT entered, we conducted tests where we held all inputs fixed except RUT. In these tests, we set the other characteristics such as loan amount, maturity, and income to the median values observed in our data and hold them constant. We set the test size to 100 observations and tested the five bank websites we scraped where IDs were requested as an input. As shown by Figure C.1, despite occasional variation in interest rates for different RUTs, the interest rates remain largely identical across a random sample of RUTs.

Similarly, four bank websites (three of which we could successfully scrape) require the phone number as an input. We conduct a similar test of whether the interest rates shown by the bank depend on the phone number (e.g., the bank might use the phone number's area code to condition the interest rate on where the consumer lives without explicitly asking the consumer where they live). We randomly sample Chilean phone number area codes when creating the phone number used as an input to the bank website and again test 100 observations holding other inputs fixed. Figure C.2 shows that interest rates do not differ by area code for any of the three banks that require phone number as an input.

We obtain the following simulated loan outcomes for each participant in our RCT from each scraped website: monthly interest rate, equivalent annual charge (*carga anual equivalente*, or CAE, which is analogous to an APR), and total cost of the loan (*costo total del crédito*, or CTC).

C.4 Comparison of Websites' Rates and Received Rates

To compare the rate an individual in our RCT actually received in the administrative data to the rate they would have seen on the same bank's website and for the same bank on the third-party comparison website, we begin by matching the interest rates we scraped from these websites to the interest rates of the loans that these individuals actually received. First, we restrict our sample to the 31,148 people in our RCT who took out a loan after participating. Next, for each of these participants, we keep one unique consumer loan from the administrative data, which is the first loan obtained after the date that the individual participated in the RCT. In the rare case that an individual obtained more than one loan on the same day (0.5% of borrowers), we take the largest of the loans they obtained that day. Then, for each individual who took up a consumer loan in the administrative banking data, we match interest rates the individual would have seen on the bank and comparison websites—based on the loan amount, loan maturity, and income they reported in the baseline survey—with the interest rate they obtained in the administrative data.

We merge at the consumer and bank level using encrypted bank identifiers in the administrative data: for example, if the individual took a loan from Bank A, we merge the interest rate they obtained with what they would have seen on Bank A's website, and what the third-party comparison website would have shown them for a loan from Bank A. If, for a given loan, there are multiple matched interest rates quotes from different sources of the same bank, e.g., one from the bank's website and one from the third-party comparison tool, we keep all quotes as distinct observations. The merged sample has 15,809 observations. We then further restrict to loans in the administrative data that were obtained during the same months in which we ran the scrapers, i.e., September–October 2023. This limits the size of the sample but ensures that any observed differences between rates from bank and third-party comparison websites and rates obtained in administrative data are not due to changes in interest rates over time.

The interest rate quotes shown by banks and comparison websites are highly inaccurate (Figures 5b and 5c). We next document possible explanations for this inaccuracy. The first and most compelling explanation is that these websites do not ask the user for key inputs: none ask for the comuna of residence, and only three out of seven ask for income, both of which are significant predictors of interest rate. Thus, they do not provide quotes conditional on all the relevant borrower characteristics that influence the interest rate. Second, the Chilean credit bureau does not provide a continuous credit score, and thus the bank simulators and third-party comparison websites do not condition on a continuous credit score. (They also do not condition on the binary flag for whether a borrower has defaulted on a loan in the past, which is a severe credit event indicating that the borrower has missed three payments and judicial proceedings have been initiated against them, which effectively shuts them out of credit markets.)

An alternative explanation for the differences in interest rates shown in Figures 5b and 5c could

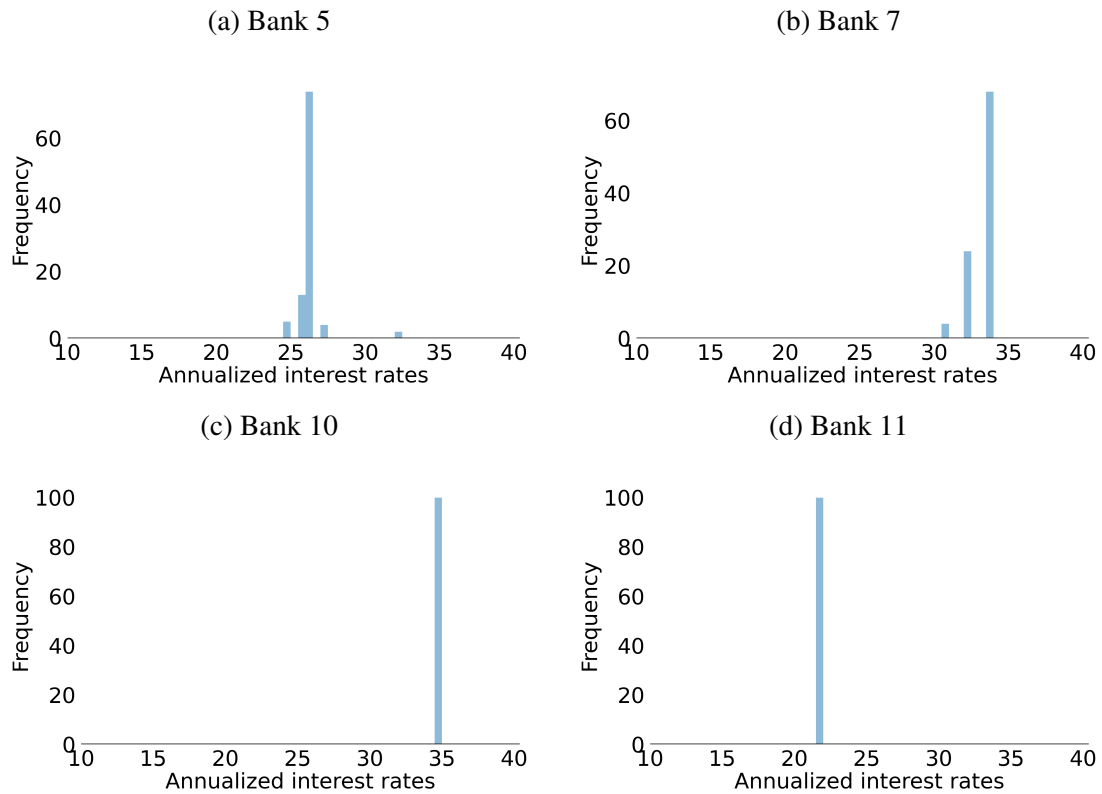
be that the loan that the participants ultimately obtained might have different terms (e.g., loan amount, maturity) than the terms the participant was initially looking for, whereas our scripts to scrape the bank and comparison websites used the loan terms the participant was initially looking for. To test this, we plot the observed difference between rates participants would have observed on the bank simulator and the rate they obtained from the same bank against the difference in loan size or maturity between the desired terms they reported on the baseline survey and the terms of the actual loan they obtained. The scatter plots are presented in Figure C.3 for bank websites and Figure C.4 for the third-party comparison tool. Loan size differences are presented in panel (a) and maturity differences in panel (b). The majority of points are clustered along or near the vertical line at zero, indicating little difference in the other loan terms (loan amount or maturity). However, there is still substantial variation in interest rates received for a given value of the difference in other loan terms. We regress the difference in interest rates on the difference in loan terms and find that the R^2 of these regressions are 0.134 for loan amount and 0.230 for maturity for bank websites, and 0.076 for loan amount and 0.081 for maturity for the third-party comparison website. This suggests that the differences between rates seen on bank or third-party comparison websites and the rates participants actually obtained cannot primarily be explained by participants obtaining loans with different loan amount or maturity than they were initially searching for.

Table C.1: Bank Website and Third-Party Comparison Tool Inputs

	Borrower characteristics						Loan characteristics			
	Name (1)	RUT (2)	Document number (3)	Income (4)	Phone number/email (5)	Comuna (6)	Loan amount (7)	Maturity (8)	First payment date (9)	Insurance options (10)
<i>Panel A: Bank websites</i>										
Bank 1		Y		Y			Y	Y	Y	Y
Bank 2		Y					Y	Y	Y	Y
Bank 3		Y		Y			Y	Y	Y	Y
Bank 4	Y	Y		Y	Y		Y	Y	Y	Y
Bank 5		Y		Y			Y	Y	Y	Y
Bank 6							Y	Y		Y
Bank 7		Y					Y	Y	Y	Y
Bank 8		Y	Y		Y		Y	Y	Y	Y
Bank 9		Y								
Bank 10		Y					Y	Y	Y	Y
Bank 11	Y	Y		Y	Y		Y	Y		Y
Bank 12		Y			Y		Y	Y		
<i>Panel B: Comparison Tools</i>										
Third party comparison tool							Y	Y		
Public comparison tool							Y	Y		

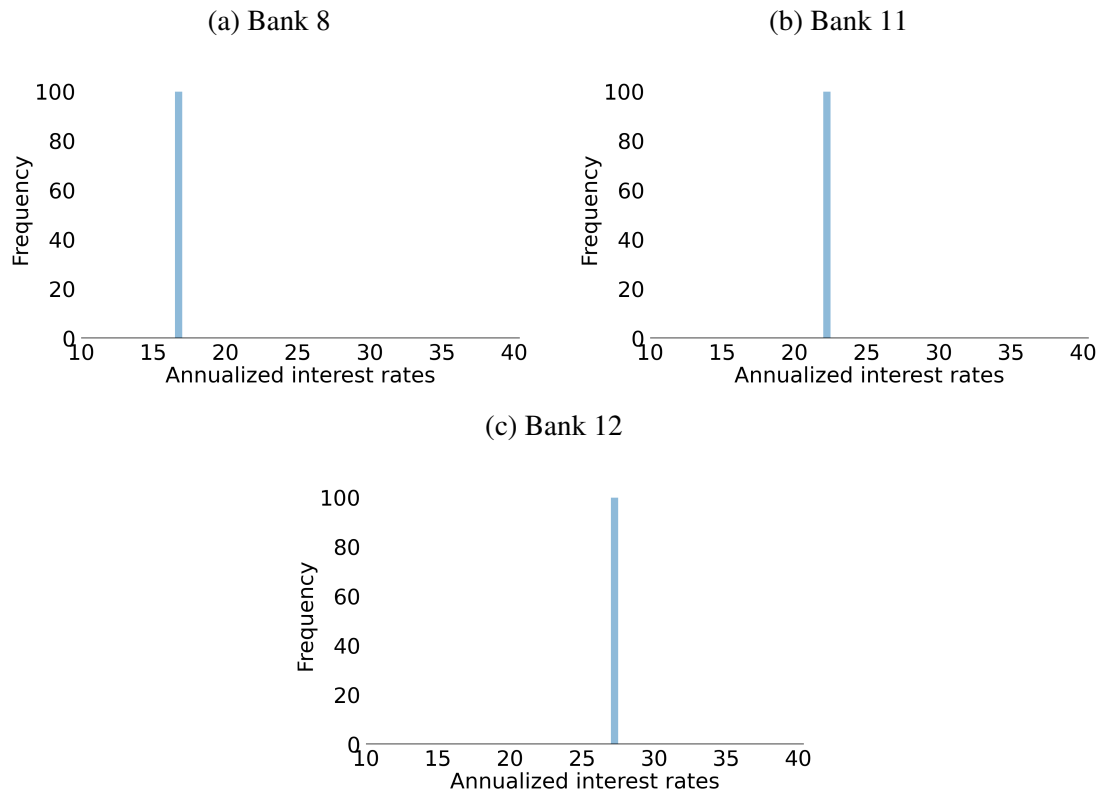
This table shows what inputs are required by each bank website and each third-party comparison tool as of April 3, 2024. Bank names are anonymized. Column (1), “Name”, refers to the name of the person searching for information. Column (2), “RUT”, refers to the *rol único tributario*, the national ID number in Chile. Column (3), “Document number”, refers to the serial number on the national identity card which is distinct from the national ID number or RUT. Column (6), “Comuna”, is a geographic area analogous to a municipality; we include this column to emphasize that no banks or third-party comparison tools request this information, despite it being an important predictor of interest rates used by banks in their algorithms. Column (9), “First payment date”, refers to a field asking the consumer when they would like to make their first payment, which sometimes allows a specific day and other times a specific month beginning in the month after the current month.

Figure C.1: Test of Whether Bank Website Terms Vary by RUT



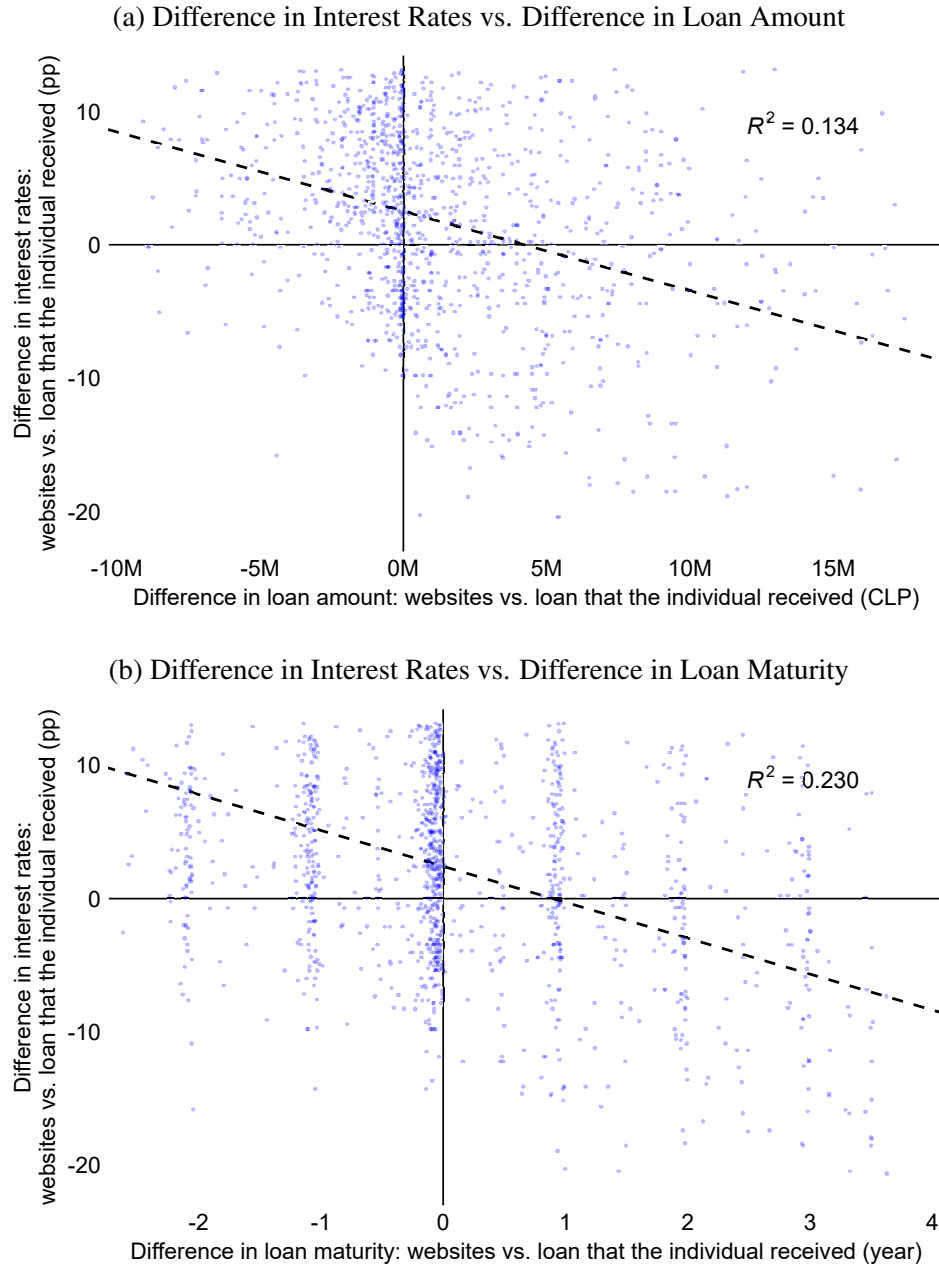
This figure shows the annualized interest rates from bank websites, derived from a test where we varied only the RUT while holding other inputs constant. Specifically, we standardized inputs such as income, loan amount, and loan maturity by using the median values from the baseline survey: a monthly income of 740,000 pesos, a loan amount of 2,500,000 pesos, and a maturity of 3 years. The x-axis represents the annualized interest rates shown by each bank's simulator. Each panel includes 100 observations.

Figure C.2: Test of Whether Bank Website Terms Vary by Phone Area Code



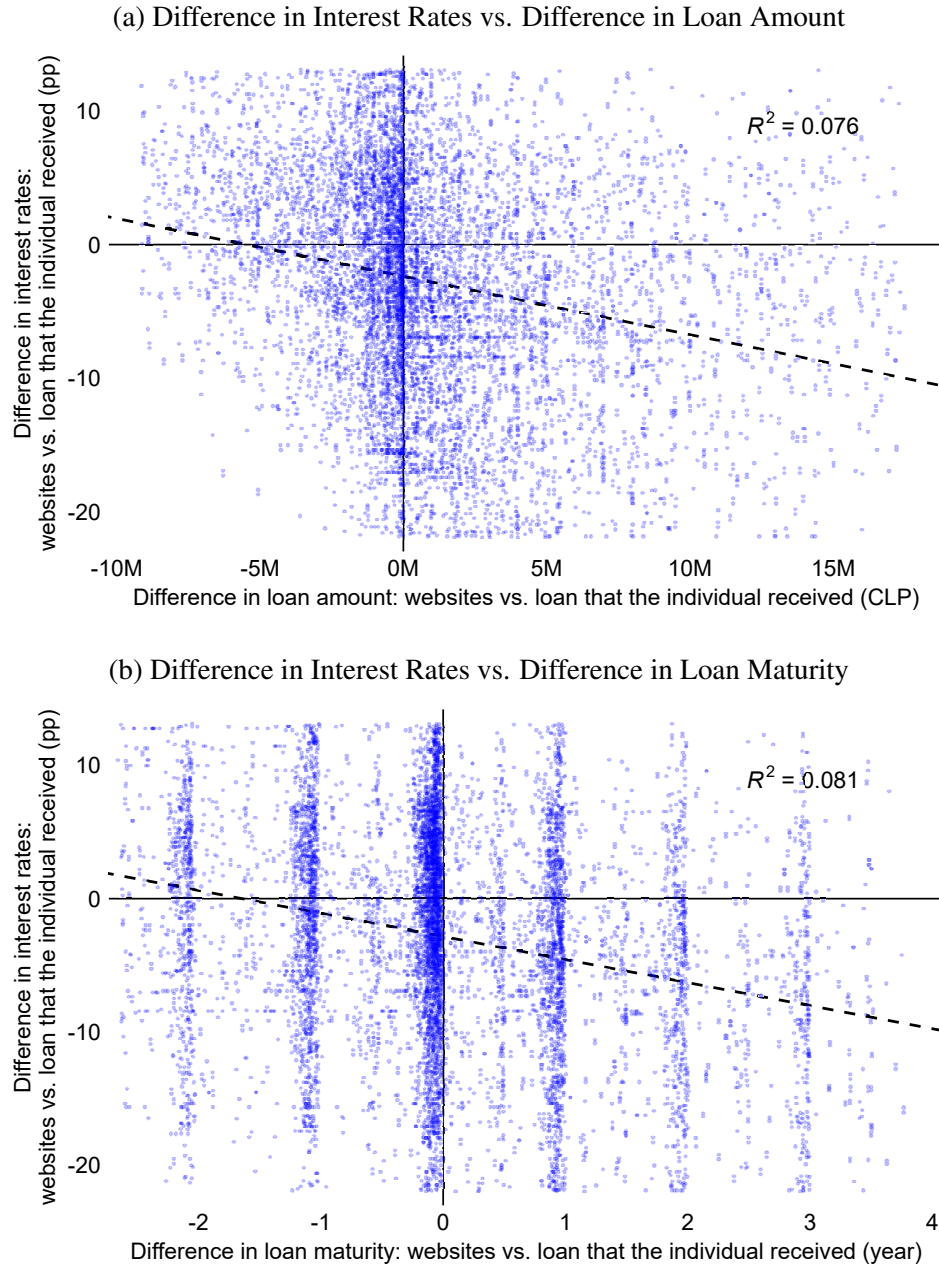
This figure shows the annualized interest rates from bank websites, derived from a test where we varied only the phone number area code while holding other inputs constant. Specifically, we standardized inputs such as income, loan amount, and loan maturity by using the median values from the baseline survey: a monthly income of 740,000 pesos, a loan amount of 2,500,000 pesos, and a maturity of 3 years. The x-axis represents the annualized interest rates shown by each bank's simulator. Each simulation consists of 100 observations.

Figure C.3: Difference in Loan Terms: Bank Websites vs. Loan the Individual Received



This figure shows scatterplots of differences in interest rates against differences in loan amounts in panel (a) and maturity in panel (b). Interest rates on bank websites are the rates we scraped from bank's simulator websites for consumer loans for the same bank at which the participant actually obtained a loan, given the inputs from the participant's baseline survey responses. The difference in interest rates is calculated as the interest rate shown on the bank website websites minus the interest rate of the loan that the individual received. The difference in loan amount or maturity is calculated similarly. All differences are calculated using raw data and then the differences are winsorized at the 5th and 95th percentiles. The dashed line and R^2 are from a linear regression of the winsorized difference in interest rates on the winsorized difference in loan amount or maturity. The number of observations is 1,592. For legibility, observations beyond the 5th and 95th percentiles of either axis are excluded from the plot.

Figure C.4: Difference in Loan Terms: Comparison Tool vs. Rate the Individual Received



This figure shows scatterplots of differences in interest rates against differences in loan amounts in panel (a) or maturity in panel (b). Interest rates on the third-party comparison website are the rates we scraped from the third-party comparison website for the same bank at which the participant actually obtained a loan, given the inputs from the participant's baseline survey responses. The difference in interest rates is calculated as the interest rate shown on the bank website websites minus the interest rate of the loan that the individual received. The difference in loan amount or maturity is calculated similarly. All differences are calculated using raw data and then the differences are winsorized at the 5th and 95th percentiles. The dashed line and R^2 are from a linear regression of the winsorized difference in interest rates on the winsorized difference in loan amount or maturity. The number of observations is 11,115. For legibility, observations beyond the 5th and 95th percentiles of either axis are excluded from both panels.

Appendix D Experimental Timeline

Pilot From November 2019 to March 2020, we conducted a randomized pilot of the loan price comparison tool. The pilot tested our online ads campaign, survey, and a prototype of the interactive tool. In our pilot, 8,916 Chilean users searching for keywords related to consumer loans clicked our ads.

Focus Groups In October and November 2020 we ran a series of focus groups and surveys, in which 81 Chilean participants provided us with qualitative comments on the user experience, design, and perceived usefulness of the price comparison tool and survey questionnaire. The results from these focus groups and individual interviews prompted us to add more guidance, tips, and directions to our tool, as some participants had trouble understanding the graphs and data.

Google ads testing From January to July 2021 we ran Google ads tests to determine the most effective ads.

RCT After making a series of improvements to the interactive price comparison tool based on our pilot experiment and focus groups, we launched the RCT in November 2021. We subsequently made monthly updates to the tool so that it always showed data based on the previous six months. One issue that arose during the RCT was that from January 24, 2022 to March 10, 2022, due to a bug in our code for the baseline survey, we mistakenly showed *all* participants the questions eliciting prior beliefs. As the timing of participation in the RCT is not random, we exclude these 4,979 observations from the regressions testing the effect of eliciting priors (Table 4), but keep these observations in regressions testing the effects of the tools. Thus, our experiment actually included 117,042 rather than 112,063 participants, with the difference representing those affected by this bug.

Appendix E Price Comparison and Simple Tool Construction

In this appendix we provide detail on how we used the administrative, loan-level data from the CMF to create the price comparison tool and simple tool that we showed to participants depending on their randomized treatment arm assignment.

E.1 Price Comparison Tool

First, we merged two data sets: a loan-level data set and a data set with borrower characteristics at the time of loan application. Both data sets contain information that banks are required to report to the CMF for each consumer loan they issue. We then bin all loan-borrower observations within a municipality (comuna) into income quartiles, loan amount quartiles, and one-year maturity bins, where the quartile cutoffs for both income and loan amount vary by comuna. We remove data points more than five standard deviations away from the interest rate mean as we consider these to

be extreme outliers.

To ensure that the loan histograms display sufficient information, we impose the following two conditions before showing a histogram to a participant assigned to the price comparison tool arm:

1. There are at least five data points in total and at least two distinct interest rates in the histogram.
2. Both the participant's income and loan amount fall within extended ranges of the data used to construct the histogram (where "extended ranges" correspond to between 75% of the minimum value to 125% of the maximum value, for both income and loan amount).

If the above two conditions are met, then the histogram is shown to the participant as in Figure 2a. The share of participants in the price comparison tool arm who saw comuna-level histograms was 82.2%.

If one of these conditions fails, then the geographic range is expanded to contiguous comunas. Loans in the participant's comuna and all contiguous comunas are again binned into income quartiles, loan amount quartiles, and one-year maturity bins. If the above two conditions are met for the histogram that corresponds to the participant's comuna and its contiguous comunas, a histogram is returned. A message also accompanies the histogram specifying that there were insufficient data for the participant's comuna, and that the participant is seeing information that includes neighboring comunas. The share of participants in the price comparison tool arm who saw first-degree neighboring comuna histograms was 4.6%.

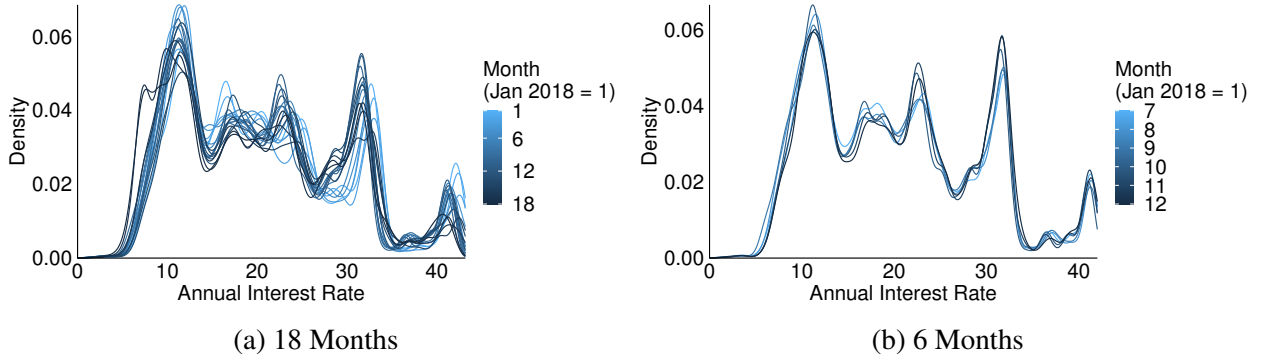
If one of the conditions still fails, then the geographic range is expanded to second-degree contiguous comunas and the steps above are repeated. The share of participants in the price comparison tool arm who saw data for their comuna and the first- and second-degree contiguous comunas was 0.6%.

If a histogram still could not be created for a participant based on the above conditions, then data for the entire region (equivalent in Chile to a US state) is used with a corresponding message that the information displayed is at the regional level. The share of participants in the price comparison tool arm who saw regional data was 8.4%.

Length of Past Data Shown In order to provide consumers with personalized data based on loans actually obtained by similar consumers, we necessarily have to use "historical" data that goes a certain distance back in time. (For example, if we only used data from the past month, for many consumers there would not be enough observations within each cell defined by consumer and loan characteristics to show the distribution of rates.) Thus, we face a trade-off between how recent the data used by the tool are—which is more relevant if the distribution of interest rates changes over time—and how many data points we show each consumer. In Figure E.1 we illustrate the first component of this trade-off: panel (a) shows the distribution of interest rates for consumer loans

in each of the 18 months between January 2018 and June 2019 to illustrate how much interest rates change over time. Clearly, using 18 months of data goes too far into the past given that the distribution of interest rates does change over time. On the other hand, panel (b) shows that the distribution of interest rates is relatively stable over six months. We conducted a formal analysis of how much the distribution of interest rates changes relative to the current month's distribution if we include data for the past X months in the tool, as well as how many data points would be shown to participants if we include data for the past X months (which varies depending on loan and borrower characteristics), and determined that showing data over the past six months was optimal. We thus refreshed the data underlying the tool each month so that the tool always showed the most recent six months of data.

Figure E.1: Distribution of Interest Rates over Time



This figure shows how the distribution of consumer loan interest rates changes over time. Panel (a) displays the distribution of interest rates over the 18 months from January 2018 to June 2019. Panel (b) shows the distribution of interest rates for the six months from July 2018 to December 2019.

E.2 Simple Tool

We use loan-level CMF data to estimate the benefits of searching at more banks. First, for a particular participant, we subset the data to consumer loans originated in the last six months in that participant's comuna, within-comuna income quartile, within-comuna loan size quartile, and maturity bin—i.e., the same data used to construct the histogram that the participant would have seen if they had been assigned to the price comparison tool. Within this subset of the data, suppose there are J banks that have originated L loans to borrowers. We then randomly draw an interest rate ℓ_0 from this set of loans and consider it the participant's "first offer." We then draw another quote ℓ_1 from the loans issued by the remaining $J - 1$ banks, excluding all loans issued by the bank that issued the first loan we drew. We then use the desired maturity and loan size to calculate the monthly interest payments corresponding to loans ℓ_0 and ℓ_1 . We then consider two cases:

1. If $\ell_1 < \ell_0$, we calculate the difference in monthly payments, and calculate the difference in total loan costs as the present value of the difference in monthly payments over the life of the loan.

2. If $\ell_1 \geq \ell_0$, we set the “benefit of searching at one additional bank” to zero, as the participant would not take out the ℓ_1 loan that has a higher interest rate than ℓ_0 .

We then repeat this drawing of two sequential quotes 1,000,000 times. We estimate the “benefit of searching at one additional bank” to be the average reduction in monthly cost and in total loan cost across these simulations. Thus, the benefit of searching is always a non-negative number, though zeros are included in the average.

To find the benefit of searching at $n \in \{2, 3, 4, 5\}$ additional banks, we simulate the process of sequentially drawing n loans from $J - n$ banks 1,000,000 times. The benefits of search are then calculated by comparing the lowest interest rate ℓ_n from any draw $n > 0$ to the interest rate from the initial draw ℓ_0 , and if $\ell_n \geq \ell_0$ for all $n > 0$ within a particular simulation, the benefit of searching is coded as zero in that simulation.

We repeat this procedure for all bins determined by loan and borrower characteristics. As in the price comparison tool, if there are fewer than five loans or fewer than two unique interest rates in the bin, we expand the bin to include comunas that border our reference comuna. If there are still less than five loans or two unique rates, we expand the bin again to include comunas that border the bordering comunas to the reference comuna. Finally, if there are still less than five loans or two unique rates, we expand the bin to include all comunas in the region (analogous to a US state).

Appendix F Data Details

F.1 Administrative Data

Winsorizing We winsorize continuous variables at 95th percentile for variables that are bounded below at 0 and winsorize at the 5th and 95th percentile for variables that can take negative values. For variables measured after treatment, we winsorize within treatment arm to avoid truncating a true treatment effect. We winsorize the final variable used in our analysis; for example, if the outcome of interest is the difference between an individual’s belief about the rate they would obtain from the baseline survey and the rate they actually obtained in the administrative data, we first take the difference of the raw variables and then winsorize the difference.

F.2 Survey Data

Survey Questionnaires English translations of our survey questionnaires are available at the following links:

1. Baseline survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/baseline_survey_en.pdf
2. Follow-up phone survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/followup_survey_en.pdf

3. Follow-up Whatsapp survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/whatsapp_survey_en.pdf

The original Spanish versions of the questionnaires are also available:

1. Baseline survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/baseline_survey_es.pdf
2. Follow-up phone survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/followup_survey_es.pdf
3. Follow-up Whatsapp survey: https://seankhiggins.com/assets/pdf/pricecomp_rct/whatsapp_survey_es.pdf

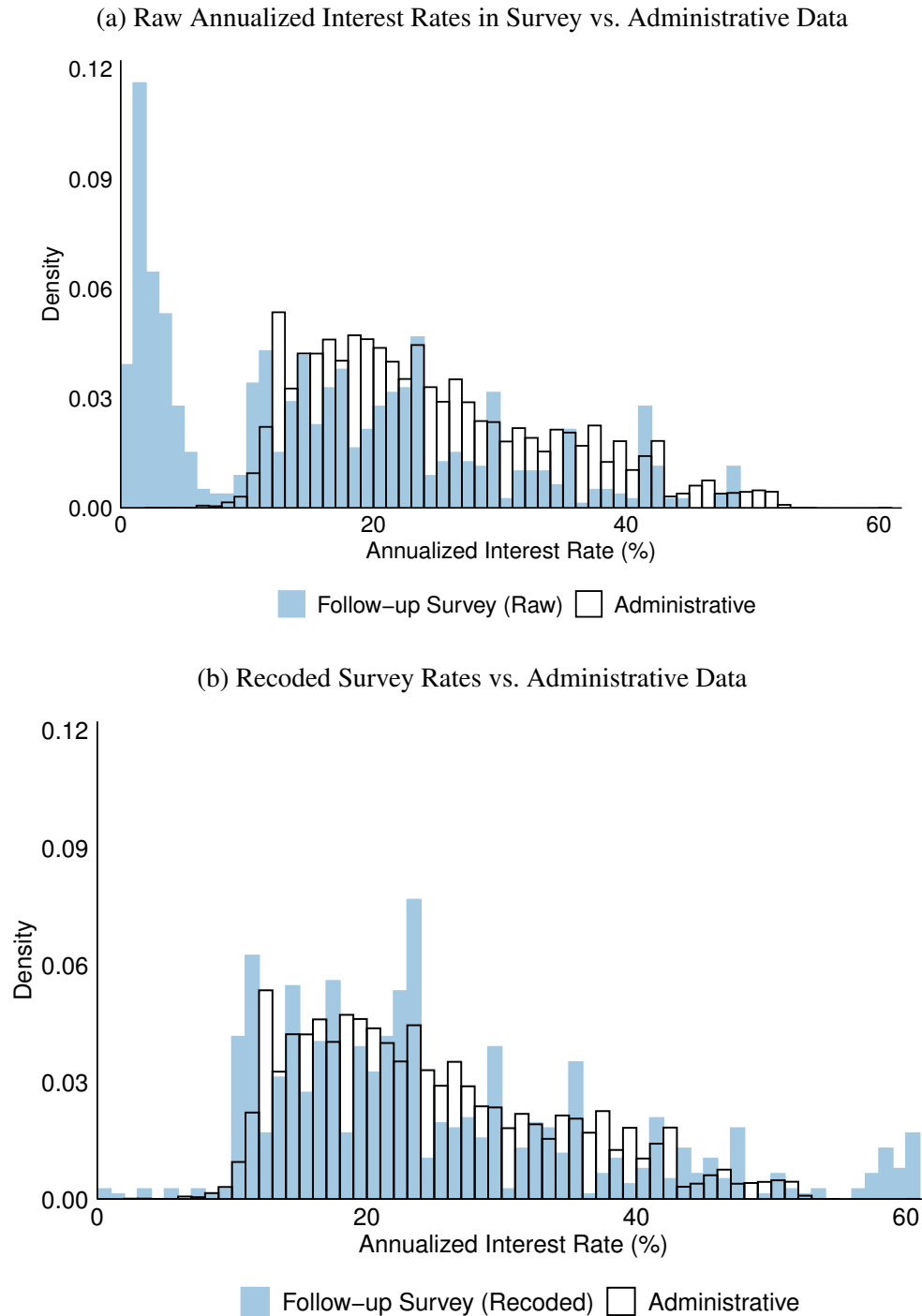
Timeline The baseline survey was conducted as we recruited participants to our RCT, from November 2021 to June 2023. The follow-up phone survey was conducted from January 2023 to March 2024.²⁴ We surveyed participants via phone at least six months after they participated in the RCT. We collected WhatsApp surveys to ask more detailed questions about the timing of negotiation from April 2024 to May 2025.

Annualizing Rates In the follow-up phone survey, participants could report interest rates as an annual or a monthly rate. About 34% of participants reported annual rates, and 66% reported monthly rates which we annualize. However, there were cases where participants appear to have responded with a monthly rate but stated that it was an annual rate. Figure F.1a shows the distribution of annualized interest rates based on participants' follow-up survey responses. About 32% of respondents gave annual rates below 10 percentage points; furthermore, the distribution of annualized reported rates is bimodal, and appears to combine one distribution of rates below 10% and another of rates above 10%. In contrast, the true distribution of annual interest rates in administrative data is unimodal and nearly all rates are above 10%. To correct for this inaccurate reporting, we assume that reported annual interest rates of 10% or lower are implausibly low, and should be recoded as monthly rates that are then converted to annual rates. Figure F.1b shows that the distribution of our corrected annualized interest rate distribution is much closer to the distribution of administrative interest rates.

²⁴We also conducted “recovery surveys” from February 2024–March 2024 for some questions that we had not initially included in the survey but realized we should add while surveys were ongoing. During these recovery surveys, we called back those who had already responded to the survey to ask a small set of additional questions. We had originally intended to conclude the follow-up survey in March 2024. We then received last-minute additional funding to continue the surveys in April 2024, but most of the surveyors had already found other jobs since the surveys were planned to end in March 2024. The survey firm hired new surveyors for the April 2024 surveys, but determined that the responses collected by the new surveyors did not pass its quality checks, and thus the survey firm stopped collecting additional surveys, dropped these observations, and provided us with a data set of survey responses collected through the initially-planned end date of March 2024.

Winsorizing We follow the same protocol for winsorizing as in administrative data.

Figure F.1: Interest Rate Distributions



This figure shows the distributions of reported annualized interest rates from our follow-up phone survey in panel (a) and the recoded rates in panel (b), both compared to the interest rates in administrative data. Participants could choose to respond to our follow-up survey question with interest rates expressed as a monthly or annual rate. The figure omits rates above the 95th percentile of the distribution for legibility.

Appendix G Robustness to Alternative Measures of Dispersion

The questions we use to measure beliefs about dispersion follow those used to elicit uncertainty about future earnings (Guiso, Jappelli, and Pistaferri, 2002), returns to schooling (Attanasio and Kaufmann, 2014), and macroeconomic growth (Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024): we ask participants the lowest and highest rates a bank could offer them and the percent of offers with rates above the midpoint of the two to capture potential asymmetry in the distribution. Our preferred measure of dispersion is the highest minus lowest rate a bank could offer them. While this measure has the advantage of simplicity, and is proportional to the standard deviation (SD) assuming rates follow a symmetric triangular distribution (as in Attanasio and Kaufmann, 2014; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024; Bassi, Rasul, Veroux, and Vitali, 2026), it also has disadvantages. First, the measure may be overly sensitive to the endpoints of the distribution—especially when we use the same measure in administrative data. Second, it does not capture potential asymmetry in the distribution: for example, the participant may think the lowest rate that a bank could offer them is quite low, but that the distribution contains a long left tail.

In this appendix, we also show robustness to using the implied SD of the belief distribution under certain functional form assumptions. We use the three belief questions about the highest rate, lowest rate, and fraction of rates above the midpoint to construct an implied standard deviation of the interest rate distribution under two functional form assumptions.²⁵ First, we assume a symmetric triangular distribution (as in Attanasio and Kaufmann, 2014; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024; Bassi, Rasul, Veroux, and Vitali, 2026). Second, we assume a split triangular distribution, exploiting our asymmetry question (as in Guiso, Jappelli, and Pistaferri, 2002; Attanasio and Kaufmann, 2014; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024).

We compare this implied standard deviation for a particular participant to the actual standard deviation of the conditional interest rate distribution faced by this participant according to administrative data (i.e., the distribution conditional on borrower and loan characteristics that the participant saw if assigned to the price comparison tool arm or would have seen if assigned to that arm). The actual standard deviation of this distribution in administrative data does not suffer from the same drawback as the highest minus lowest rate, namely that it may be overly sensitive to the endpoints of the distribution. We find that an even higher fraction of participants underestimate interest rates using these alternative measures of dispersion (Figure B.9): 78.6% and 82.5% underestimate the second moment assuming a symmetric and split triangular distribution, respectively. Furthermore, the price comparison tool still has a large effect on beliefs using these alternative

²⁵Attempting to elicit a more detailed distribution led to higher dropout in our pilot, consistent with Weber, D’Acunto, Gorodnichenko, and Coibion (2022).

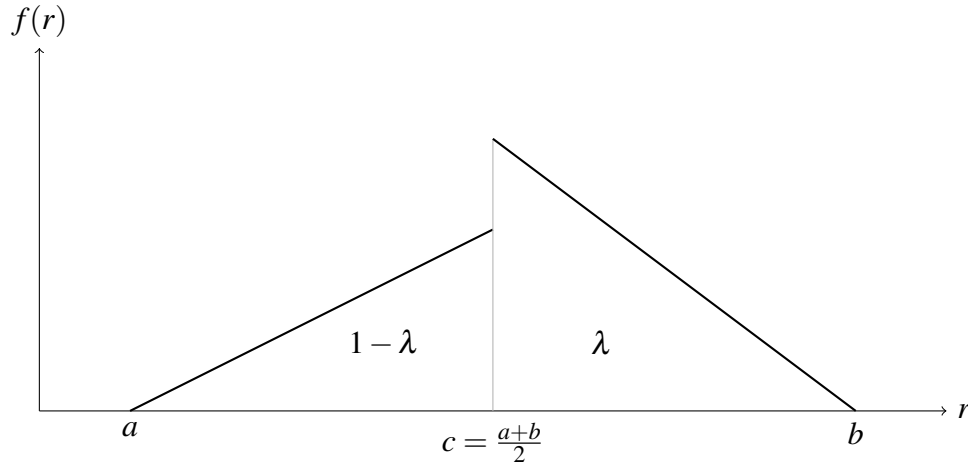
measures (Table B.10).

Below, we detail the calculation of the implied standard deviation of the belief distribution, assuming a split or symmetric triangular distribution.

G.1 Split Triangular Distribution

We begin with a split triangular distribution, as the symmetric triangular distribution is a nested case with equal mass on each side of the distribution. Following Guiso, Jappelli, and Pistaferri, 2002, we model the belief distribution as a split triangular distribution whose support is divided at its midpoint. Figure G.1 illustrates the parametrization. In the figure, a denotes the lowest interest rate the respondent believes a bank could offer, b denotes the highest interest rate they believe a bank could offer, $c = (a + b)/2$ is the midpoint of the support, and λ represents the fraction of offers the respondent believes lie above the midpoint.

Figure G.1: Parametrization of the Split Triangular Distribution



The split triangular distribution is fully characterized by the parameters (a, b, λ) , which are the parameters we elicit beliefs about in the baseline survey. To recover the dispersion of beliefs, we derive the standard deviation of the distribution as a function of these parameters.

The first moment is given by

$$\mathbb{E}[r] = \int_a^c r f(r) dr + \int_c^b r f(r) dr = \frac{1-\lambda}{3}(2a+b) + \frac{\lambda}{3}(a+2b).$$

Similarly, the second moment is

$$\begin{aligned} \mathbb{E}[r^2] &= \int_a^c r^2 f(r) dr + \int_c^b r^2 f(r) dr \\ &= \frac{1-\lambda}{24} (11a^2 + 10ab + 3b^2) + \frac{\lambda}{24} (3a^2 + 10ab + 11b^2). \end{aligned} \tag{9}$$

We then plug these into the formula for standard deviation to obtain the implied standard deviation of the distribution: $\sigma = \sqrt{\mathbb{E}[r^2] - \mathbb{E}[r]^2}$.

G.2 Symmetric Triangular Distribution

The symmetric triangular distribution is a special case of the split triangular distribution with $\lambda = 1/2$. In this case, $\sigma = (b - a)/\sqrt{24}$, which is proportional to our preferred measure of the difference between the highest and lowest rates.

Appendix H Google Search

H.1 Obtaining Data from Google Search Results

To obtain data on interest rates that participants would have likely seen on Google, we wrote a script that mimics users searching using the same search terms that led them to our experiment. Because Google search results can vary by the location the user is searching from, our script also specified where the user was searching from, based on the distribution of our participants across municipalities (comunas). Specifically, for each search, we first randomly selected a search phrase–comuna pair. The set of search terms from which we sampled comes from the set of exact search phrases that participants typed into Google prior to clicking on our ad and participating in our experiment. When drawing from this set of search phrases, we weight by the frequency of each phrase. The set of comunas corresponds to the comunas where our participants live based on our baseline survey, and we again weight our sampling by the number of participants from each comuna. During each automated search, we changed the geolocation parameter to match the selected comuna and searched Google using the selected search phrase. We scraped all available information from each result on the first page of search results, which includes both ads and regular search results. The information we scraped includes the content provider, link, title, text snippet, and the position of the result on the page. Our script ran from October 19, 2023, to February 5, 2024, resulting in 6,677,889 Google search results from 101,852 search phrase–comuna pairs.

We scraped these results using a desktop emulator, mimicking what a user would see if they opened Google on a desktop computer. Ideally, we would have also scraped the same results using mobile emulation to check if people see anything different when searching Google on mobile phones. However, we were unable to manipulate location information with mobile emulation. This is because Google adopts different functions to determine the user’s location, with the geolocation parameters of the browser used on desktops, and the user’s IP address used on mobile phones. It was not operationally feasible to fake the IP addresses of each Chilean comuna, so we were unable to scrape location-specific mobile results.

We next used OpenAI’s Assistant API to extract and classify variables from the raw text scraped from Google search results. We also tested traditional text processing techniques to extract these

variables from the Google search results, but found that employing an advanced large language model (LLM) yielded higher accuracy for this complicated natural language processing task. We used the “gpt-4-turbo-preview” model of the Assistant API, the state-of-art text processing model at the time, along with its built-in “retrieval” tool.²⁶ The key variables to extract were interest rate, bank, language, country, loan type (consumer loan, credit card, mortgage, or other product), whether the interest rate is a monthly or annual rate, and whether the interest rate excluded fees or was an APR including fees. We keep only Spanish-language Google results that included an interest rate for consumer loans in Chile, and we annualize all interest rates.

H.2 Comparison of Rates Seen on Google and Received Rates

To compare rates that participants in our RCT would have seen on Google to rates that they actually received, we merge our scraped Google Search data with administrative data. We restrict our sample to the 31,148 people in our RCT who took out a loan after participating. Next, for each of these participants, we keep one unique consumer loan from the administrative data, which is the first loan obtained after the date that the individual participated in the RCT. In the rare case that an individual obtained more than one loan on the same day (0.5% of borrowers), we take the largest of the loans they obtained that day. We further restrict to loans obtained during the months in which we ran the script, i.e., October 2023 to February 2024. While this limits the sample size, it mitigates the concern that any observed differences could be due to changes in interest rates over time. For each individual who took out a consumer loan, we merge the interest rates they would have seen on Google with the interest rate they received in the administrative data, where the merge is done based on the comuna where the participant lives and the bank at which they obtained the loan. (We cannot merge on the exact search phrase used by the consumer, as Google only reports aggregate data search phrases for privacy reasons.)

Appendix I Predictors of Biased Beliefs

In this appendix, we assess whether baseline characteristics predict how biased participants’ beliefs are. To do this in a data-driven way that guards against overfitting with many covariates, we estimate a series of regularized regression models using elastic net (Friedman, Hastie, and Tibshirani, 2010). The outcome variables measure how biased an individual’s beliefs are, compared to the distribution of interest rates that they face conditional on borrower and loan characteristics. For the first moment, we compute

$$r_i^{bias} \equiv r_i^{belief} - r_i^{admin}, \quad (10)$$

²⁶Given that we used a closed-source LLM, the results generated by our code may not be perfectly reproducible in the future due to the stochastic nature of the model and model updates.

where r_i^{belief} is the rate the participant expected to obtain in the baseline survey, and r_i^{admin} is the median of the conditional distribution they face. We compare to the median of the conditional distribution as in Figure B.8, rather than to the rate they actually obtained as in Figure 3, to avoid losing sample as the majority of participants did not obtain a loan within one year after participating in the RCT. For the second moment, we take the range between the highest and lowest rates they thought a bank could offer them as their belief about dispersion and subtract the corresponding range from the conditional distribution in administrative data, as in Figure 4:

$$dispersion_i^{bias} \equiv \left(\bar{r}_i^{belief} - \underline{r}_i^{belief} \right) - \left(\bar{r}_i^{admin} - \underline{r}_i^{admin} \right). \quad (11)$$

For each measure r_i^{bias} and $dispersion_i^{bias}$, we then create four measures: (i) the measure as defined in equations (10) and (11), which can take positive or negative values; (ii) its absolute value; (iii) a binary indicator for underestimated; and (iv) a binary indicator for overestimated.²⁷ We regress these outcomes on all baseline characteristics from Table 2; we impute missing values with sample means and include “missingness indicators”—equal to one if the original (pre-imputation) covariate had a missing value—as additional covariates.²⁸

For each model, we use 20-fold cross-validation to identify the best combination of the regularization strength (λ) and mixing parameters (α) to minimize the mean squared error (MSE), where α interpolates between RIDGE and LASSO penalties. We then extract the estimated coefficients from the model corresponding to the optimal (λ, α) pair for each outcome. This approach allows us to isolate the most predictive features while reducing overfitting. Table I.1 presents the coefficient estimates from the selected elastic net models that render the minimum MSE.

²⁷For the binary indicators for under- or overestimated, we consider small biases within ± 1 pp of the value in administrative data to be accurate rather than under- or overestimated.

²⁸Among the education categories in the baseline characteristics, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity.

Table I.1: What Predicts Biased Beliefs?

	Prior minus administrative data							
	Interest Rate				Dispersion			
	Level (1)	Abs. value (2)	Under (3)	Over (4)	Level (5)	Abs. value (6)	Under (7)	Over (8)
<i>Personal characteristics</i>								
Age	-0.304	-0.217	0.002	-0.001	-0.485	-0.295	0.003	-0.003
log(Income)	-0.966	-4.852	-0.031	0.014	-3.024	-2.577	0.028	-0.027
Incomplete high-school	11.663	11.119	-0.072	0.061	9.453	8.718	-0.042	0.048
Complete high-school	3.593	4.286	-0.025	0.034	3.963	4.226	-0.021	0.028
Complete 2-year program	1.012	1.907	0.001		0.925	1.259		0.011
<i>Financial products</i>								
Bank account	-0.606	-1.513			-2.585	-2.559	0.021	-0.027
Any loan	-1.315	-2.303		-0.000	-2.137	-1.963	0.004	-0.011
<i>Desired loan characteristics</i>								
log(Loan Amount)	-0.867	-5.866	-0.021	0.006	-0.697	-6.241		-0.007
log(Maturity (years))	0.551		0.004		-2.541	-1.564	0.026	-0.024
Observations	14,401	14,401	14,401	14,401	13,677	13,677	13,677	13,677

This table shows coefficient estimates from regularized regression models using elastic net. For each outcome variable capturing the level of bias in beliefs about the first and second moments of the interest rate distribution, we regress the outcome on a common set of covariates. The model includes baseline characteristics from Table 2 and missingness indicators, with missing values imputed by sample means (the coefficients for the missingness indicators are omitted from the table for legibility). Among the education categories in the baseline characteristics, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity. For each outcome, we use 20-fold cross-validation to select the best combination of the regularization strength (λ) and mixing parameters (α), with the objective of minimizing mean squared error. The displayed coefficients correspond to the estimates at the optimal (λ, α) pair for each outcome. Coefficients shrunk to zero are left blank.

Appendix J Testing for Heterogeneous Treatment Effects

To test for heterogeneous treatment effects of the price comparison tool in a disciplined manner, we implement the machine-learning methodology proposed by Chernozhukov, Demirer, Duflo, and Fernández-Val (2025). We focus on whether there is predictable heterogeneity in the treatment effect of the price comparison tool on belief updating and on search, negotiation, and loan terms as a function of the same set of baseline characteristics as in Appendix I. We measure belief updating the same way as in equation (2), i.e., $Posterior_i - Prior_i$ for beliefs about the rate the participant expects to obtain, the lowest and highest rates a bank could offer them, and the dispersion in rates.

Following Chernozhukov, Demirer, Duflo, and Fernández-Val (2025), we consider the conditional average treatment effect (CATE):

$$s_0(Z) := E[Y(1) | Z] - E[Y(0) | Z],$$

where $Y(1)$ is the potential outcome if treated with the price comparison tool, $Y(0)$ is the potential outcome if assigned to the control group, and Z is a vector of baseline characteristics. The best linear predictor (BLP) of $s_0(Z)$ given $S(Z)$ is the solution to

$$\min_{b_1, b_2} E [s_0(Z) - b_1 - b_2 S(Z)]^2,$$

which, if it exists, is

$$\text{BLP}[s_0(Z) | S(Z)] = \beta_1 + \beta_2 (S(Z) - \mathbb{E}[S(Z)]),$$

where $\beta_1 = \mathbb{E}[s_0(Z)]$ and $\beta_2 = \text{Cov}[s_0(Z), S(Z)] / \text{Var}[S(Z)]$. If $S(Z)$ is a perfect proxy for $s_0(Z)$, then $\text{Cov}[s_0(Z), S(Z)] = \text{Var}[S(Z)]$ and thus $\beta_2 = 1$. In general, $\beta_2 \neq 1$ due to noisy predictions $S(Z)$. If $S(Z)$ is purely noise and uncorrelated with $s_0(Z)$, then $\beta_2 = 0$. Furthermore, if there is no heterogeneity, i.e., $s_0(Z) = s$, then $\beta_2 = 0$ because $\text{Cov}[s_0(Z), S(Z)] = \text{Cov}[s, S(Z)] = 0$. Rejecting the hypothesis that $\beta_2 = 0$ therefore indicates that both (i) $S(Z)$ is a relevant predictor and, more importantly for our purposes, (ii) there is heterogeneity in $s_0(Z)$. Therefore, by estimating the β_2 coefficient of $\text{BLP}[s_0(Z) | S(Z)]$ and testing whether the coefficient is significantly different from zero, we can empirically test for heterogeneity in the treatment effect across individuals.²⁹

In the ML step, we consider support vector machine, LASSO, and random forest, and choose the model that maximizes $\Lambda \equiv |\beta_2|^2 \text{Var}(S(Z)) \propto \text{Corr}^2(s_0(Z), S(Z))$ for the BLP analysis.³⁰ Maximizing Λ is equivalent to maximizing the correlation between the ML proxy $S(Z)$ and the CATE $s_0(Z)$, or equivalent to maximizing the R^2 in the regression of $s_0(Z)$ on $S(Z)$. The models that maximize Λ for our analysis on $\text{Posterior}_i - \text{Prior}_i$ are as follows: LASSO for expected rate, lowest rate, and the highest rate; and random forest for dispersion. The models that maximize Λ for our analysis on search, negotiation, and loan outcomes are as follows: LASSO for number of offers and $\text{Pr}(\text{take loan})$ measured in survey data; support vector machine for number of institutions searched, $\text{Pr}(\text{negotiate})$, and log interest rate offered (all measured in survey data), as well as $\text{Pr}(\text{take loan})$ measured in administrative data; and random forest for number of institutions applied and log interest rate taken (all measured in survey data), as well as log interest taken measured in administrative data.

J.1 Belief Updating

Table J.1 shows that we reject the null hypothesis of no heterogeneity in treatment effects of the tool for belief updating about the rate participants expect to obtain and the highest rate a bank

²⁹The coefficient estimates will depend on the data split that takes place before training the machine learning models. The coefficient estimates, p -values, and lower and upper bounds of their confidence intervals that we report in the tables in this appendix are the medians of the estimates and confidence intervals from 1,000 such distinct splits, as recommended by Chernozhukov, Demirer, Dufo, and Fernández-Val (2025).

³⁰More specifically, since each of the 1,000 data splits leads to a different Λ , we maximize the median Λ from the 1,000 data splits.

could offer them, but we fail to reject the null hypothesis of no heterogeneity for the lowest rate a bank could offer them and dispersion.

Table J.1: Test for Heterogeneous Treatment Effects on Belief Updating

	Expected rate (1)	Lowest rate (2)	Highest rate (3)	Dispersion (4)
ATE	16.05*** [0.000] (13.18, 18.96)	10.86*** [0.000] (8.56, 13.15)	30.16*** [0.000] (24.61, 35.70)	15.86*** [0.000] (12.24, 19.46)
HTE	0.57*** [0.001] (0.23, 0.91)	0.34 [0.145] (-0.12, 0.83)	0.56*** [0.001] (0.22, 0.91)	0.15 [0.200] (-0.08, 0.38)
Observations	4,699	4,651	4,578	4,302
Control Mean Posterior	29.22	22.65	47.45	23.18
Control Median Posterior	18	12	25	10.68
Bin Density FEs	Yes	Yes	Yes	Yes

This table shows average treatment effect (ATE) and heterogeneous treatment effect (HTE) coefficient estimates from the Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) machine learning procedure described in Appendix J. We conduct this test for the outcomes in Table 5. p -values testing the null hypothesis that the parameter is equal to zero are in square brackets, while 95% confidence intervals are in parentheses. The bin density fixed effects (FEs) are deciles of how many loans are in the price comparison tool histogram that would have been shown to the participant based on their borrower and loan characteristics, had they been assigned to the price comparison tool arm. If any of the participant's loan amount, maturity, or income is outside of the support of the administrative data, the bin density variable would be a missing value; for these individuals, we create an additional fixed effect category that we include in the regression, i.e., we include an additional dummy equal to 1 if the bin density is a missing value. Among the education categories in the baseline characteristics, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity. The smaller sample size compared to Table 5 is due to the exclusion of participants treated with the simple tool. Coefficients, p -values, and lower and upper bounds of confidence intervals are medians over 1,000 data splits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Given that we detected heterogeneous treatment effects for updating about the expected and highest interest rates, we follow Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) to determine which characteristics predict treatment effects of the tool on belief updating. To do this, we split the data into training and estimation samples, and after training the ML model on the training sample, we use the model to estimate predicted individual treatment effects for each individual in the estimation sample. We then group individuals in the estimation sample into quintiles based on predicted individual treatment effects and estimate “group average treatment effects” (GATES) for each quintile. Next, we estimate differences in the average characteristics Z across the groups with the highest and lowest predicted treatment effects (i.e., the most-affected and least-affected groups) in a classification analysis (CLAN).

For GATES and CLAN, we choose the model that maximizes $\bar{\Lambda} \equiv \sum_{k=1}^K \gamma_k^2 \mathbb{P}(S \in I_k)$, where $\gamma_k \equiv \mathbb{E}[s_0(Z) \mid G_k]$ is the sorted group average treatment effect for group k , and $G_k \equiv \{S \in I_k\}$ is the group defined by the interval $I_k \equiv [l_{k-1}, l_k]$; in our case, these intervals correspond to quintile bins of S . Maximizing $\bar{\Lambda}$ is equivalent to maximizing the R^2 in a regression of $s_0(Z)$ on $\bar{S}(Z)$, where $\bar{S} = \sum_{k=1}^K \gamma_k \mathbb{1}(S(Z) \in I_k)$. The models that maximize $\bar{\Lambda}$ are LASSO for all four outcomes.

Table J.2 presents the GATES analysis, estimating treatment effects for the quintiles with the lowest and highest predicted treatment effects. Consistent with our findings in Table J.1, we find that the difference in treatment effects of the tool for the most- and least-affected quintiles is statistically significant (column (3)). Interestingly, although there are statistically significant heterogeneous treatment effects, the tool leads participants to increase their expectations about the rate they will obtain and dispersion even in the least-affected quintile (by 10.3 pp, compared to 24.7 pp in the most-affected quintile). In contrast, and again consistent with the results in Table J.1, we find that the differences in the treatment effects between the most affected and the least affected quintiles for belief updating about the lowest rate and dispersion are not statistically significant at the 5% level. In Table J.3, we present a CLAN analysis to understand which individual traits are associated with the largest impacts on belief updating for the two outcomes for which we observe a statistically significant heterogeneous treatment effect of the tool.

Table J.2: Heterogeneous Treatment Effects: GATES Analysis

	Most affected quintile (1)	Least affected quintile (2)	Difference (3)	Observations (4)
Expected rate	24.67*** [0.000] (18.28, 30.97)	10.26*** [0.002] (3.80, 16.68)	14.38*** [0.002] (5.27, 23.47)	4,699
Lowest rate	15.09*** [0.000] (10.05, 20.17)	9.05*** [0.001] (3.94, 14.15)	6.22* [0.090] (-0.96, 13.40)	4,651
Highest rate	45.08*** [0.000] (32.94, 57.33)	19.29*** [0.002] (6.87, 31.65)	25.65*** [0.004] (8.39, 43.06)	4,578
Dispersion	20.10*** [0.000] (12.04, 28.18)	13.25*** [0.001] (5.19, 21.39)	6.46 [0.263] (-4.91, 17.91)	4,302

The table shows sorted group average treatment effect coefficients for the outcomes in Table 5. The sample is divided into 5 groups, based on the quintiles of the machine learning proxy predictor $S(Z)$. The smaller sample size compared to Tables 3 and 4, panel B, is due to the exclusion of participants treated with the simple tool. Coefficients, p -values, and lower and upper bounds of confidence intervals are medians over 1,000 data splits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table J.3: Heterogeneous Treatment Effects: Classification Analysis (CLAN)

	Expected rate (1)	Highest rate (2)
<i>Personal characteristics</i>		
Age	-11.29*** [0.000] (-12.39, -10.17)	-8.94*** [0.000] (-10.10, -7.72)
log(Income)	-1.68*** [0.000] (-1.88, -1.49)	-1.68*** [0.000] (-1.88, -1.49)
Incomplete high-school	0.12*** [0.000] (0.09, 0.15)	0.11*** [0.000] (0.08, 0.14)
Complete high-school	0.54*** [0.000] (0.49, 0.59)	0.47*** [0.000] (0.41, 0.52)
Complete 2-year program	0.03 [0.173] (-0.01, 0.07)	0.06** [0.013] (0.01, 0.10)
<i>Financial products</i>		
Bank account	-0.47*** [0.000] (-0.52, -0.42)	-0.42*** [0.000] (-0.47, -0.36)
Any loan	-0.32*** [0.000] (-0.37, -0.26)	-0.34*** [0.000] (-0.39, -0.29)
<i>Desired loan characteristics</i>		
log(Loan Amount)	-2.54*** [0.000] (-2.69, -2.39)	-2.65*** [0.000] (-2.81, -2.51)
log(Maturity (years))	-0.60*** [0.000] (-0.67, -0.52)	-0.55*** [0.000] (-0.63, -0.48)
Observations	4,699	4,578

This table shows a classification analysis for the outcomes in Table 5 that exhibit significant HTE coefficients in Table J.1: expected rate and highest rate. The sample is divided into 5 groups, based on the quintiles of the CATE proxy predictor $S(Z)$. We report the mean difference in the baseline characteristics from Table 2 between the highest and lowest $S(Z)$ quintiles. For example, for age, the coefficient is the difference between the mean age of the lowest $S(Z)$ quintile and the mean age of the highest $S(Z)$ quintile. Among the education categories in the baseline characteristics, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity. The smaller sample size compared to Tables 3 and 4, panel B, is due to the exclusion of participants treated with the simple tool. Coefficients, p -values, and lower and upper bounds of confidence intervals are medians over 1,000 data splits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

J.2 Search, Negotiation, and Loan Terms

Across all search, negotiation, and loan term outcomes, we find no detectable heterogeneity in treatment effects (Table J.4). This suggests that the treatment effects of the tool on negotiation, the probability of taking out a loan, and interest rates tend to be spread evenly across participants.

Table J.4: Test for Heterogeneous Treatment Effects on Search, Negotiation, and Loan Terms

	Survey Data						Administrative Data	
	N of inst. searched (1)	N of inst. applied (2)	Pr(negotiate) (3)	N of offers (4)	Pr(take loan) (5)	Log interest rate (6)	Pr(take loan) (7)	Log interest rate (8)
ATE	0.012 [0.904] (-0.185, 0.208)	0.024 [0.733] (-0.117, 0.166)	0.038* [0.057] (-0.001, 0.076)	0.072 [0.119] (-0.019, 0.163)	0.036 [0.232] (-0.023, 0.095)	-0.119 [0.120] (-0.270, 0.031)	0.008 [0.183] (-0.004, 0.021)	0.013 [0.298] (-0.011, 0.037)
HTE	0.172 [0.417] (-0.248, 0.596)	0.311* [0.087] (-0.046, 0.661)	0.254 [0.474] (-0.454, 0.960)	-0.007 [0.979] (-0.554, 0.540)	-0.001 [0.998] (-0.542, 0.542)	0.230 [0.355] (-0.258, 0.711)	0.289 [0.121] (-0.073, 0.680)	0.207 [0.117] (-0.052, 0.466)
Observations	2,172	2,095	2,065	2,084	2,081	354	30,718	5,982

This table shows average treatment effect (ATE) and heterogeneous treatment effect (HTE) coefficient estimates from the Chernozhukov, Demirer, Dufo, and Fernández-Val (2025) machine-learning procedure described in Appendix J. We conduct this test for the outcomes in Tables 3 and 4. p -values testing the null hypothesis that the parameter is equal to zero are in square brackets, while 95% confidence intervals are in parentheses. Missing baseline characteristics are imputed with sample means and missingness indicators are included as additional regressors. Among the education categories in the baseline characteristics, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity. The smaller sample size compared to Tables 3 and 4, panel B, is due to the exclusion of participants treated with the simple tool. Coefficients, p -values, and lower and upper bounds of confidence intervals are medians over 1,000 data splits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix K Effects of Tool on Where Participants Searched

In this appendix, we test whether the price comparison tool affected *where* participants searched, despite not having an average treatment effect on the number of institutions searched or formal applications. Specifically, we test whether the tool induced participants to be more or less likely search or apply at their home bank, at banks other than their home bank, at the banks they expected to search at according to the baseline survey, or at banks they did not list among those where they expected to search. (In the baseline survey, we asked about the first and second bank at which they expected to search, only to those assigned to the elicit beliefs arm.) For each respondent we construct four binary outcomes for each of search and formal applications: whether they searched/applied at their home bank, at a non-home bank, at an expected bank, and at an unexpected bank. Table K.1 reports the results: we find no statistically significant effects of the price comparison tool on where participants searched.

Next, we test whether respondents change the particular banks at which they search. We use the follow-up survey questions in which they listed all banks at which they searched, as well as all banks at which they applied, to run a respondent-bank level regression with bank fixed effects. The

Table K.1: Effects of Tools on Where Participants Searched

	Home bank (1)	Non-home bank (2)	Expected bank (3)	Unexpected bank (4)
<i>Panel A: Searched</i>				
Simple Tool	0.019 (0.018)	−0.019 (0.016)	0.000 (0.028)	−0.055* (0.029)
Price Comparison Tool	−0.010 (0.018)	0.000 (0.016)	0.014 (0.028)	−0.033 (0.029)
Control Mean (in levels)	0.776	0.846	0.639	0.552
Observations	3,123	3,124	1,783	1,783
<i>Panel B: Applied</i>				
Simple Tool	0.024 (0.027)	−0.013 (0.027)	−0.015 (0.038)	0.025 (0.032)
Price Comparison Tool	−0.001 (0.027)	0.005 (0.027)	−0.016 (0.038)	0.034 (0.032)
Control Mean (in levels)	0.628	0.623	0.503	0.756
Observations	1,912	1,912	1,054	1,054

This table reports estimates of treatment effects on individual-level search and application outcomes among consumer-loan applicants. Panels A and B use the follow-up survey questions on where the respondent searched or applied (respectively) to create a dummy variable equal to 1 if the respondent searched or applied at any home bank in column (1), any non-home bank in column (2), any bank they had named as the first or second bank where they expected to search in the baseline survey in column (3), or any bank they had not named at baseline in column (4). Columns (3) and (4) have a smaller sample because these baseline survey questions were only asked to those assigned to the elicit beliefs treatment. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

outcome y_{ib} is a binary indicator equal to one if respondent i reported searching at (or applying to) bank b in the follow-up survey. We then estimate

$$\begin{aligned}
y_{ib} = & \delta_b + \theta \mathbb{1}(\text{Simple Tool})_i + \sum_j \kappa_j \mathbb{1}(\text{Simple Tool})_i \times \mathbb{1}(b = j) \\
& + \beta \mathbb{1}(\text{Price Comparison Tool})_i + \sum_j \gamma_j \mathbb{1}(\text{Price Comparison Tool})_i \times \mathbb{1}(b = j) + \varepsilon_{ib},
\end{aligned} \tag{12}$$

where $\mathbb{1}(\text{Simple Tool})_i$ and $\mathbb{1}(\text{Price Comparison Tool})_i$ are dummies equal to one if the participant was assigned to that treatment arm and δ_b are bank fixed effects. We cluster standard errors at the individual level, since that is the level of treatment. The price comparison tool's treatment effect on search at a particular bank b is $\beta + \gamma_b$ (while the treatment effect for the omitted reference bank is β). We test the joint null that the price comparison tool's treatment effects on search at each bank are zero: $\beta = \beta + \gamma_1 = \beta + \gamma_2 = \dots = 0$. The p -value is 0.438 for search and 0.350 for formal applications. We therefore cannot reject the null that the price comparison tool has no effect on the

specific institutions where respondents searched or applied.

Appendix L Separating Likely Always-Takers from Likely Compliers

Given that the price comparison tool had an effect on the probability of taking out a loan, in this section we attempt to isolate likely “always-takers” (those who would have taken out a loan regardless of treatment) from likely “compliers” (those who only take out a loan if treated). Our goal is to understand whether the lack of an average treatment effect of the price comparison tool on interest rates in the admin data—in contrast with the effect in survey data—is due to a negative treatment effect for always-takers being offset by a selection effect where compliers are more marginal borrowers receiving higher interest rates. If this were the case, the selection effect leading compliers to be included in the regression only in the treatment group would pull the treatment effect towards zero.

We attempt to isolate always-takers and compliers by again implementing Chernozhukov, Demirer, Duflo, and Fernández-Val (2025), using loan take-up as the outcome $Y(\cdot)$. In each iteration, we divide the sample into two equally sized subsamples. Using the training half of the sample, we train two distinct machine learning models that predict $E[Y(1) | Z]$ and $E[Y(0) | Z]$; each model’s hyperparameter optimization uses two-fold cross-validation across 100 evaluations. Following Chernozhukov, Demirer, Duflo, and Fernández-Val (2025), we test various learners: elastic net, random forest, neural nets, and support vector machine. Using the trained models, we predict $E[Y(1) | Z]$ and $E[Y(0) | Z]$ for each observation in the other half of the data. This follows the “honest” approach (Athey and Imbens, 2016) where the model is trained on one subsample while the predictions are obtained using a different subsample.

We use a larger set of characteristics compared to Appendices I and J, as we find these additional covariates are helpful to isolate compliers whereas they make little difference for the analyses in those appendices. In addition to the characteristics from Table 2, we include variables from administrative and baseline survey data quantifying whether the participant is a first-time consumer loan borrower according to administrative data; whether the participant obtained each of a set of other loan products in the past (auto loans, cash advances, credit cards, educational loans, etc.) according to the baseline survey; past search behavior (whether they have searched for a loan in the past, number of loan offers received in the past, etc.) according to the baseline survey; beliefs (interest rate they expect to obtain on the loan they take out, lowest and highest rates banks could offer, biased beliefs about both the first and second moments compared to the conditional distribution in administrative data, etc.); search expectations (how long they expect to search, number of banks they expect to search, expected savings from searching at an additional bank, etc.); financial literacy (using the questions from Lusardi and Mitchell, 2011); self-reported behavioral biases

(procrastination, etc.); and cognitive ability (using the questions from Frederick, 2005).³¹

Always-takers are defined as participants with $E[Y(1) | Z] = E[Y(0) | Z] = 1$, while compliers are defined by $E[Y(1) | Z] = 1, E[Y(0) | Z] = 0$. Thus, an estimand that we find performs well for isolating compliers is

$$\frac{E[Y(1) | Z]}{E[Y(0) | Z]} = \frac{P(Y = 1 | T = 1, Z)}{P(Y = 1 | T = 0, Z)}, \quad (13)$$

as this measure increases as $E[Y(1) | Z] \rightarrow 1$ and as $E[Y(0) | Z] \rightarrow 0$. Furthermore, we find it performs better than a more standard estimand of the treatment effect $E[Y(1) | Z] - E[Y(0) | Z]$, because this difference can be large even for groups in which $E[Y(0) | Z]$ is not close to 0.

We thus plug the individual-level potential outcome estimates $\hat{E}[Y(1) | Z]$ and $\hat{E}[Y(0) | Z]$ from the Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) method into (13) to obtain

$$\frac{\hat{E}[Y(1) | Z]}{\hat{E}[Y(0) | Z]}. \quad (14)$$

We iterate this process 250 times and, for each individual, take the median of (14) across iterations.³²

We then sort observations by their median values of (14), and group observations into quantiles. Across the various learners tested, we choose the model with the highest estimate of $\bar{\Lambda}$ (defined in Appendix J). The learner with the highest estimate of $\bar{\Lambda}$ was support vector machine. We then tested model performance under different quantile groupings (deciles, quintiles, quartiles, and terciles). We cannot compare these by their estimates of $\bar{\Lambda}$ since this statistic mechanically increases in the number of groups. Thus, we assess the first stage for each quantile grouping by looking at treatment effects on loan take-up by quantile, and choose the quantile grouping that best separates likely always-takers from likely compliers. The quantile grouping that demonstrated the strongest first stage was terciles. We classify individuals in the lowest tercile as likely always-takers, as the tool's treatment effect on loan take-up is zero for this tercile, and those in the second and third terciles as likely compliers, as the tool's treatment effect on loan take-up is large and statistically significant for each of these terciles. In other words, we find that likely compliers are spread across the second and third terciles of (14).

³¹As in Appendices I and J, for education level, we omit the indicator for whether a participant completed a 5-year degree program or higher to avoid perfect collinearity. To retain the full sample, we impute missing characteristics with sample means and include missingness indicators for all variables as additional covariates.

³²This is the same number of iterations used by Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) in their empirical application. Since the initial sample split divides the data in half, on average 125 estimates of (14) are computed for each individual across the 250 iterations, but there is variation in this across individuals due to the random nature of the sample splits.

We then estimate

$$y_i = \alpha + \beta_1 \mathbb{1}\{\text{Price Comparison Tool}\}_i + \beta_2 \mathbb{1}\{\text{Complier}\}_i + \beta_3 \mathbb{1}\{\text{Price Comparison Tool}\}_i \times \mathbb{1}\{\text{Complier}\}_i + \varepsilon_i, \quad (15)$$

where y_i is either loan take-up—to test whether our method successfully separates likely always-takers from likely compliers—or log interest rates. $\mathbb{1}\{\text{Complier}\}_i$ is an indicator variable equal to 1 when the observation is labeled as a likely complier based on their tercile of (14). The estimate $\hat{\beta}_1$ can be interpreted as the treatment effect for likely always-takers, while $\hat{\beta}_1 + \hat{\beta}_3$ is the estimated coefficient on treatment for compliers (which is contaminated by any selection effect).

Table 4, panel C, presents the treatment effect of the price comparison tool for likely always-takers ($\hat{\beta}_1$), the selection-contaminated effect for likely compliers ($\hat{\beta}_1 + \hat{\beta}_3$), and the p -value of a test of whether these differ (the p -value of the test $\beta_3 = 0$). These results are discussed in the main text.

Appendix M Inverse-U Test

In order to test whether the inverse-U shape in Figure 6 is statistically significant, we follow the approach described in Wolak (1989). Based on Prediction 2 and the empirical results in Figure 6, we test for the presence of an inverse-U shape with its peak at bin $(-10, 0]$ by jointly testing $\beta_{\leq -30} < \beta_{(-30, -20]} < \beta_{(-20, -10]} < \beta_{(-10, 0]}$ and $\beta_{(-10, 0]} > \beta_{(0, 10]} > \beta_{(10, 20]} > \beta_{> 20}$, where subscripts indicate bins of bias about the first moment, as used in the x-axis of Figure 6.

Following Wolak (1989), we conduct two complementary tests. First, we test whether the data are inconsistent with the hypothesized inverse-U shape by treating the inequality constraints above as the null hypothesis versus an unrestricted model. We fail to reject the null ($p = 0.887$), implying that the data do not contradict the inverse U-shape. Second, we test whether the constraints are jointly satisfied by testing a null hypothesis that the difference between $\hat{\beta}$ coefficients following our inverse-U shape is jointly zero.³³ We reject this null hypothesis at the 10% significance level ($p = 0.089$).

Taken together, these two tests recommended by Wolak (1989) suggest that the non-monotonic inverse-U-shaped treatment effect of the price comparison tool on the probability of negotiating—as predicted by Prediction 2 and seen visually in Figure 6—is statistically significant.

³³In both tests, we estimate the covariance matrix of our test statistic via 5,000 bootstrap iterations resampling at the individual level and re-estimating the bin-specific regressions in each iteration. Both statistics follow a weighted sum of χ^2 distributions. We estimate these weights via 100,000 Monte Carlo draws. The inverse-U shape hypothesis encodes six consecutive constraints: $\beta_{(-30, -20]} - \beta_{\leq -30} \geq 0$, $\beta_{(-20, -10]} - \beta_{(-30, -20]} \geq 0$, $\beta_{(-10, 0]} - \beta_{(-20, -10]} \geq 0$ on the increasing side, and $\beta_{(-10, 0]} - \beta_{(0, 10]} \geq 0$, $\beta_{(0, 10]} - \beta_{(10, 20]} \geq 0$, $\beta_{(10, 20]} - \beta_{> 20} \geq 0$ on the decreasing side. The first test takes these six inequalities as the null hypothesis. The second test takes as its null that all six differences are exactly zero, testing against the alternative that they are all non-negative.

Appendix N Mechanism Testing

To test whether the mechanism underlying the treatment effect of the price comparison tool on negotiation is that the tool shifted beliefs about the interest rate distribution, we conduct a Kwon and Roth (2026) mechanism test. Specifically, Kwon and Roth (2026) propose a test of the sharp null that the effect of treatment T on outcome Y operates entirely through a specific mediator or mechanism M . In our context, we test whether the the observed effect of the price comparison tool (T) on whether an individual negotiated with a financial institution (Y) is fully driven by the effect of the tool on beliefs about interest rates (M).³⁴

We consider bias about both moments of the interest rate distribution, where we measure bias in the first moment using equation (10) and bias about the second moment using equation (11). However, since our mechanism M should be measured post-treatment, we now use the individual's *posterior* beliefs based on the survey questions after viewing the price comparison tool or control video to measure r_i^{belief} , $\bar{r}_i^{belief}$, and $\underline{r}_i^{belief}$ in equations (10) and (11). We exclude participants who skipped all three posterior beliefs questions.

Using these two measures of posterior beliefs, we partition individuals according to the degree of posterior bias, given that the Kwon and Roth (2026) test requires M to take on discrete values with a sufficient number of observations per category. To maintain a sufficient number of observations per category, we create groups based on our theoretical model, and order these groups for the test such that the monotonicity assumption should be satisfied according to our theoretical model. Specifically, we create four groups and order them in ascending order of the probability of successfully negotiating, where it is worth noting that the descriptions below are based on whether participants are more or less likely to negotiate according to their *posteriors* after seeing the price comparison tool or control video, which should not be confused with our model predictions about heterogeneous treatment effects of the tool based on participants' *priors*:

1. Those who responded “don’t know” to any of the three posterior interest rate questions are unlikely to successfully negotiate given that they do not have quantifiable beliefs about the distribution.³⁵
2. Those whose posteriors vastly underestimate or overestimate the first moment of the distribution and underestimate the second moment are less likely to successfully negotiate than those with different beliefs, for the reasons discussed in Section 1.
3. Those who vastly underestimate or overestimate the first moment and *overestimate* the second moment are more likely to negotiate than group 2, given that they overestimate the

³⁴Thus, we keep the price comparison tool and control groups for this test and drop the simple tool group.

³⁵Those who responded “do not want to answer” or skipped the question are also counted as “don’t know” responses, unless they abandoned the entire survey at this module in which case they are excluded.

second moment and thus have a lower reservation rate. Similarly, those who *somewhat underestimate* the first moment and *underestimate* the second moment are also more likely to negotiate than group 2, given that they somewhat underestimate—rather than vastly underestimate or overestimate—the first moment, again for the reasons discussed in Section 1. Because the model does not indicate which of these two subgroups is more or less likely to negotiate than the other, we group them together.

4. Those who somewhat underestimate the first moment and overestimate the second moment are the most likely to successfully negotiate.

Based on the empirical results from Figure 6, we define vastly underestimating the first moment as having a posterior bias less than (i.e., more negative than) -30 pp, and somewhat underestimating as having a posterior bias between -30 and 0 pp. Here, we do not assume that those whose beliefs are within 1 pp of the true first or second moment have accurate beliefs, given the need to have a sufficient number of observations per group and thus to define a small number of groups.

The Kwon and Roth (2026) test assumes monotonicity, i.e., that if T shifts M it can move consumers from lower-numbered to higher-numbered groups, but not the reverse. There are three reasons that this assumption is likely satisfied by the groups and ordering we impose. First, “don’t know” is the first group as the price comparison tool should shift individuals who would have otherwise responded “don’t know” in the absence of seeing the tool to a well-defined posterior belief, but not from a well-defined posterior to responding “don’t know.” Second, if seeing the tool leads participants to update by shifting their beliefs closer to the distribution shown in the tool—as we document empirically—consumers will either remain in the same group (e.g., from underestimating the second moment to underestimating it to a lesser extent) or will shift from vastly underestimating the first moment to somewhat underestimating it, which always implies moving to a higher-numbered group in the ordering above. Third, the groups above are in ascending order of the probability of negotiating based on participants’ posterior beliefs (according our theoretical model), and we know that the tool led to an increase in negotiation, so the tool is likely shifting people to higher-numbered groups consistent with the monotonicity assumption.

The test also assumes independence, which require treatment assignment to be independent of potential outcomes and mediators. Given the randomized design of our experiment, this condition is satisfied.

Results Our results indicate that we cannot reject the hypothesis that belief updating is the sole mechanism driving the effect of the tool on negotiation behavior ($p = 1.000$).

Placebo test To ensure that our failure to reject the sharp null—that the effect of the price comparison tool on negotiation is fully explained by its effect on beliefs—is not an artifact of our test being underpowered, we conduct a placebo test. Specifically, we define the mediator (M) as search

(the number of institutions searched by the individual). We group and order individuals according to the following categories based on the number of institutions searched, in order to ensure that no group has too few observations and to match the number of groups in our main test above: (i) 0–1, (ii) 2, (iii) 3, and (iv) 4 or more institutions. We use the same sample as in our main test so that any differences in results cannot be attributed to changes in the sample. If we failed the placebo test, we would conclude that the Kwon and Roth (2026) test is underpowered in our setting, as the effect of the price comparison tool on negotiation cannot be fully explained by more than one mechanism. However, we reject the null hypothesis at the 10% significance level, implying that search behavior alone cannot explain the treatment-induced increase in negotiation ($p = 0.081$).

We conclude that the effect of the price comparison tool on negotiation is largely explained by its effect on beliefs about the interest rate distribution. This is useful to rule out that other potential mechanisms such as salience, attention, or consumers directly showing the tool to lenders as credible information that they could get a better rate elsewhere explain most of the tool’s effect on negotiation.

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